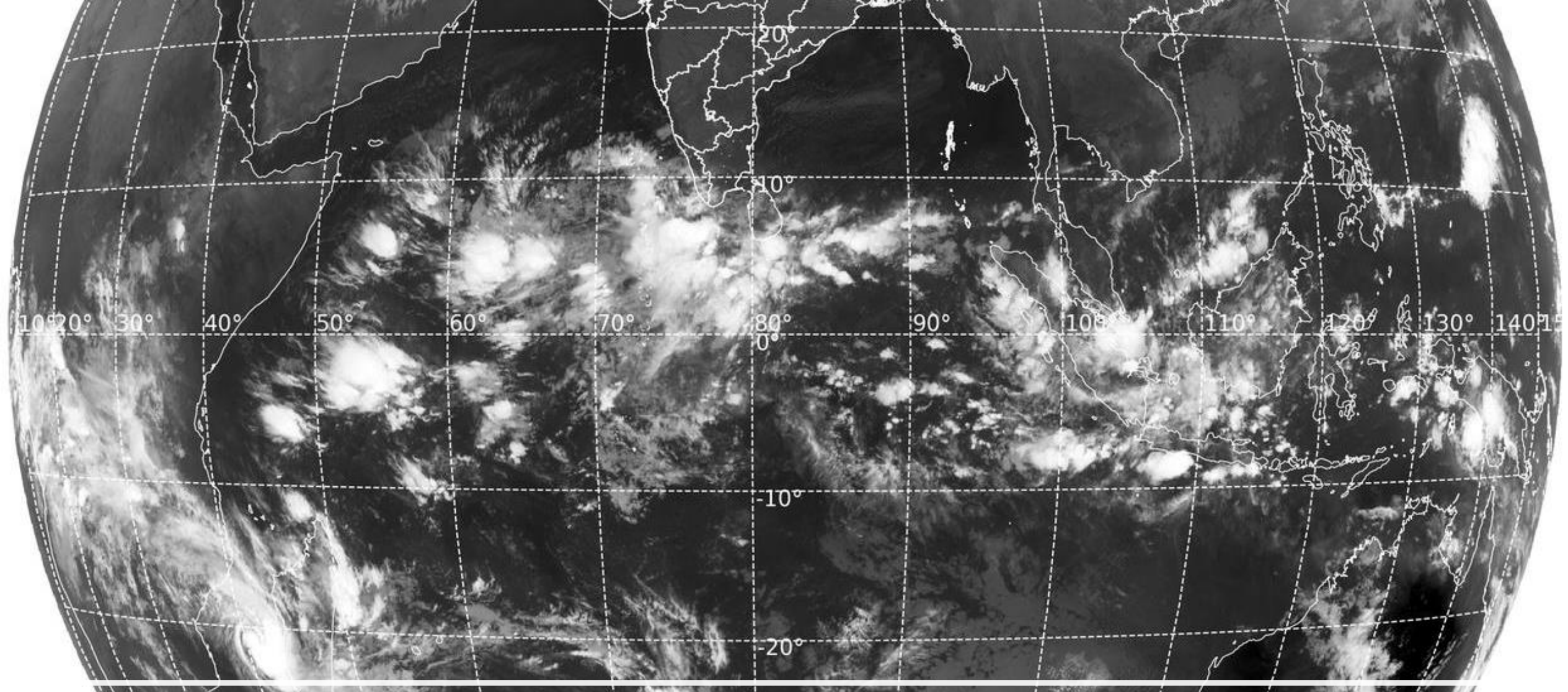




Remote Sensing and Geographic Information Science

Note: In this ppt, figures/text are adopted from various sources; this is only for class explanation purpose.

Course Content: CE 3420 and CE 6610

- Segments 1-2 : Remote Sensing (RS)
- Segments 3-4: Digital Image Processing (DIP)
- ~~Segments 3-4: Geographic Information Science (GIS)~~
- Segments 1-4: Geographic Information Science (GIS) (Thursday's : Bring your laptops)
- Segments 5-6 (CE 6610) : Dr. Satish Regonda

Course Evaluation

- Two announced Quiz – 20% each = 40 % (PG Course: 30%)
 - Quiz-I: RS and GIS
 - Quiz- II: DIP and GIS
- End Semester: 30% (PG Course : 20%)
- Assignments/Practical component: Google Earth Engine and Quantum GIS (QGIS) : 15% (PG Course: 5%)
 - Pop Quiz (Numbers not defined) + Proxy for attendance : 15% (PG Course: 5%)
- 5 min class presentations (volunteer basis): 5 % Bonus
- Segments 5-6 (CE 6610) : Dr. Satish Regonda (35%)

Reference Books

Remote Sensing and Digital Image Processing

- Rees, W.G., Physical Principles Of Remote Sensing, Cambridge University Press, 2001.
- Sabins F., Remote Sensing : Principles And Interpretation, New York 1997
- Lillesand T.M., And Kiefer R.M., Remote Sensing And Image Interpretation, Fourth Edition, Wiley, 1999.
- Jensen J.R., Remote Sensing Of Environment: An Earth Resource Perspective, Prentice Hall, 2000.
- F.M. Henderson And A.J. Lewis (Ed.), Principles And Applications Of Imaging Radar, Manual Of Remote Sensing, 3rd Edition, Vol. 2, ASPRS, 1998
- R. C. Gonzalez, R. E. Woods, 2008, Digital Image Processing, Prentice Hall.
- R. C. Gonzalez, R. E. Woods, W. L. Eddins, 2009, Digital Image Processing using Matlab, Pearson

GIS

- Burrough P.A. and McDonnell R.A., “Principles of Geographical Information Systems”, Oxford University Press, 2006.
- Ian Heywood Sarah, Cornelius and Steve Carver, “An Introduction to Geographical Information Systems”. 3rd Edition, Pearson Education. New Delhi, 2006.
- Michael Worboys and Matt Duckham, GIS: A Computing Perspective -2nd edition, CRC Press, Boca Raton, 2004.
- Lo, C.P. and Yeung, Albert K.W., Concepts and Techniques of Geographic Information Systems Prentice Hall, 2002.

Remote Sensing

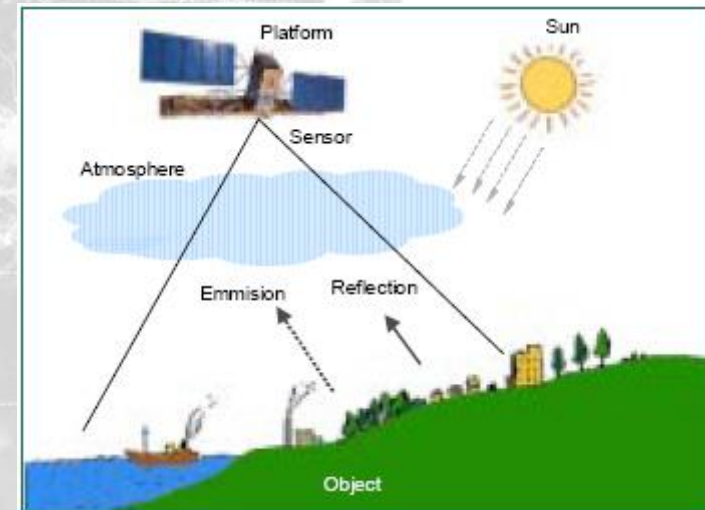
The art and science of obtaining information about an object or feature without physically coming in contact with that object or feature

Remote sensing can be used to measure

- Variations in acoustic wave distributions ([Details](#))
- Variations in force distributions (e.g., gravity meter)
- Variations in electromagnetic energy distributions

Remotely collected data through various sensors

may be analyzed to obtain information about the objects or features under investigation



<http://geoportal.icimod.org>

Electromagnetic Energy

Electromagnetic energy or electromagnetic radiation (EMR) : [Details](#)

- Energy propagated in the form of an advancing interaction between electric and magnetic fields (Sabbins, 1978)
- Travels with the velocity of light
- Visible light, ultraviolet rays, infrared, heat, radio waves and x-rays are different forms

Expressed either in terms of frequency (f) or wave length (λ) of radiation

$$E = h.f \quad \text{or} \quad h.c / \lambda$$

h = Planck's constant (6.626×10^{-34} Joules-sec)

c = Speed of light (3×10^8 m/sec)

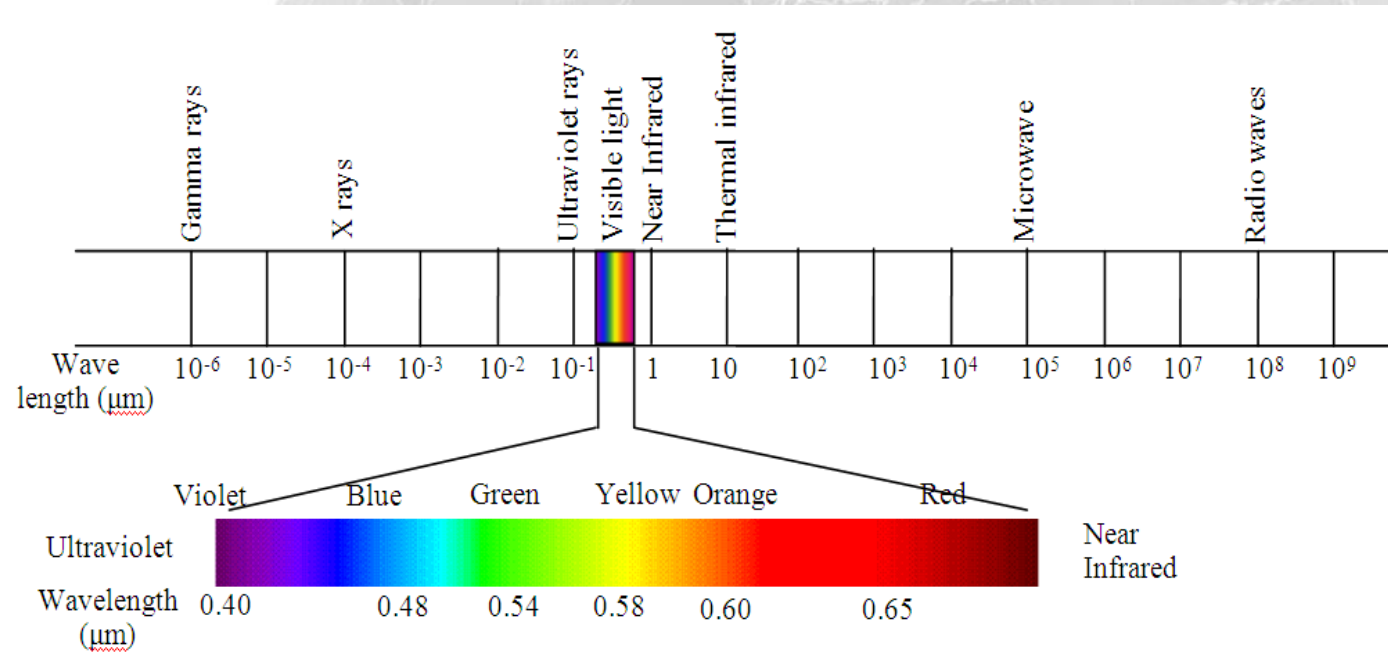
f = Frequency expressed in Hertz

λ = wavelength in micro meters (μm)

Shorter wavelengths have higher energy content and longer wavelengths have lower energy content

Electromagnetic Energy...

EMR spectrum : Distribution of the continuum of energy plotted as a function of wavelength (or frequency)



In remote sensing terminology, electromagnetic energy is generally expressed in terms of wavelength, λ .

Principles of Remote Sensing

Different objects reflect or emit different amount of energy in different bands of the electromagnetic spectrum

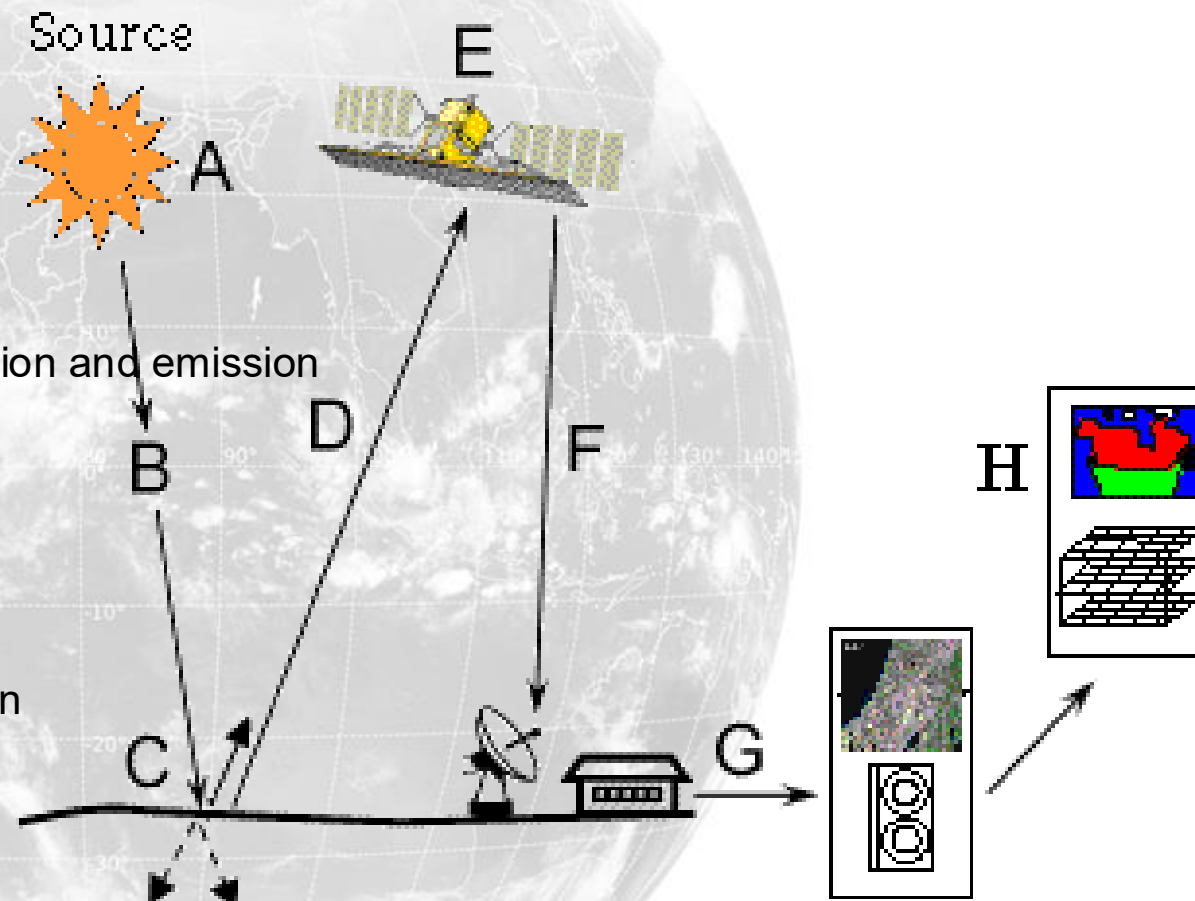
- Depends on the properties of
 - The target material
 - The incident energy (angle of incidence, intensity and wavelength)
- Uniqueness of the reflected or emitted electromagnetic radiation is used to detect and discriminate the objects or surface features

Sensor & Platform in remote sensing

- Sensor: A device used to detect the reflected or emitted electromagnetic radiation
 - Cameras and scanners
- Platform: A vehicle used to carry the sensor
 - Aircrafts and satellites

Stages in Remote Sensing

- A. Emission of electromagnetic radiation
 - The Sun or an EMR source located on the platform
- B. Transmission of energy from the source to the object
 - Absorption and scattering of the EMR while transmission
- C. Interaction of EMR with the object and subsequent reflection and emission
- D. Transmission of energy from the object to the sensor
- E. Recording of the energy at the sensor
 - Photographic or non-photographic
- F. Transmission of the recorded information to ground station
- G. Processing of the data into digital or hard copy image
- H. Analysis of data



Passive/ Active Remote Sensing

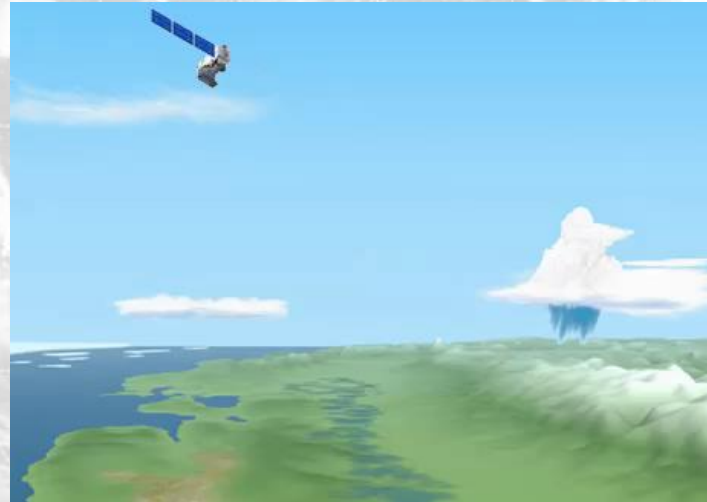
A simple analogy:

Passive remote sensing is similar to taking a picture with an ordinary camera

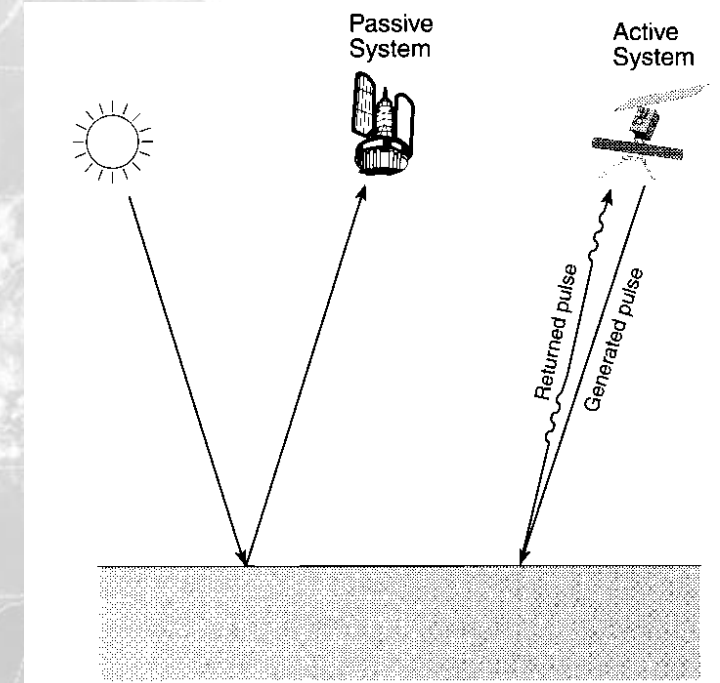
Active remote sensing is analogous to taking a picture with camera having built-in flash



Active Remote Sensing



Passive Remote Sensing



Passive Remote Sensing

Passive remote sensing: Source of energy is that naturally available

- Solar energy
- Energy emitted by the Earth etc.

Most of the remote sensing systems work in passive mode using solar energy : **Optical Remote Sensing**

- Solar energy reflected by the targets at specific bands are recorded using sensors
- For ample signal strength received at the sensor, wavelengths capable of traversing through the atmosphere without significant loss, are generally used

The Earth will also emit some radiation since its ambient temperature is about 300° K.

- Passive sensors can also be used to measure the Earth's radiance
- Challenge is that the energy content is very low

Active Remote Sensing

- Active remote sensing: Energy is generated and emitted from a sensing platform towards the targets
- Energy reflected back by the targets are recorded
- More generally: Longer wavelength bands are used

Example: Active microwave remote sensing (radar)

- Pulses of microwave signals are sent towards the target from the radar antenna located on the air / space-borne platform
- The energy reflected back (echoes) are recorded at the sensor

Remote Sensing Platforms

Ground level remote sensing

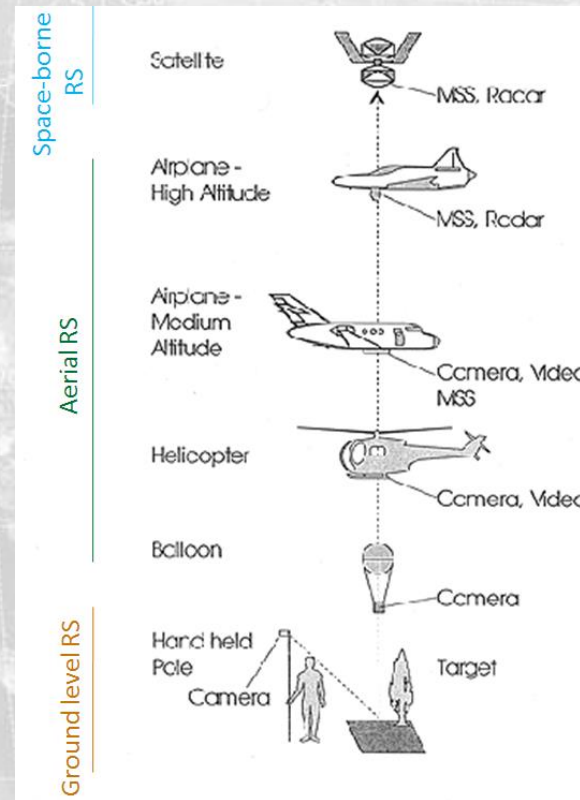
- Very close to the ground (e.g., Hand held camera)
- Used to develop and calibrate sensors for different features on the Earth's surface

Aerial remote sensing

- Low altitude aerial remote sensing
- High altitude aerial remote sensing

Space-borne remote sensing

- Space shuttles
- Polar orbiting satellites
- Geo-stationary satellites



Modified from

<http://www.ilmb.gov.bc.ca/risc/pubs/aquatic/aerialvideo/assets/figure1.gif>

Air-borne Remote sensing

- Downward or sideward looking sensors mounted on aircrafts are used to obtain images
- Very high spatial resolution images (20 cm or less) can be obtained

Drawbacks:

- Less coverage area and high cost per unit area of ground coverage
- Mainly intended for one-time operations, whereas space-borne missions offer continuous monitoring of the earth features
- LiDAR, analog aerial photography, thermal imagery and digital photography are commonly used in airborne remote sensing

Space-borne Remote sensing

Sensors are mounted on space shuttles or satellites orbiting the Earth

- Geostationary and Polar orbiting satellites

- Example: Landsat satellites, Indian remote sensing (IRS) satellites, IKONOS, SPOT satellites, AQUA and TERRA (NASA), and INSAT satellite series

Advantages:

- Large area coverage, less cost per unit area of coverage
- Continuous or frequent coverage of an area of interest
- Automatic/ semi-automatic computerized processing and analysis.

Drawback: Lower resolution

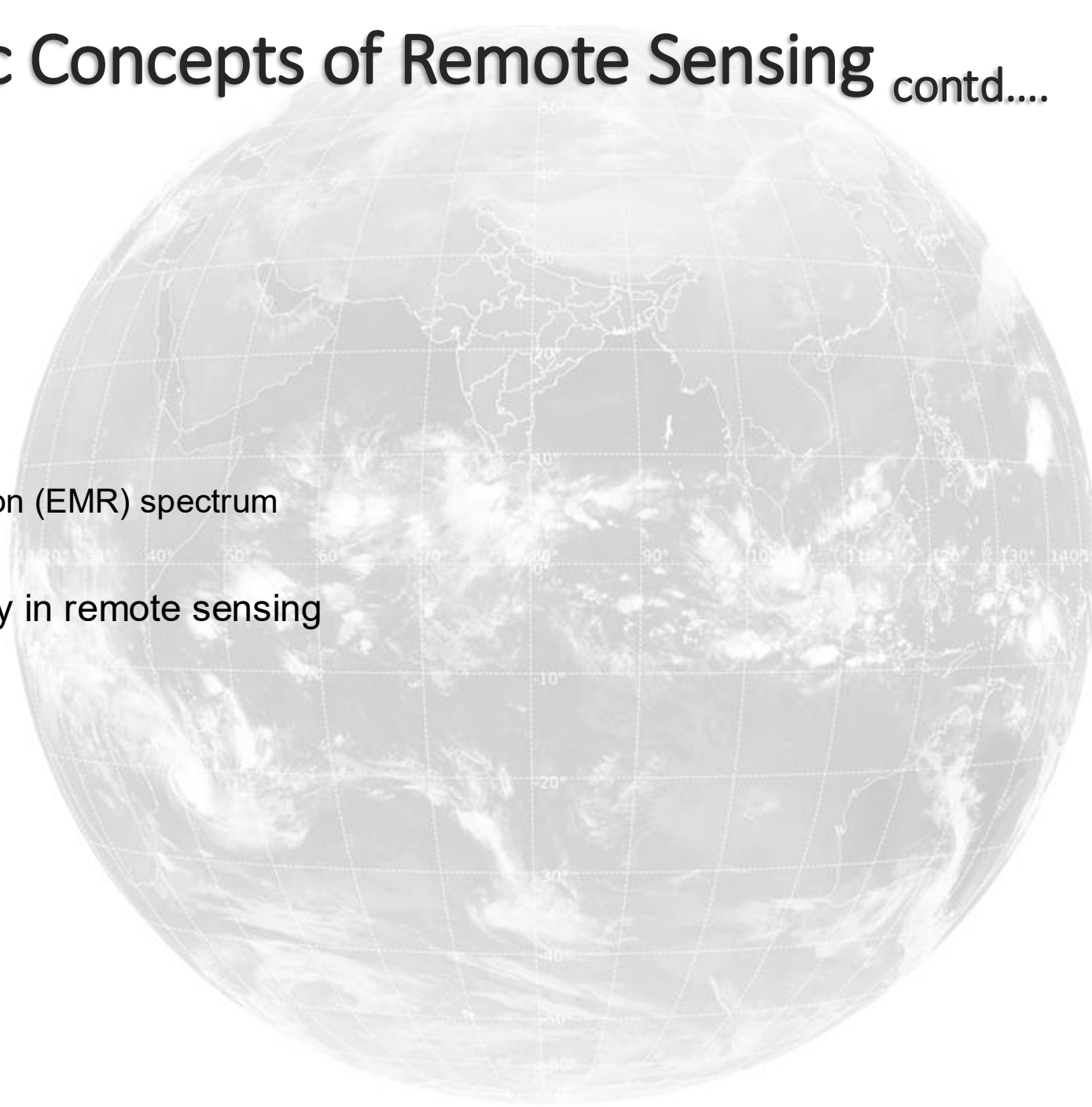
Basic Concepts of Remote Sensing contd....

What is meant by

Electromagnetic energy

Electromagnetic radiation (EMR) spectrum

Source of radiation/energy in remote sensing



Electromagnetic Energy

Electromagnetic energy: All energy moving in a harmonic sinusoidal wave pattern with a velocity equal to that of light

- Harmonic pattern means waves occurring at frequent intervals of time.

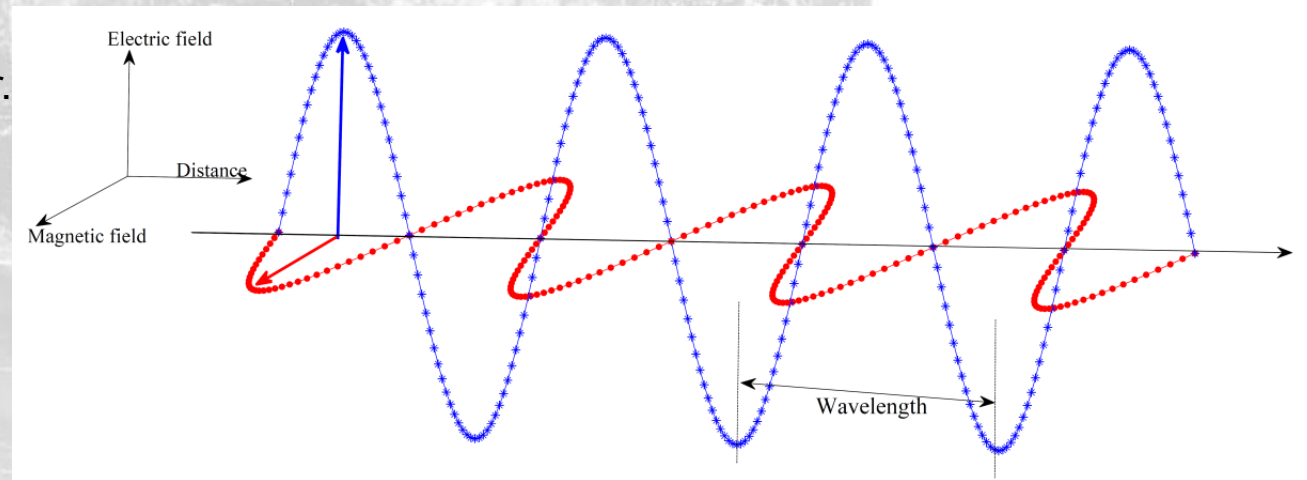
Contains both electric and magnetic components which oscillate

- Perpendicular to each other and

- Perpendicular to the direction of energy propagation

It can be detected only through its interaction with matter.

- Example: Light, heat etc.



Electromagnetic Energy...

Characteristics of electromagnetic (EM) energy – Wave theory

- Velocity (c)

- EM waves travel at the speed of light (3×10^8 m/s.)

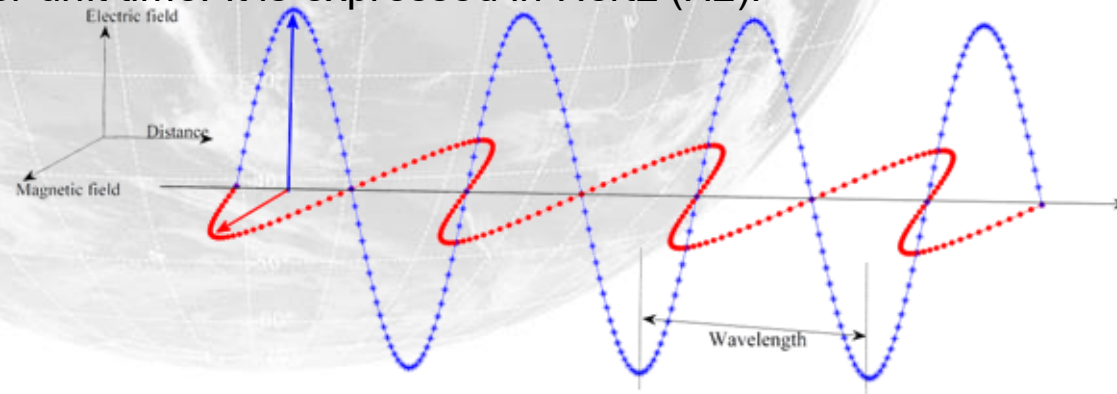
- Wavelength (λ)

- Distance from any point of one wave to the same position on the next wave
 - The wavelengths commonly used in remote sensing are very small
 - It is normally expressed in micrometers ($1 \mu\text{m} = 1 \times 10^{-6}$ m)
 - In remote sensing EM waves are categorized in terms of their wavelength location in the EMR spectrum

- Frequency (f)

- Number of waves passing a fixed point per unit time. It is expressed in Hertz (Hz).

$$c = \lambda f$$



Electromagnetic Energy...

Characteristics of electromagnetic (EM) energy – Particle theory

- Electromagnetic radiation is composed of discrete units
- These discrete units are called Photons or Quanta
- Photons are the basic units of EM energy

$$E = h.f \quad \text{or} \quad h.c / \lambda$$

h = Planck's constant (6.626×10^{-34} Joules-sec)

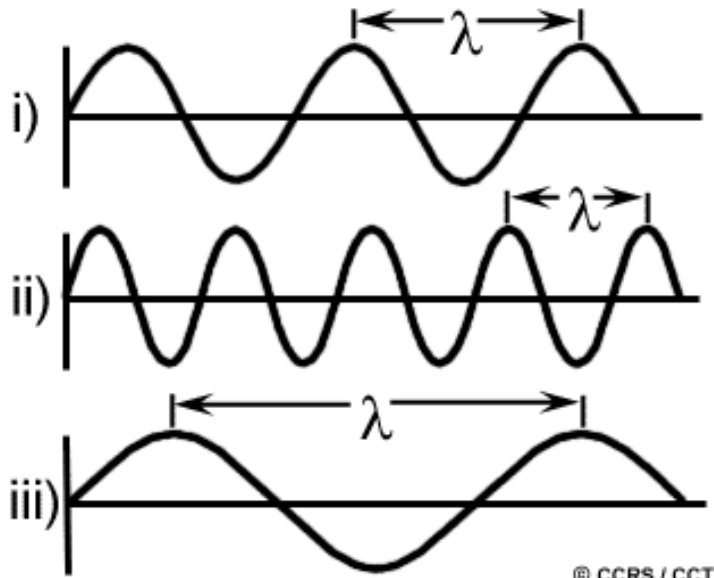
c = Speed of light (3×10^8 m/sec)

f = Frequency expressed in Hertz

λ = wavelength in micro meters (μm)

Electromagnetic Radiation (EMR)

- Electromagnetic waves can be described by their wavelengths, energy, and frequency



© CCRS / CCT

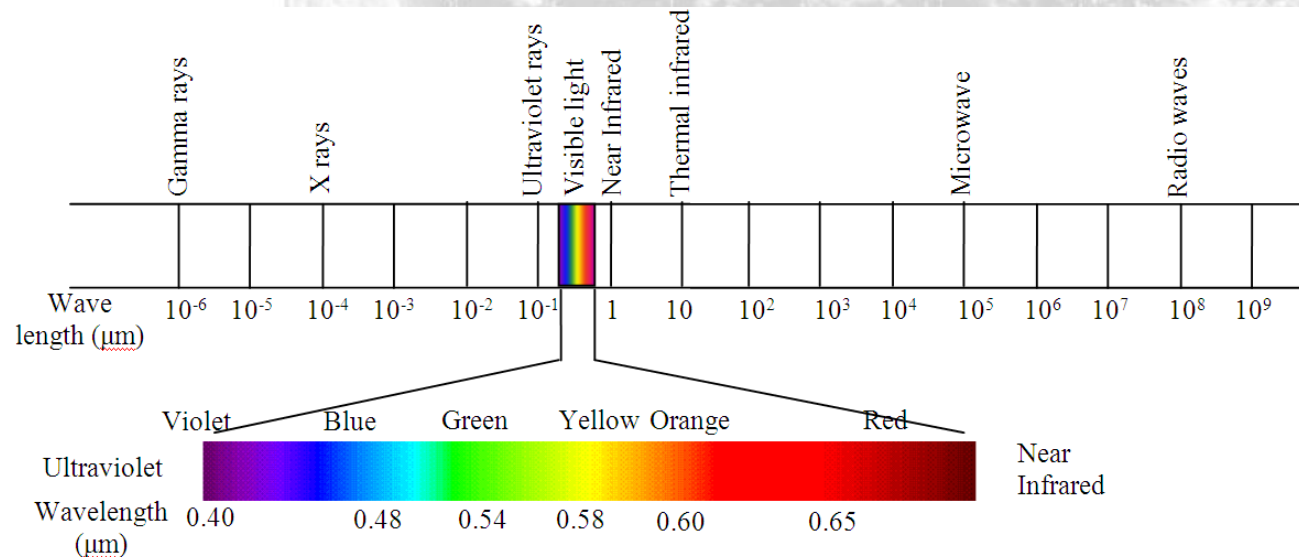
$f * \lambda = c$	$E = h * f$
f = frequency	E = energy
λ = wavelength	h = Planck's constant
c = speed of light (3×10^8 m/sec)	f = frequency

Higher the Wavelength → Lower the Frequency → Lower the energy

EMR Spectrum

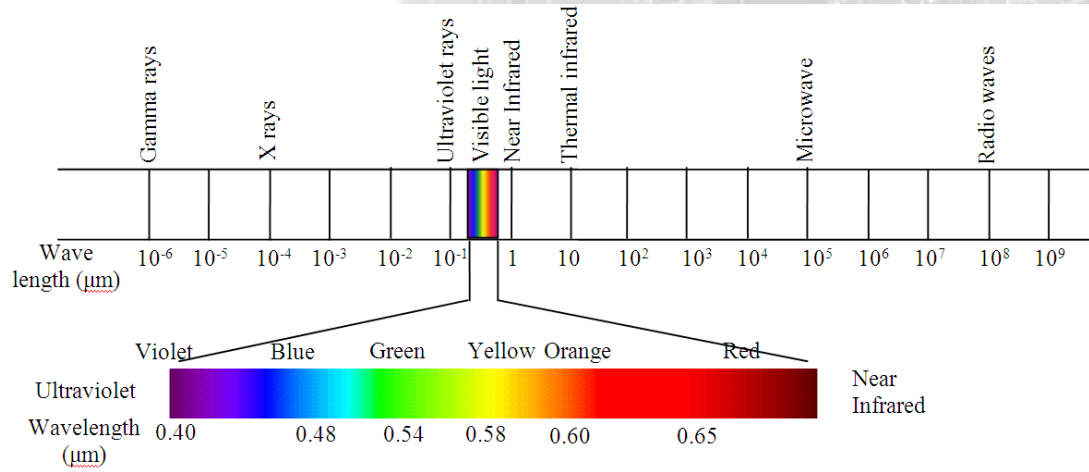
EMR Spectrum: Electromagnetic radiation (EMR) spectrum

- Distribution of the continuum of radiant energy plotted as a function of wavelength (or frequency)
- Divided into regions or intervals
- No strict dividing line between one spectral region and its adjacent one



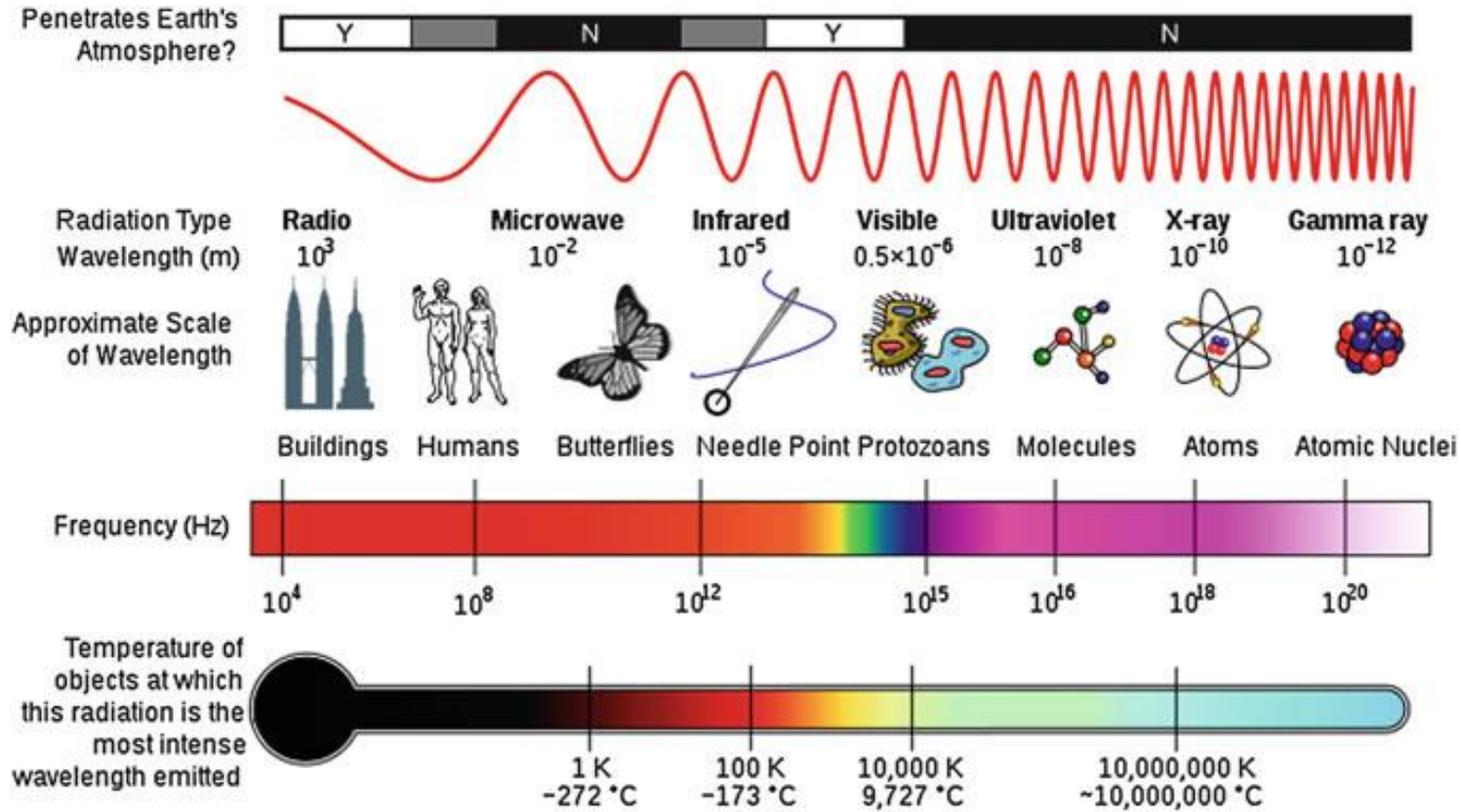
EMR Spectrum...

Ranges from gamma rays (very short) to radio waves (long wavelengths)



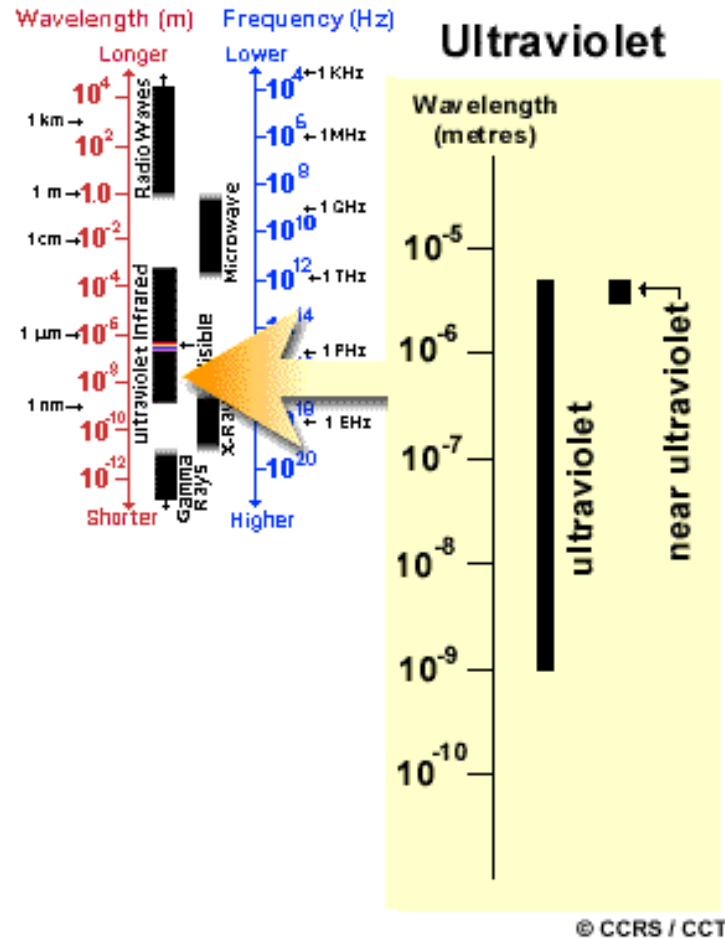
- Gamma rays, X-rays and most of the UV rays
 - Mostly absorbed by the earth's atmosphere and hence not used in remote sensing
- Most of the remote sensing systems operate in visible, infrared (IR) and microwave regions
- Some systems use the long wave portion of the UV spectrum

Electromagnetic Spectrum



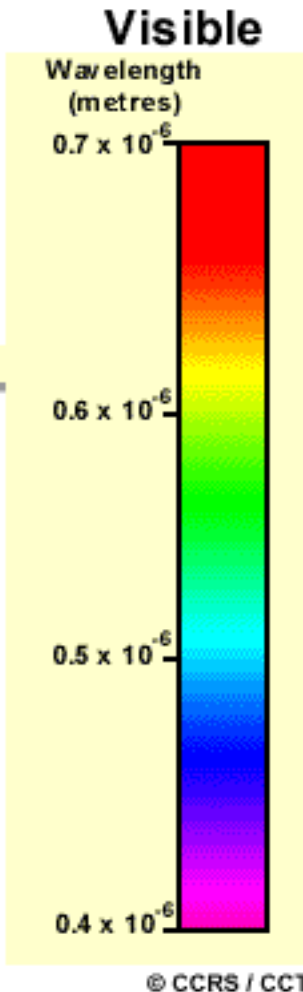
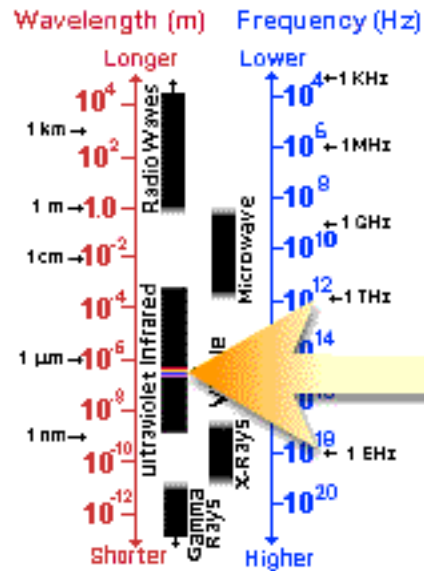
- The Sun produces a continuous spectrum of energy from gamma rays to radio waves
- The earth's atmosphere, clouds, and surface continuously emit radiation throughout the electromagnetic spectrum
- Various parts of the EM spectrum may be differentiated using wavelength

Electromagnetic Spectrum



- For most purposes, the ultraviolet or UV portion of the spectrum has the shortest wavelengths which are practical for remote sensing.
- This radiation is just beyond the violet portion of the visible wavelengths, hence its name.
- Some Earth surface materials, primarily rocks and minerals, fluoresce or emit visible light when illuminated by UV radiation.

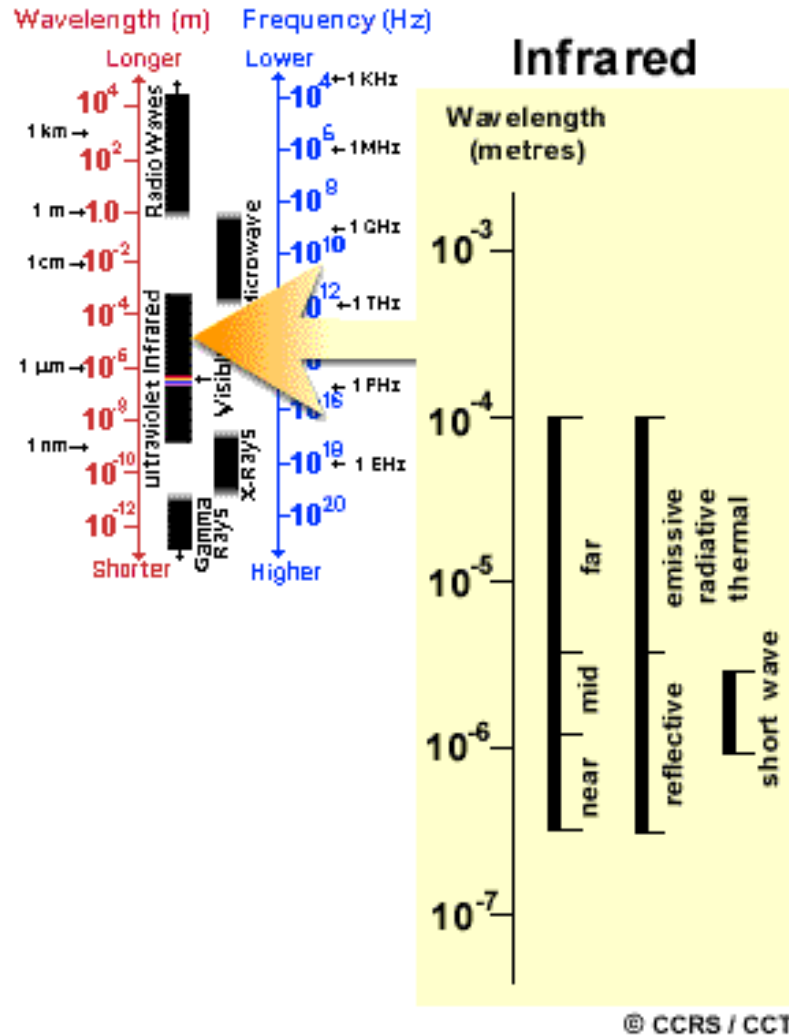
Electromagnetic Spectrum



Violet: 0.4 - 0.446 μm
Blue: 0.446 - 0.500 μm
Green: 0.500 - 0.578 μm
Yellow: 0.578 - 0.592 μm
Orange: 0.592 - 0.620 μm
Red: 0.620 - 0.7 μm

- The light which our eyes - our "remote sensors" - can detect is part of the visible spectrum.
- It is important to recognize how small the visible portion is relative to the rest of the spectrum.
- There is a lot of radiation around us which is "invisible" to our eyes, but can be detected by other remote sensing instruments and used to our advantage.
- The visible wavelengths cover a range from approximately 0.4 to 0.7 μm .
- The longest visible wavelength is red and the shortest is violet.
- Common wavelengths of what we perceive as particular colors from the visible portion of the spectrum are given below.
- It is important to note that this is the only portion of the spectrum we can associate with the concept of colors.

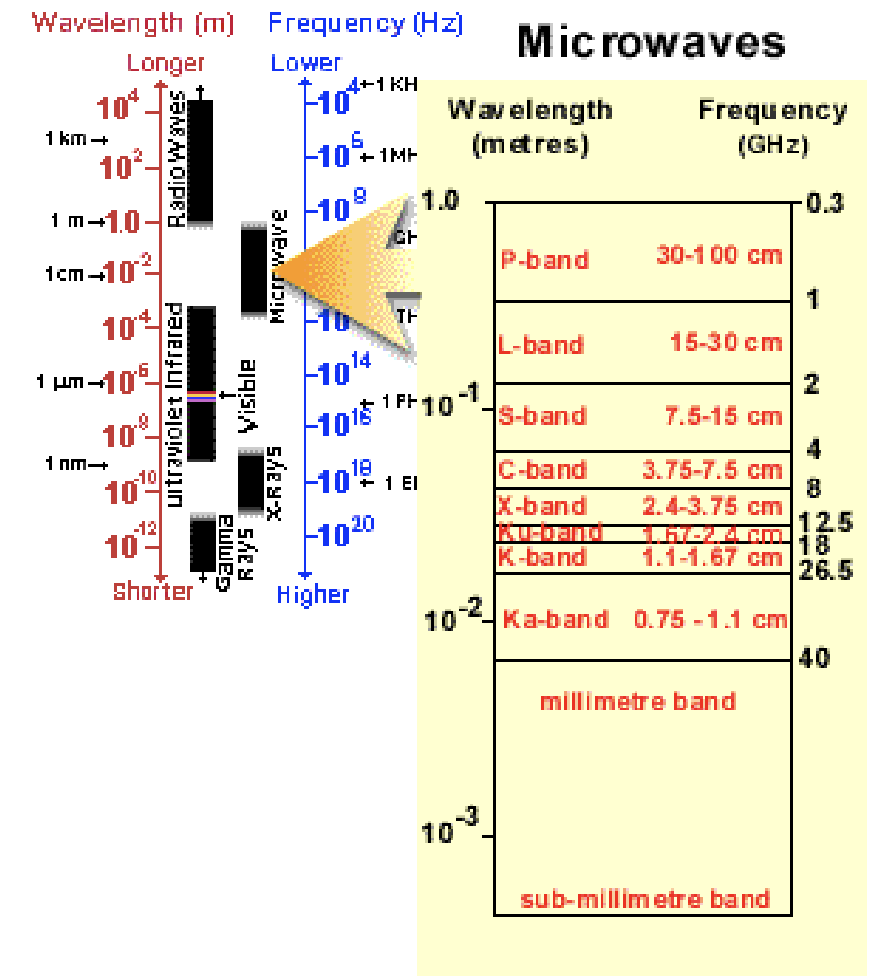
Electromagnetic Spectrum



- The infrared (IR) region covers the wavelength range from approximately $0.7 \mu\text{m}$ to $100 \mu\text{m}$ - more than 100 times as wide as the visible portion!
- The infrared region can be divided into two categories based on their radiation properties - the reflected IR, and the emitted or thermal IR.
- Radiation in the reflected IR region is used for remote sensing purposes in ways very similar to radiation in the visible portion.
- The reflected IR covers wavelengths from approximately $0.7 \mu\text{m}$ to $3.0 \mu\text{m}$.
- The thermal IR region is quite different than the visible and reflected IR portions, as this energy is essentially the radiation that is emitted from the Earth's surface in the form of heat.
- The thermal IR covers wavelengths from approximately $3.0 \mu\text{m}$ to $100 \mu\text{m}$.

Electromagnetic Spectrum

- The portion of the spectrum of more recent interest to remote sensing is the microwave region from about 1 mm to 1 m.
- This covers the longest wavelengths used for remote sensing.
- The shorter wavelengths have properties similar to the thermal infrared region while the longer wavelengths approach the wavelengths used for radio broadcasts.



© CCRS / CCR

EMR Spectrum...

Region	Wavelength (μm)	Remarks
Gamma rays	$< 3 \times 10^{-5}$	Not available for remote sensing. Incoming radiation is absorbed by the atmosphere
X-ray	$3 \times 10^{-5} - 3 \times 10^{-3}$	
Ultraviolet (UV) rays	0.03 - 0.4	Wavelengths < 0.3 are absorbed by the ozone layer. Wavelengths between 0.3- 0.4 μm are transmitted and termed as “Photographic UV band”.
Visible	0.4 - 0.7	Detectable with film and photodetectors.
Infrared (IR)	0.7 - 100	Specific atmospheric windows allows maximum transmission. Photographic IR band (0.7-0.9 μm) is detectable with film. Principal atmospheric windows exist in the thermal IR region (3 - 5 μm and 8 - 14 μm)
Microwave	$10^3 - 10^6$	Can penetrate rain, fog and clouds. Both active and passive remote sensing is possible. Radar uses wavelength in this range.
Radio	$> 10^6$	Have the longest wavelength. Used for remote sensing by some radars.

Energy Sources and Radiation Principle

-Solar Radiation

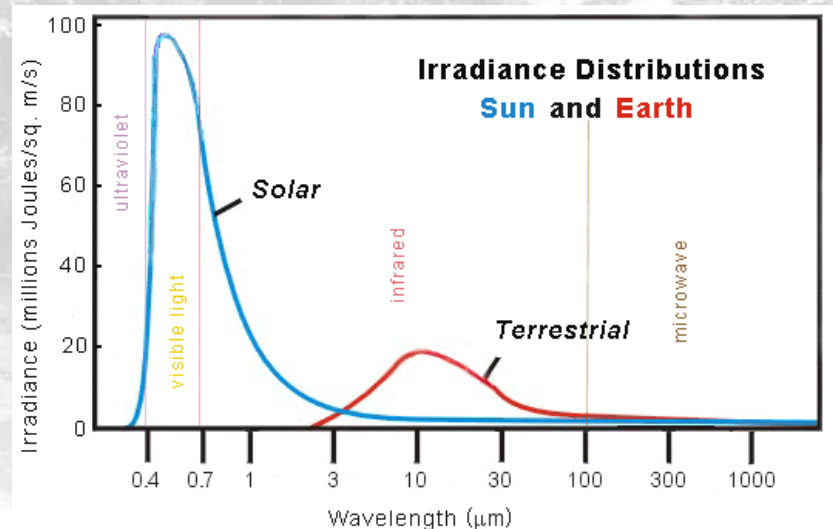
Sun is the primary source of energy that illuminates features on the Earth surface

Solar radiation

- Solar radiation (insolation) arrives at the Earth at different wavelengths
- The amount of energy it produces is not uniform across all wavelengths
 - Almost 99% is within the range of 0.28-4.96 μm
 - Within this range, 43% is radiated in the visible region between 0.4-0.7 μm
 - Maximum energy (E) is available at 0.48 μm wave length (visible green)

Irradiance: Power of electromagnetic radiation per unit area incident on a surface

Irradiance distribution of Sun and Earth
(<http://www.csulb.edu>)



Solar Radiation...

From particle theory: Energy of a quantum (Q) is proportional to the frequency

$$Q = h f$$

h = Plank's constant (6.626×10^{-34} J Sec)
 f = Frequency

From wave theory of electromagnetic radiation

$$c = \lambda f$$

c = Velocity (3×10^8 m/Sec)
 λ = Wavelength (μm)

Therefore Energy of a quantum (Q) is

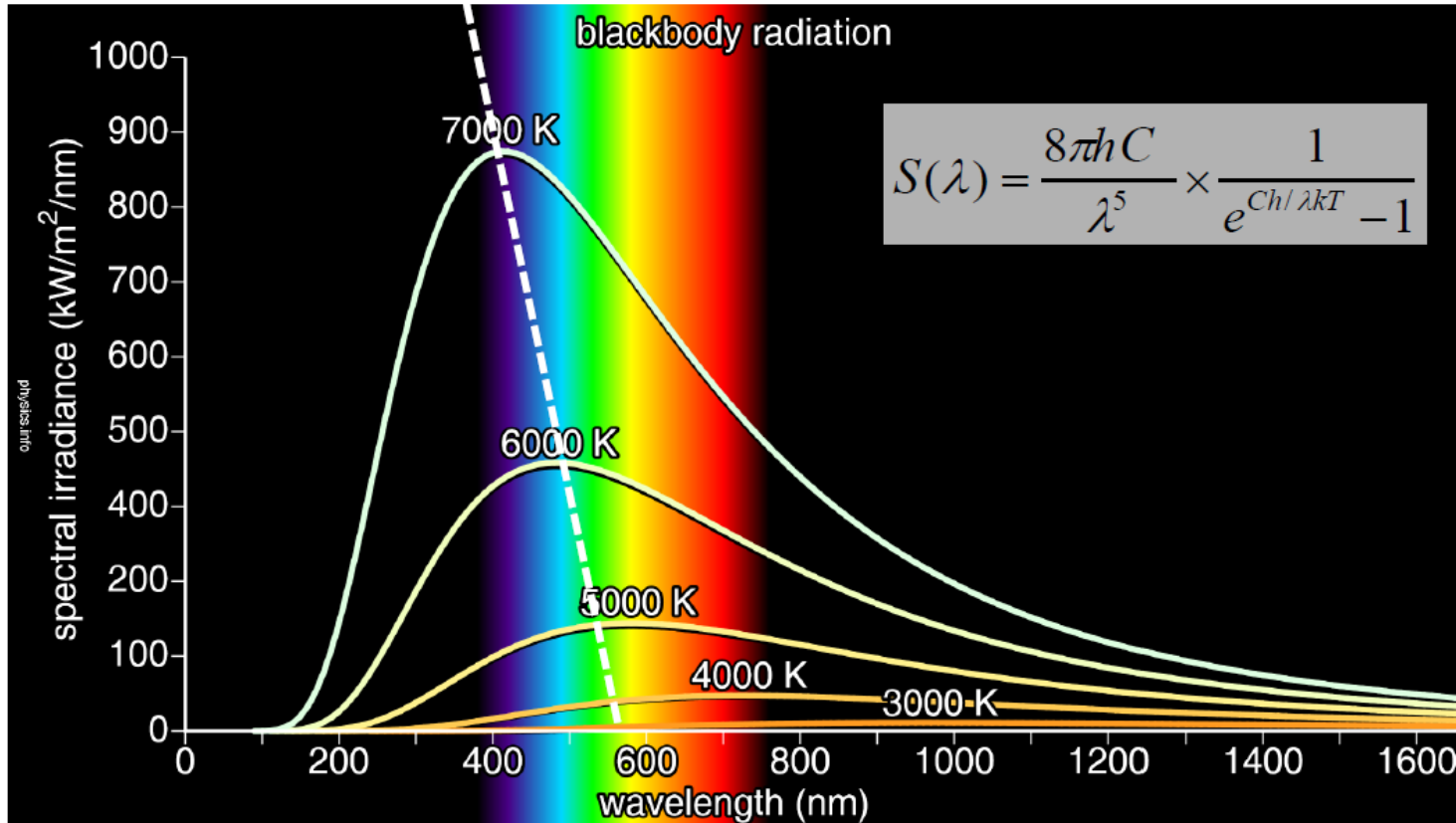
$$Q = h c / \lambda$$

The energy per unit quantum is inversely proportional to the wavelength

- Shorter wavelengths are associated with higher energy compared to the longer wavelengths
- Lower energy for microwave radiations compared to the IR regions
- For remote sensing with long wavelength radiations, the coverage area should be large enough to obtain a detectable signal

Brightness Temperature

- Brightness temperature is a convenient unit to represent radiance measurements



Planck's law

Allows conversion of radiance to brightness temperature

-the object's temperature if it were behaving as a blackbody

Brightness temperature

$$T_b = e T$$

C: Speed of light; h: Plank's Constant; K: Boltzman's Constant, e = emissivity

Energy Sources and Radiation Principle

-Radiation from Earth

Earth and the terrestrial objects also emit electromagnetic radiation

- All matter at temperature above absolute zero (0°K or -273°C) emit electromagnetic radiations continuously

Stefan-Boltzmann law

- The amount of radiation from such objects is a function of the temperature of the object
- Applicable for objects that behave as a blackbody

$$M = \sigma T^4$$

M = Total radiant exitance from the source (Watts / m²)

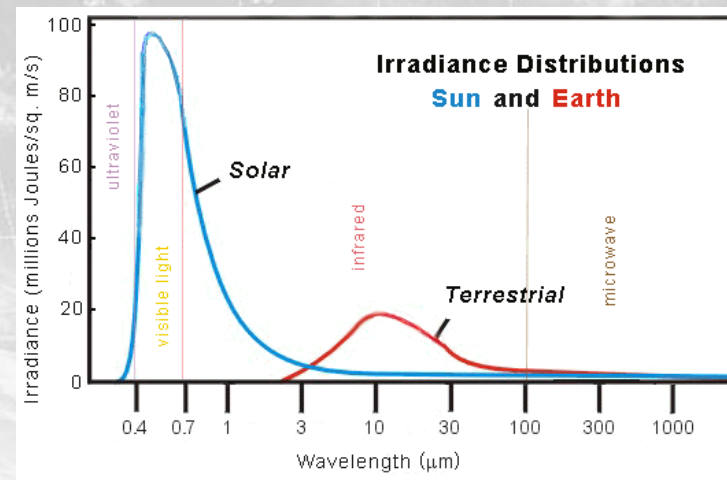
σ = The Stefan-Boltzmann constant (5.6697×10^{-8} Watts m⁻² K⁻⁴)

T = Absolute temperature of the emitting material in Kelvin.

Ambient temperature of the Earth ~ 300K

- Emits thermal IR radiation
- Maximum exitance in the region of 9.7 μ m
- Can be sensed using scanners and radiometers.

Irradiance distribution of Sun and Earth
(<http://www.csulb.edu>)



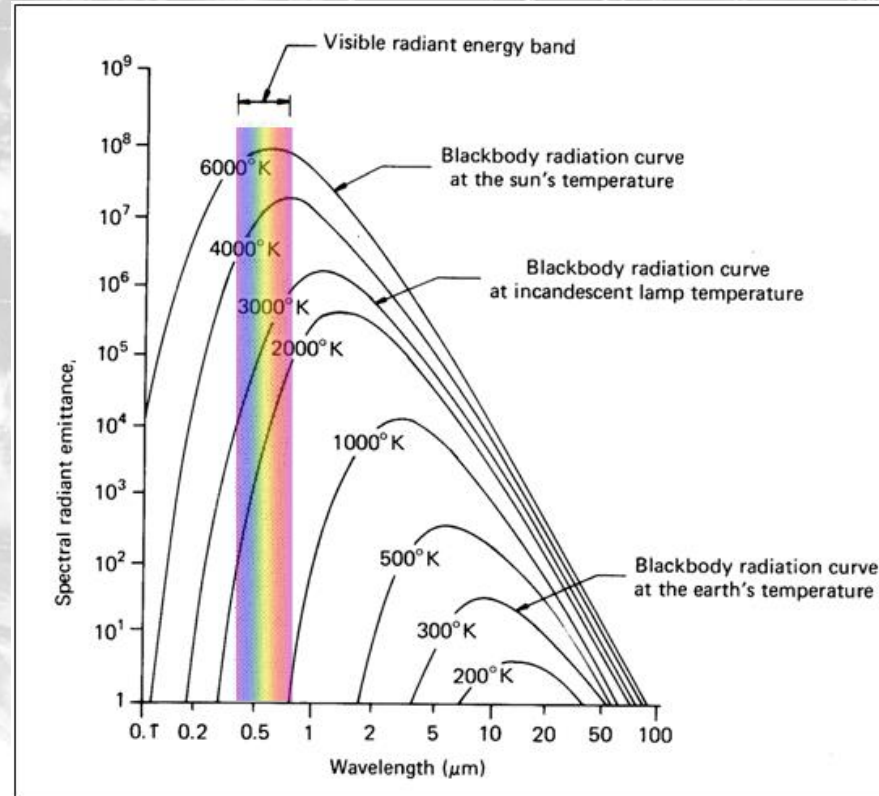
Radiation Principle

-Black Body Radiation

Blackbody : A hypothetical, ideal radiator that absorbs and re-emits the entire energy incident upon it

Spectral distribution or spectral curve : Energy distribution over different wavelengths for different temperature

- Area under the spectral curve for any temperature = Total radiant exitance at that temperature
- As the temperature increases total radiant exitance increases and hence the area under the curve
- Represents the Stefan-Boltzman's law graphically



Black Body Radiation...

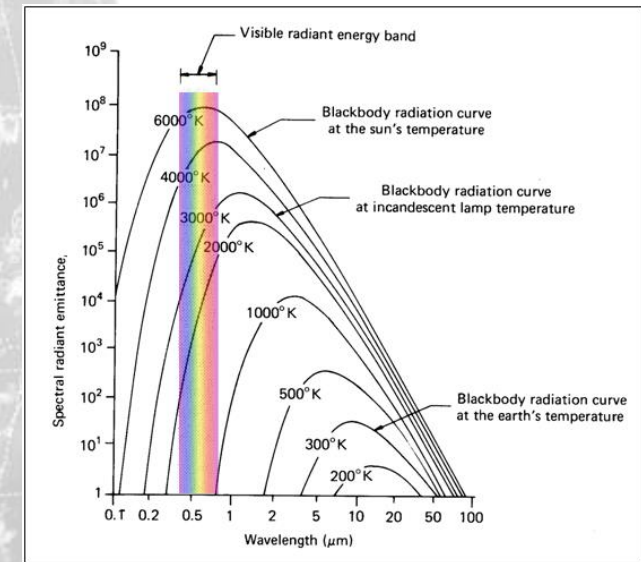
- Peak of the radiant exitance varies with wavelength
- With increase in temperature, the peak shifts towards left
- Wien's displacement law
 - Dominant wavelength at which a black body radiates λ_m is inversely proportional to the absolute temperature of the black body (in K)

$$\lambda_m = A / T$$

$A = 2898 \mu\text{m K}$, a constant

Solar radiation

- Sun's temperature is around 6000 K
- In the spectral curve at 6000K visible part of the energy (0.4-0.7 μm) dominates



Spectral energy distribution of blackbody at various temperatures

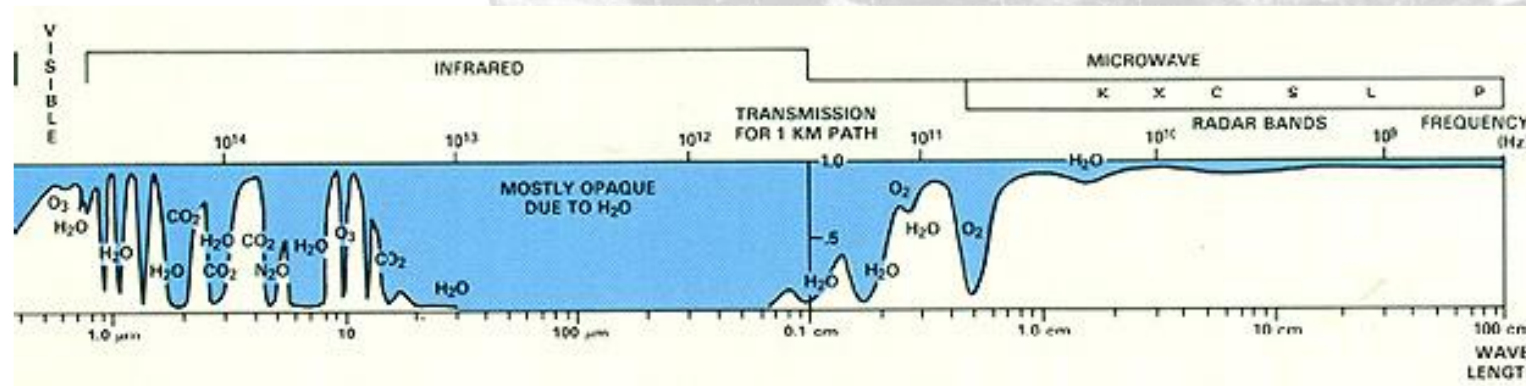
Remote Sensing of Electromagnetic Radiation

Selective wavelength bands are used in remote sensing

Electromagnetic energy interacts with the atmospheric gases and particles

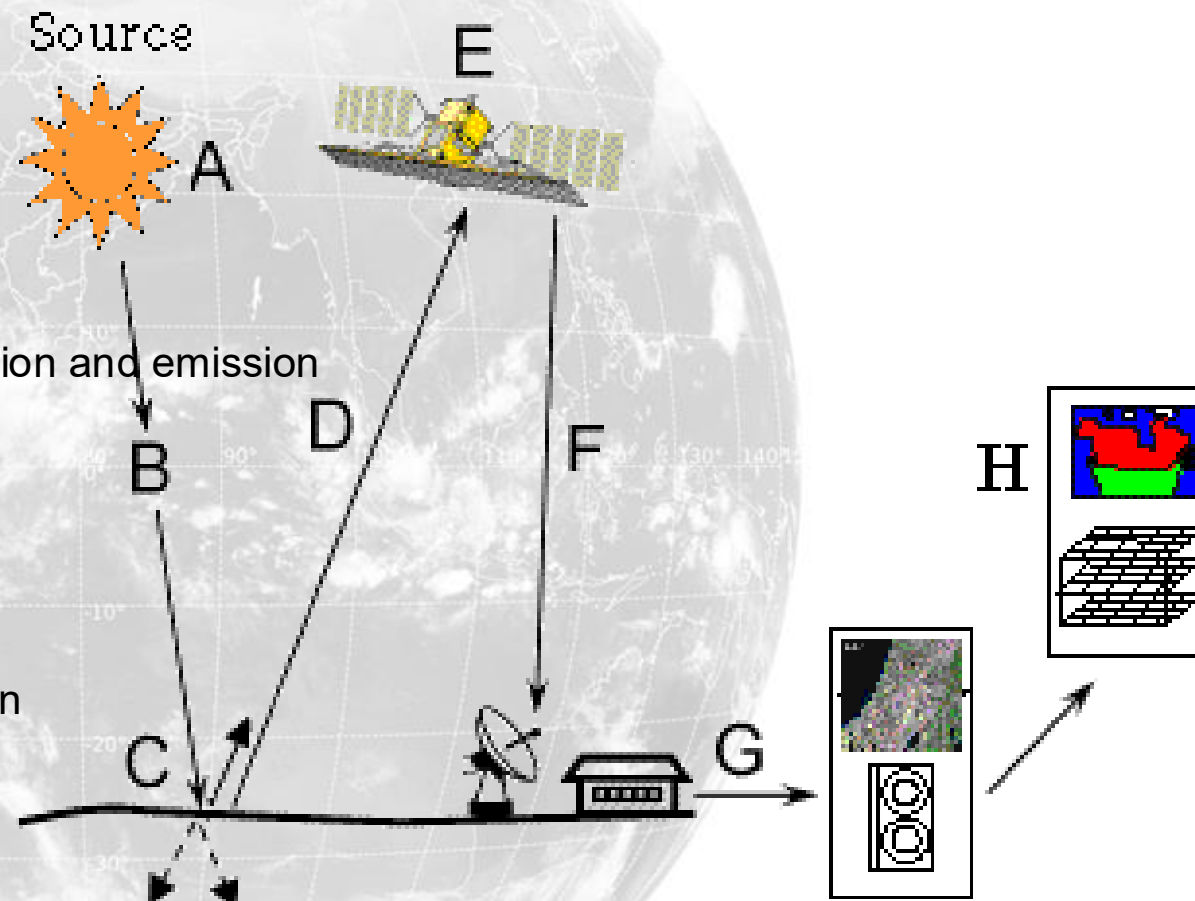
- Scattering and Absorption
- Atmosphere absorbs / backscatters a fraction of the energy and transmits the remainder

Atmospheric windows : Wavelength regions through which most of the energy is transmitted through atmosphere



Stages in Remote Sensing

- A. Emission of electromagnetic radiation
 - The Sun or an EMR source located on the platform
- B. Transmission of energy from the source to the object
 - Absorption and scattering of the EMR while transmission
- C. Interaction of EMR with the object and subsequent reflection and emission
- D. Transmission of energy from the object to the sensor
- E. Recording of the energy at the sensor
 - Photographic or non-photographic
- F. Transmission of the recorded information to ground station
- G. Processing of the data into digital or hard copy image
- H. Analysis of data



<https://tinyurl.com/pop1-quiz1>

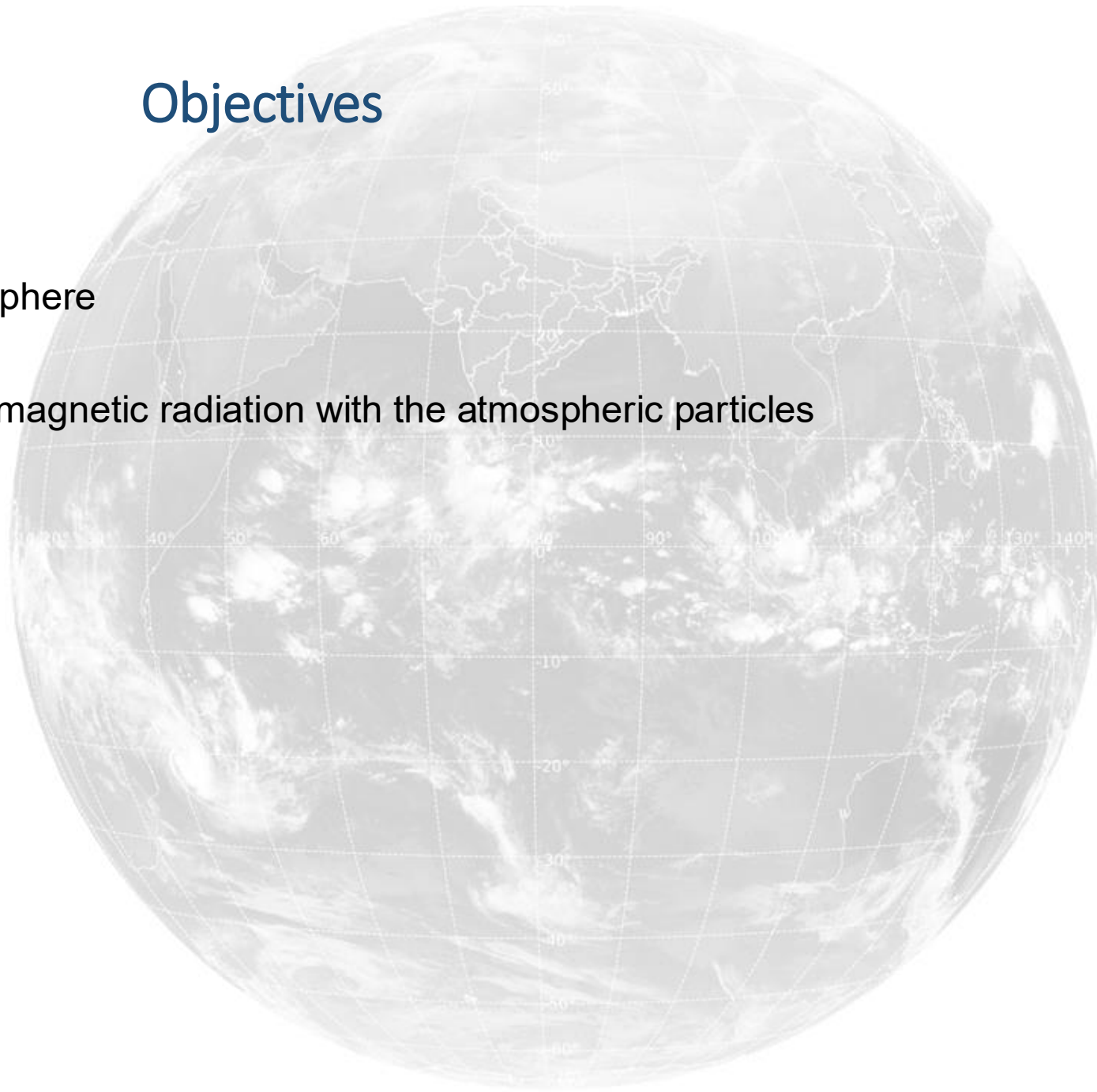


Objectives

Composition of the atmosphere

Interactions of the electromagnetic radiation with the atmospheric particles

- Scattering
- Absorption



Composition of the Atmosphere

Atmosphere : Gaseous envelop that surrounds the Earth's surface

- Much of the gases are concentrated within the lower 100km of the atmosphere
- Only 3×10^{-5} percent of the gases are found above 100 km (Gibbson, 2000)

Composition of the Earth's atmosphere (from Gibbson, 2000)

Component	Percentage
Nitrogen (N ₂)	78.08
Oxygen (O ₂)	20.94
Argon	0.93
Carbon Dioxide (CO ₂)	0.0314
Ozone (O ₃)	0.00000004

Composition of the Atmosphere...

- Oxygen and Nitrogen

- Present in the ratio 1:4
- Both together add to 99 percent of the total gaseous composition

- Ozone

- Present in very small quantities
- Mostly concentrated in the atmosphere between 19 and 23km

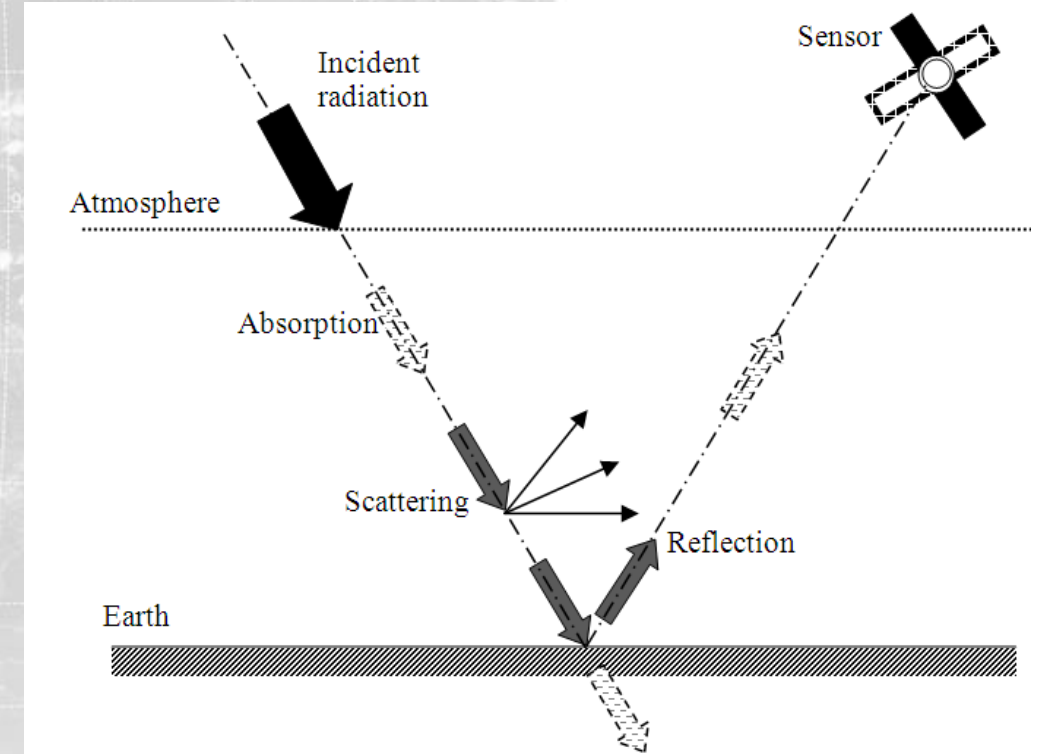
- The atmosphere also contains water vapor, methane, dust particles, pollen from vegetation, smoke particles etc.

- Dust particles and the pollen form about 50% of the total particles
- Size of these particles varies from approximately $0.01\mu\text{m}$ to $100\mu\text{m}$

- The gases and the particles present in the atmosphere cause scattering and absorption of the electromagnetic radiation passing through it

Energy Interactions

- From the source to the sensor, the radiation passes through the atmosphere
- Path length: The distance traveled by the radiation through the atmosphere
 - Varies depending on the remote sensing techniques and sources
 - Space photography using solar energy
 - Path length = 2x Thickness of the earth's atmosphere
 - Airborne thermal sensors using emitted energy from the objects on the earth
 - Path length = One way distance from the earth's surface to the sensor



Energy Interactions...

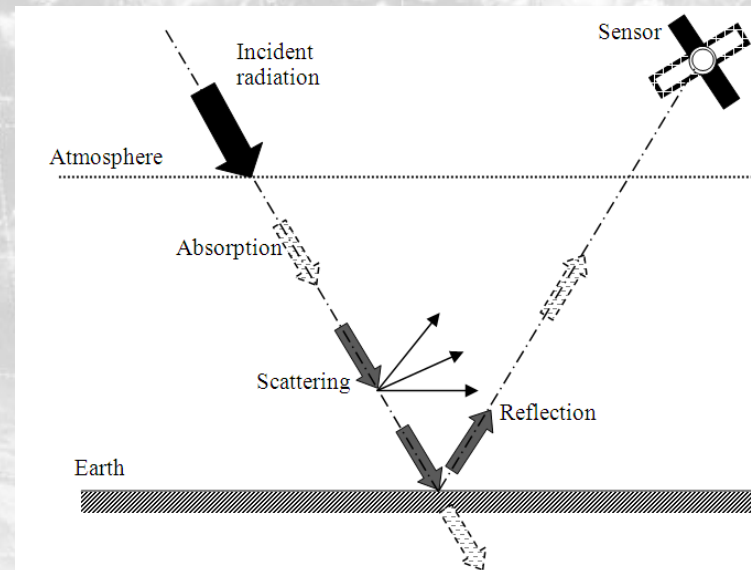
The intensity and the spectral composition of the incident radiation are altered by the atmospheric effects

Atmospheric interaction depends on the

- Properties of the radiation such as magnitude and wavelength
- Atmospheric conditions
- Path length

Interaction with the atmospheric particles

- Scattering
- Absorption



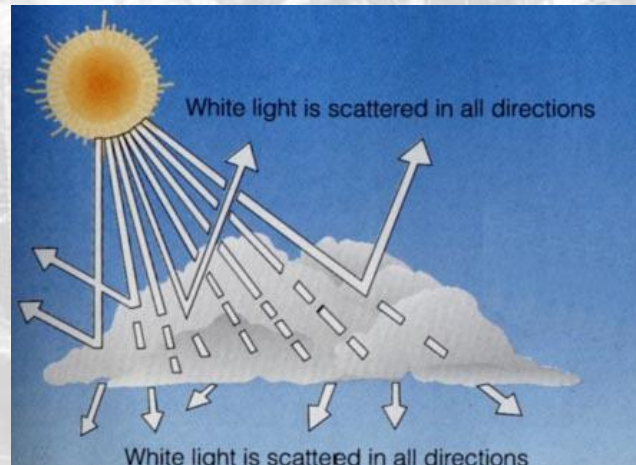
Scattering

Process by which small particles in the atmosphere diffuse a portion of the incident radiation in all directions

- There is no energy transformation during scattering
- Spatial distribution of the energy is altered

Types of scattering

- Rayleigh scattering
- Mie scattering
- Non-selective scattering



http://www.geog.ucsb.edu/~joel/g110_w08/lecture_notes/radiation_atmosphere/radiation_atmosphere.html

Scattering

-Rayleigh Scattering ([Details](#))

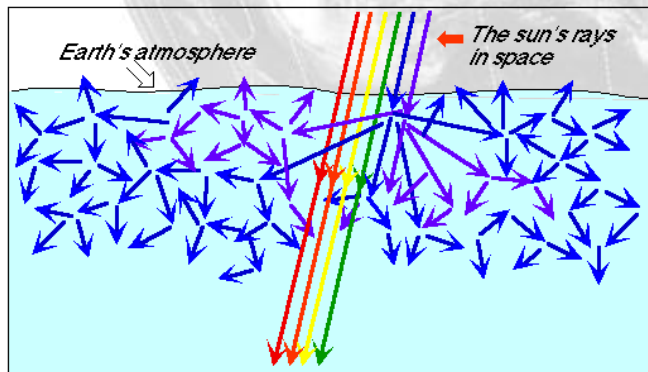
- Scattering caused by the atmospheric molecules and other **tiny particles**
- Also known as selective scattering or molecular scattering
- Dependent on the wavelength

Occurs when particles are much smaller than the wavelengths of the radiation

- Particle size less than $(1/10)^{\text{th}}$ of the wavelength

Intensity of the scattered light is inversely proportional to the fourth power of wavelength

- Shorter wavelengths are scattered more than longer wavelengths



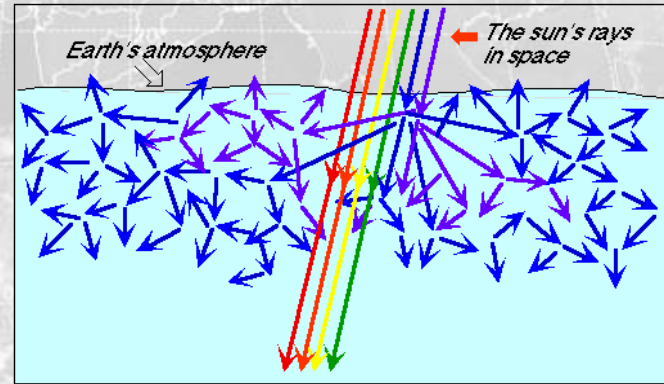
<http://home.comcast.net/~vinelandrobotics/>

Rayleigh Scattering of the Visible Part of the EM Energy

Scattering of the visible bands is caused mainly by the molecules of Oxygen and Nitrogen

■ Blue (shorter wavelength) is scattered more

- Blue light is scattered around four times the red light
- UV light is scattered about 16 times the red light
- A "blue" sky is a manifestation of Rayleigh scatter



■ Orange or red colour during sunrise and sunset

- Sun rays have to travel a longer path
- Complete scattering (and absorption) of shorter wavelength radiations
- Only the longer wavelength (orange and red) which are less scattered are visible

■ Other examples

- The haze in imagery
- Bluish-grey cast in a color image when taken from high altitude

Scattering

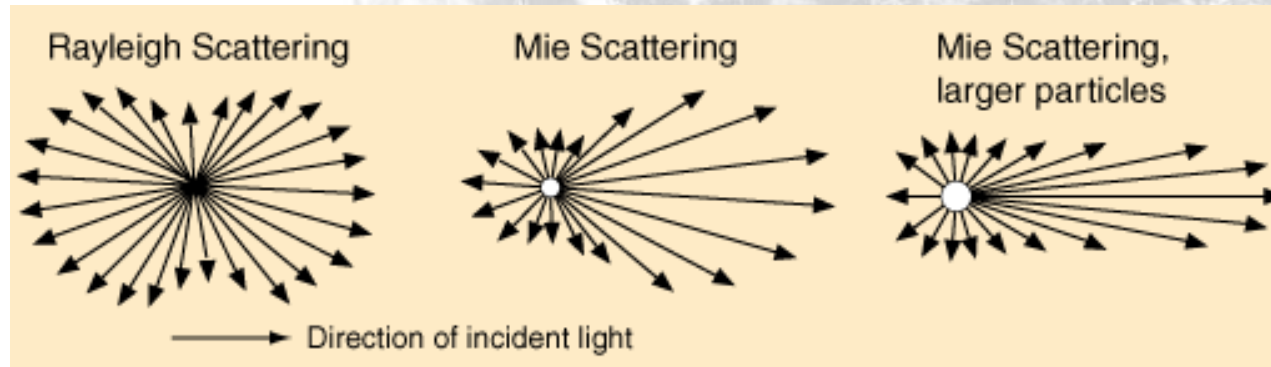
-Mie Scattering ([Details](#))

Occurs when the wavelengths of the energy is **almost equal to the diameter** of the atmospheric particles

- Usually caused by the aerosol particles such as dust, smoke and pollen
- Gas molecules are too small to cause Mie scattering of the radiation commonly used for remote sensing

Longer wavelengths also get scattered compared to Rayleigh scatter

Intensity of the scattered light varies approximately as the inverse of the wavelength



Source: <http://hyperphysics.phy-astr.gsu.edu>

Scattering

-Non-selective Scattering

When the diameters of the atmospheric particles are much larger

- Diameter is greater than 10 times the wavelengths being sensed
- Particles such as pollen, cloud droplets, ice crystals and raindrops can cause non-selective scattering of the visible light.

Non-selective scattering of visible light (of wavelength 0.4-0.7 μ)

- Generally caused by water droplets (5 to 100 μ m diameter)
- All visible and IR wavelengths get scattered equally
- Gives white or grey color to the clouds

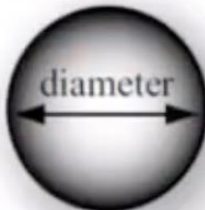
Summary of scattering

Atmospheric Scattering

Rayleigh Scattering

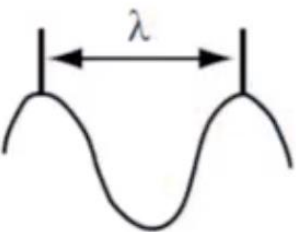
a.  Gas molecule

Mie Scattering

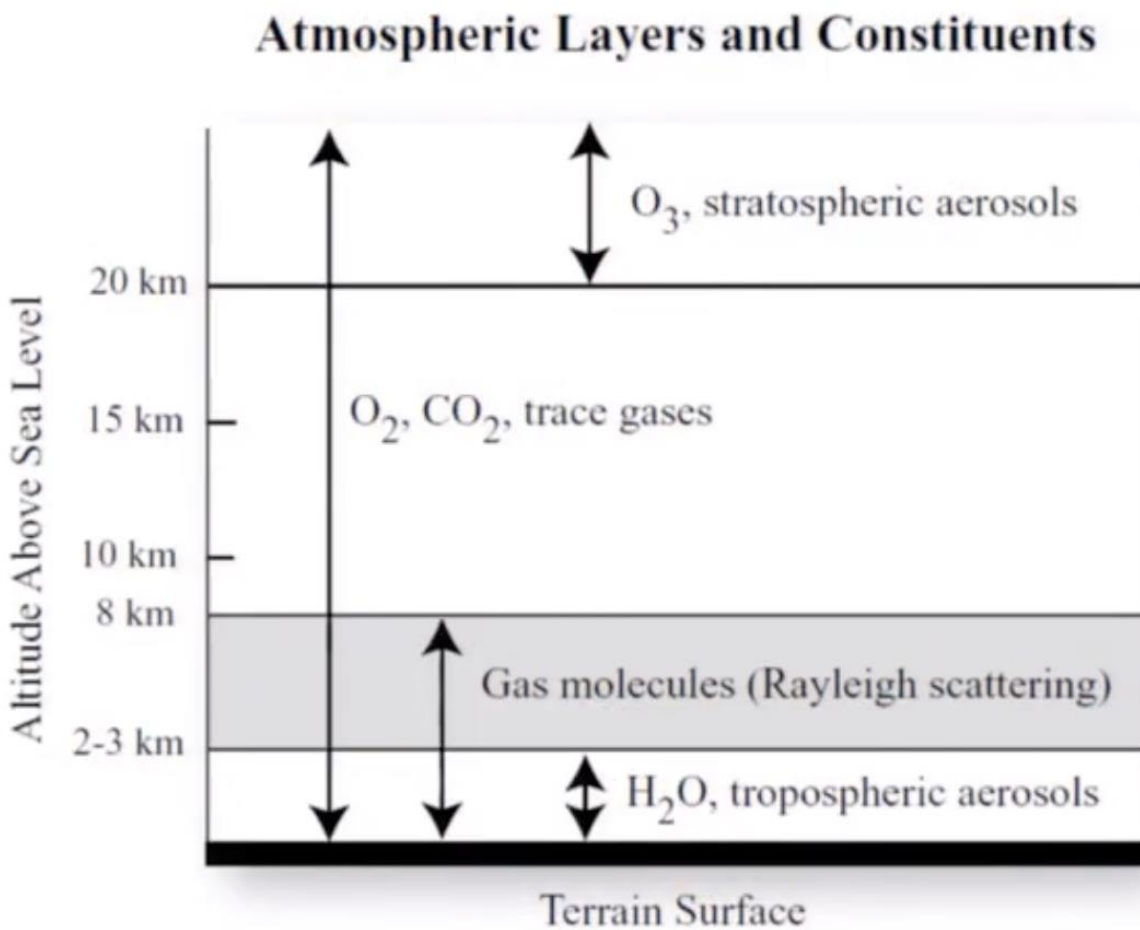
b.  Smoke, dust

Nonselective Scattering

c.  Water vapor



Photon of electromagnetic energy modeled as a wave



Red colour is widely used in signals to indicate danger/stop sign as it get scattered less and can be seen from a longer distance.

Absorption

Absorption : Process in which the incident energy is retained by particles in the atmosphere

- Energy is transformed into other forms
- Unlike scattering, atmospheric absorption causes an effective loss of energy
- Absorption depends on
 - Wavelength of the energy
 - Atmospheric composition
 - Arrangement of the gaseous molecules and their energy level
- The absorbing medium will not only absorb a portion of the total energy, but will also reflect, refract or scatter the energy.
- The absorbed energy may also be transmitted back to the atmosphere.

Absorption...

The most efficient absorbers of solar radiation are

- Water vapour, carbon dioxide, and ozone

Gaseous components are selective absorbers of the electromagnetic radiation

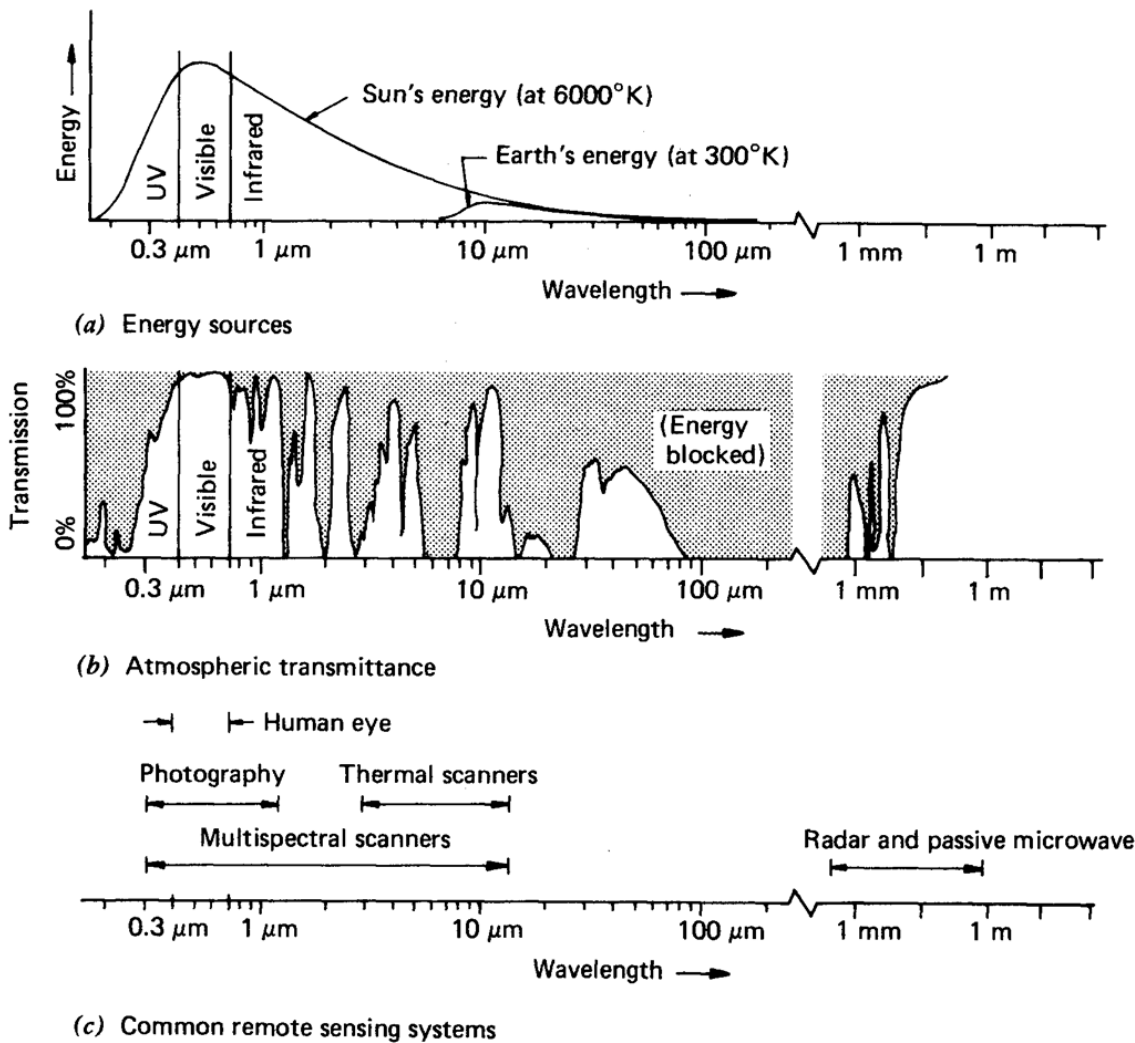
- Absorb electromagnetic energy in specific wavelength bands
- Depends on the arrangement of the gaseous molecules and their energy levels

Atmospheric window

The ranges of wavelength that are partially or wholly transmitted through the atmosphere

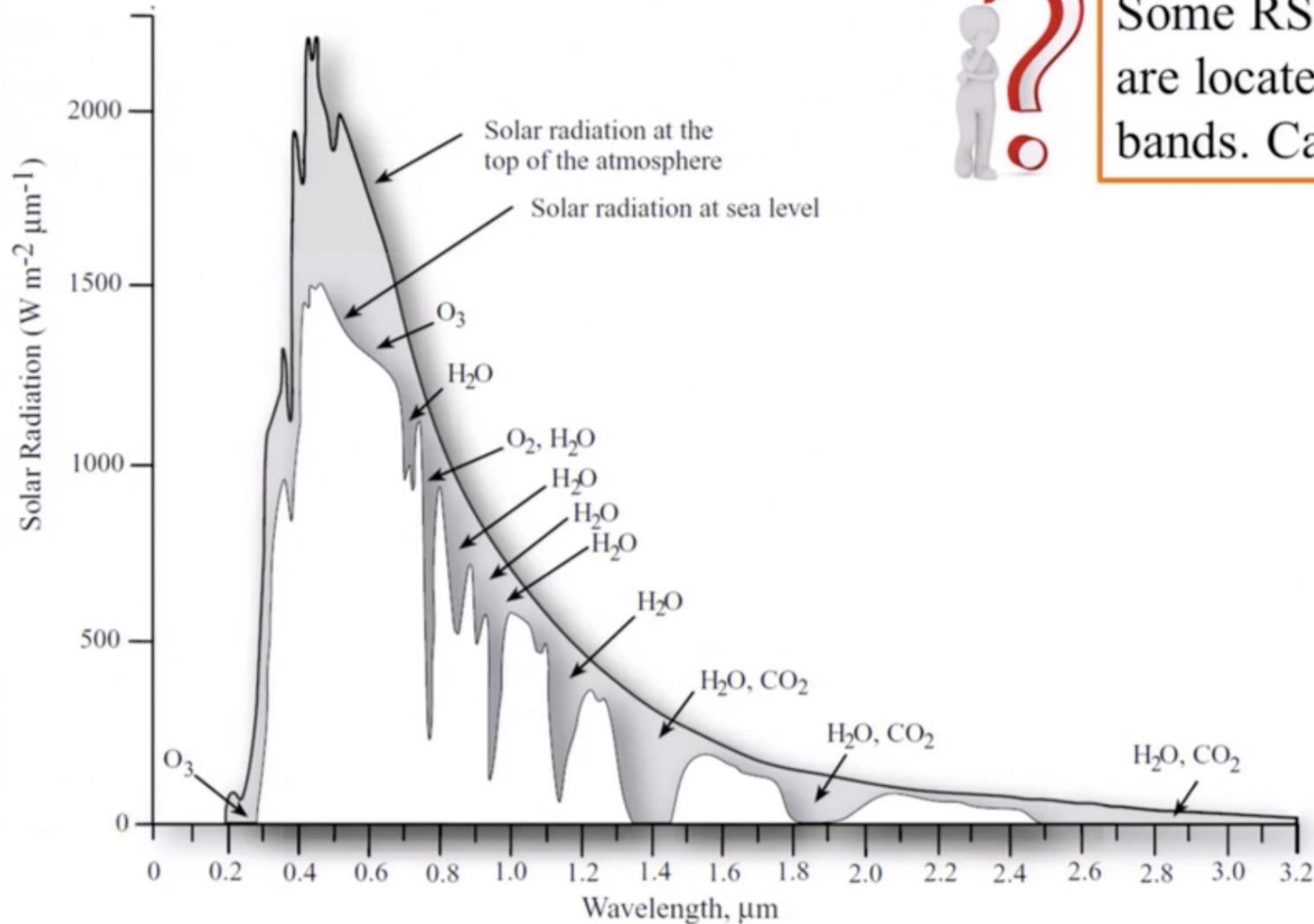
Land Remote sensing data acquisition is limited through these atmospheric windows

Atmospheric Window



- Wavelengths shorter than 0.1 μm
 - Absorbed by Nitrogen and other gaseous components
- Wavelengths shorter than 0.3 μm (X-rays, Gamma rays and part of ultraviolet rays)
 - Mostly absorbed by the ozone (O₃)
- Visible part of the spectrum
 - Little absorption occurs
- Oxygen in the atmosphere causes absorption centered at 6.3 μm.
- Infrared (IR) radiation
 - Mainly absorbed by water vapour and carbon dioxide molecules
- Far infrared region
 - Mostly absorbed by the atmosphere
- Microwave region
 - Absorption is almost nil

Absorption



Some RS systems consists of bands that are located in atmospheric absorption bands. Can you guess why?



Shadows are not completely dark and we can see objects within shadows. How?



Absorption...

The most common sources of energy are

- Incident solar energy
 - Maximum energy in the visible region
- Radiation from the Earth
 - Maximum energy in the thermal IR region
 - Two atmospheric windows
 - at 3 to 5 μm and at 8 to 14 μm

Radar & Passive microwave systems operate through a window in the region 1 mm-1 m

Major atmospheric windows used in remote sensing and their characteristics

Atmospheric window	Wavelength band (μm)	Characteristics
Upper ultraviolet, Visible and photographic IR	0.3-1 apprx.	95% transmission
Reflected infrared	1.3, 1.6, 2.2	Three narrow bands
Thermal infrared	3.0-5.0 8.0-14.0	Two broad bands
Microwave	>5000	Atmosphere is mostly transparent

Sensor Selection for Remote Sensing

•Criteria for sensor selection

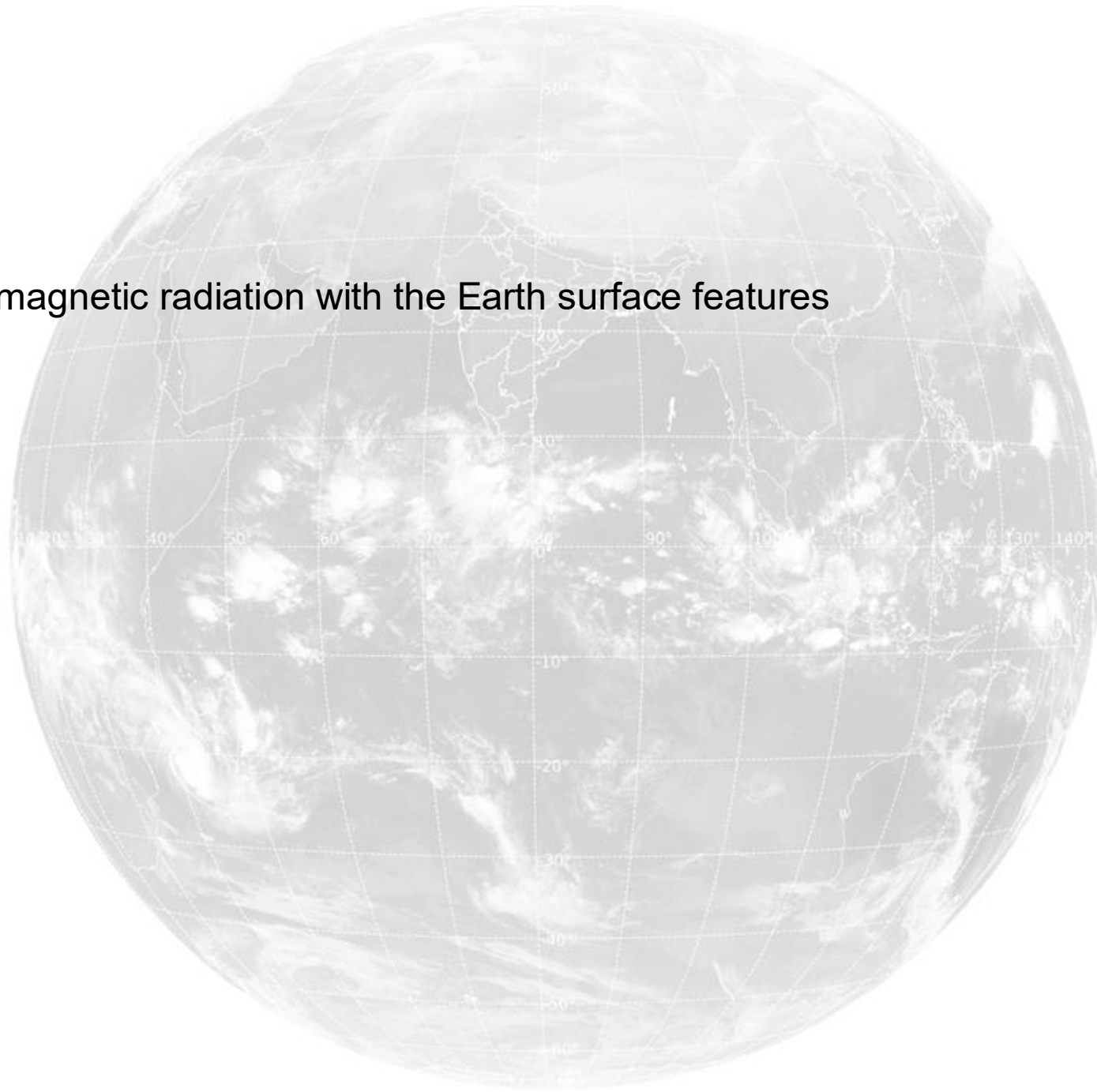
- The spectral sensitivity of the available sensors
- The available atmospheric windows in the spectral range(s) considered.
- The source, magnitude, and spectral composition of the energy available in the particular range.
- Energy interactions with the features under investigation in the selected bands

Multi Spectral Sensors sense simultaneously through multiple, narrow wavelength ranges

Objectives

Interactions of the electromagnetic radiation with the Earth surface features

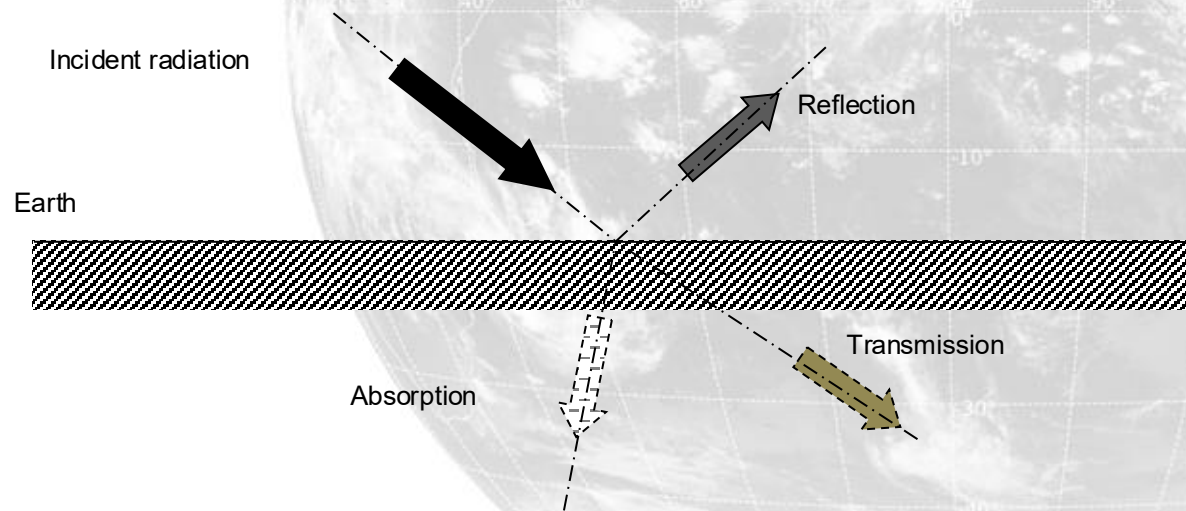
- Reflection
- Absorption
- Transmission



Energy Interactions

- Electromagnetic energy interactions with the surface features

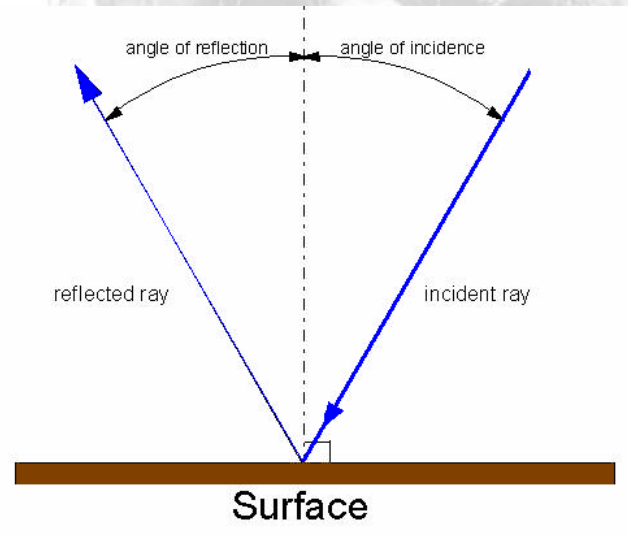
- Reflection
- Absorption
- Transmission



Energy Interactions...

Reflection

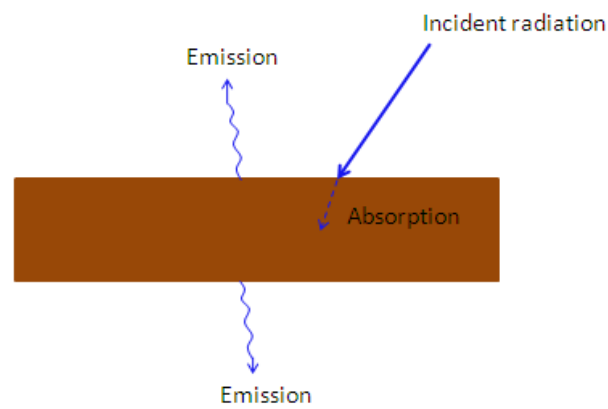
- Radiation is redirected after hitting the target
- Angle of incidence = Angle of reflectance



Energy Interactions...

- **Absorption**

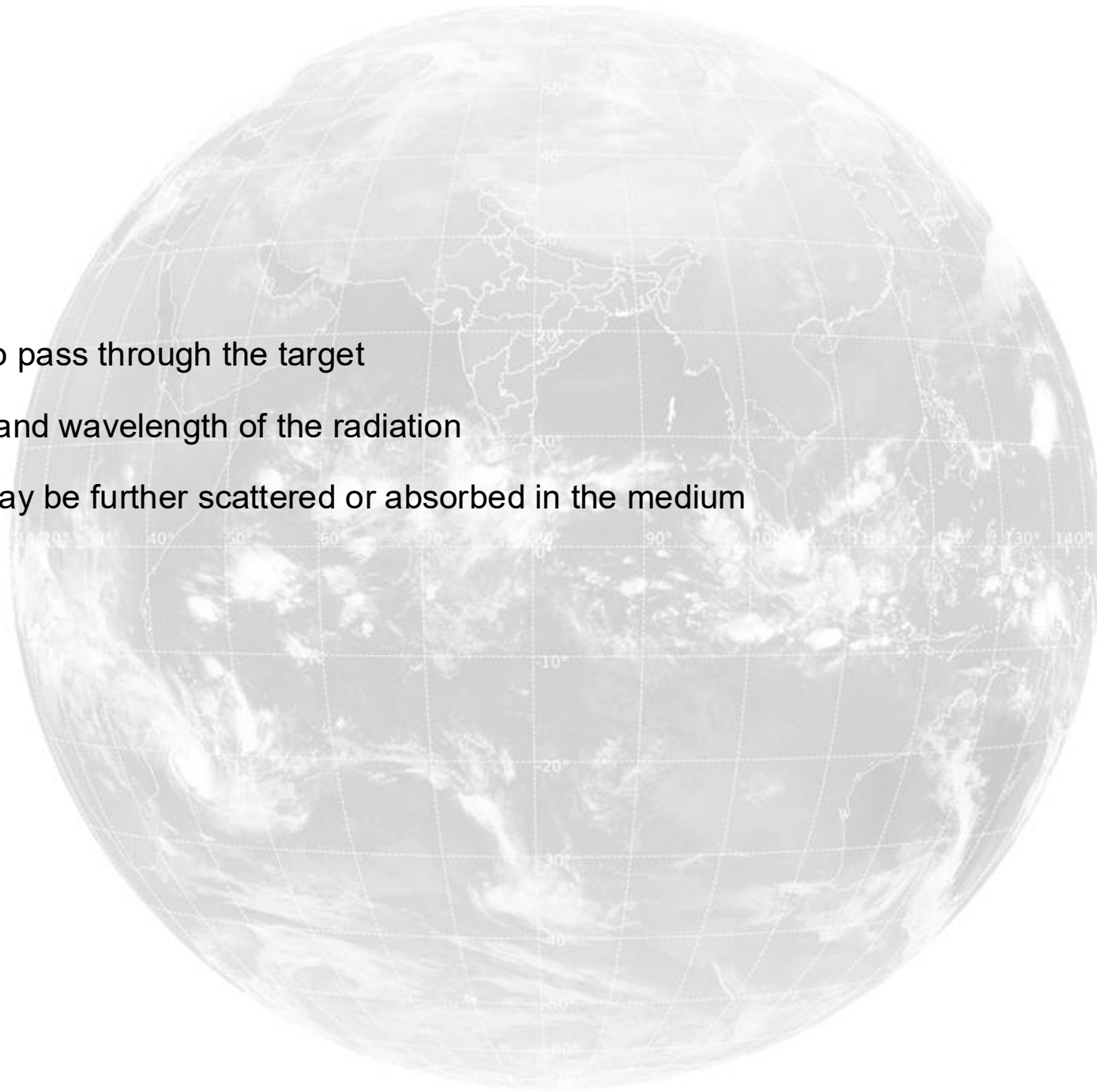
- Radiation is absorbed by the target
- A portion absorbed by the Earth's surface is available for emission as thermal radiation



Energy Interactions...

Transmission

- Radiation is allowed to pass through the target
- Changes the velocity and wavelength of the radiation
- Transmitted energy may be further scattered or absorbed in the medium



Energy Interactions...

Reflection, Absorption or Transmission ?

- Energy incident on a surface may be partially reflected, absorbed or transmitted
- Which process takes place on a surface depends on the following factors:
 - Wavelength of the radiation
 - Angle at which the radiation intersects the surface
 - Composition and physical properties of the surface

Relationship between reflection, absorption and transmission

- Principle of conservation of energy as a function of wavelength

$$E_I(\lambda) = E_R(\lambda) + E_A(\lambda) + E_T(\lambda)$$

OR

$$E_R(\lambda) = E_I(\lambda) - E_A(\lambda) - E_T(\lambda)$$

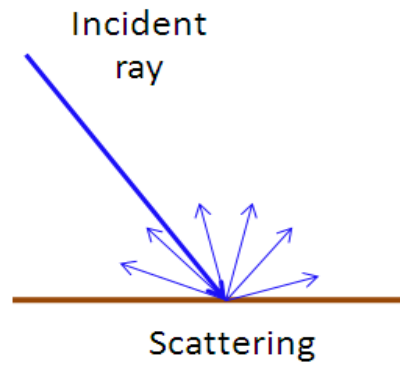
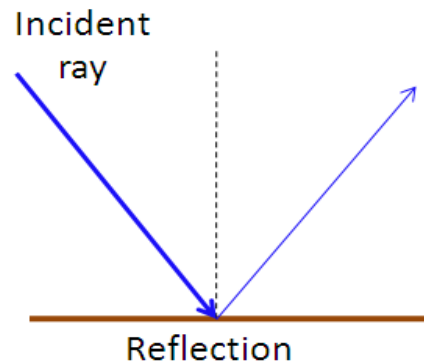
E_I = Incident energy

E_R = Reflected energy

E_A = Absorbed energy

E_T = Transmitted energy

Reflection Vs Scattering



Reflection

Incident energy is redirected

Angle of incidence = Angle of reflection

- The reflected radiation leaves the surface at the same angle as it approached
- Often called **Specular Reflection**

Scattering

- A special type of reflection
- Incident energy is diffused in many directions
- Often called **Diffuse Reflection**

Reflection Vs Scattering...

Reflection or Scattering?

Depends on the roughness of the surface with respect to the incident wavelength

Roughness of the surface $<$ Incident wavelength $\rightarrow$ Smooth surface $\rightarrow$ Reflection

Roughness of the surface $>$ Incident wavelength $\rightarrow$ Rough surface $\rightarrow$ Scattering

Roughness of the surface controls how the energy is reflected

Mainly two types

- Specular reflection
- Diffuse (Lambertian) reflection

Specular Reflection

Incident energy is completely reflected in one direction

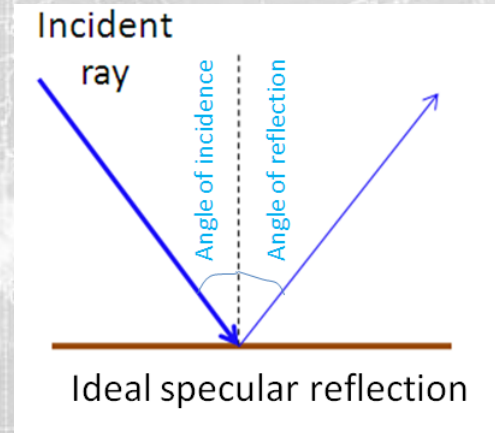
Angle of reflection is equal to the angle of incidence

- Reflection is maximum along the angle of reflectance
- Reflection is negligible in other directions

Reflectance varies with sensor location and incidence angle

Occurs when the surface is smooth and flat

Wavelength > Roughness → Specular reflection dominates



Diffuse or Lambertian Reflection

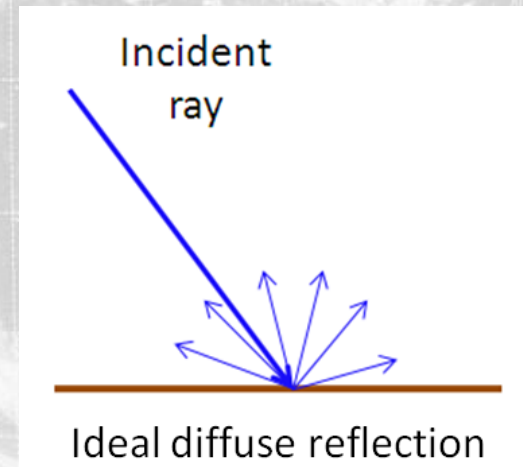
Incident energy is reflected uniformly in all directions

Occurs when the surface is rough

Wavelength < Roughness → Diffuse reflection

Important in remote sensing

- Reflectance is same irrespective of the sensor location and incidence angle
- Diffuse reflection contains the colour information of the surface



Specular and Diffuse Reflection

Depends on the wavelength of the incident energy

Example

Visible bands

- Shorter wavelengths
- Even fine materials such as sand appear as rough- cause diffuse reflection

In long wavelength radio range

- Even larger features are less than the wavelength
- Even a rocky terrain may appear smooth to incident energy - cause specular reflection

Type of Reflectors

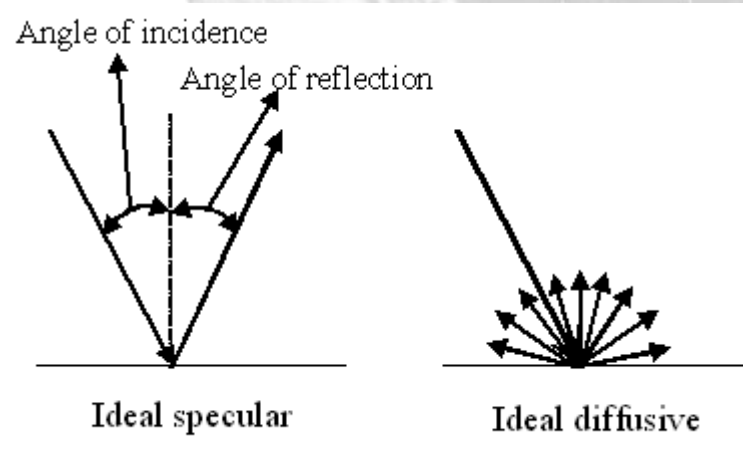
Based on the nature of reflection

Ideal Specular Reflector

- Completely reflects the incident energy with angle of reflection equal to the angle incidence

Ideal Lambertian Reflector

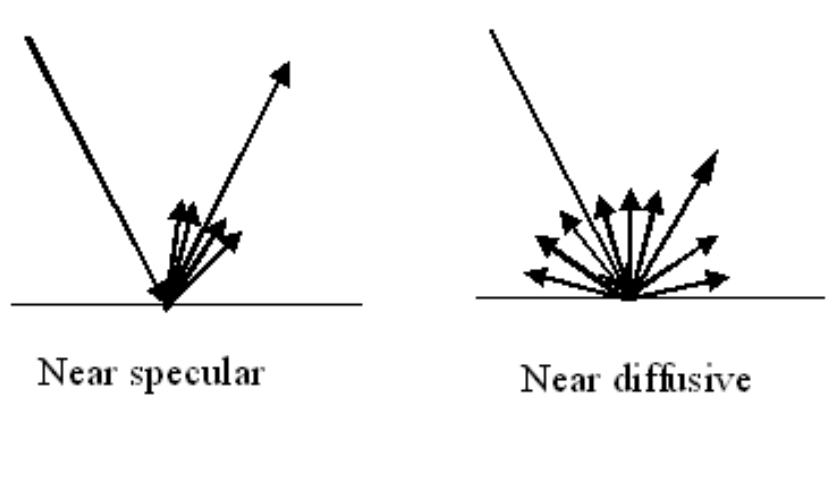
- Scatters all the incident energy equally in all the directions



Type of Reflectors...

Most of the real surface features are not perfect specular or diffuse reflectors

- Near Specular Reflector
- Near Lambertian reflector



Type of Reflectors...

Lambertian reflectors

- Ideal for remote sensing
- Reflection will be the same irrespective of the location of the sensor
- Most natural surfaces observed using remote sensing are approximately Lambertian at visible and IR wavelengths

Specular reflector

- Maximum brightness will be obtained only at one location
- Variation in the spectral signature for the same feature affects the interpretation of the remote sensing data.
- Water provides specular reflection
 - Generally gives a dark tone in the image
 - Specular reflection provides a pale tone when the sensor is located along the angle of reflection

Spectral Reflectance

- Represents the reflectance characteristics of earth surface features
- Ratio of energy reflected by the surface to the energy incident on the surface
- Measured as a function of wavelength
- Also known as **Albedo**
- Mathematical representation of spectral reflectance or albedo

$$R_{\lambda} = \frac{E_R(\lambda)}{E_I(\lambda)}$$
$$= \frac{\text{Energy of wavelength } \lambda \text{ reflected from the object}}{\text{Energy of wavelength } \lambda \text{ incident on the object}} \times 100$$

Albedo of Earth Surface Features

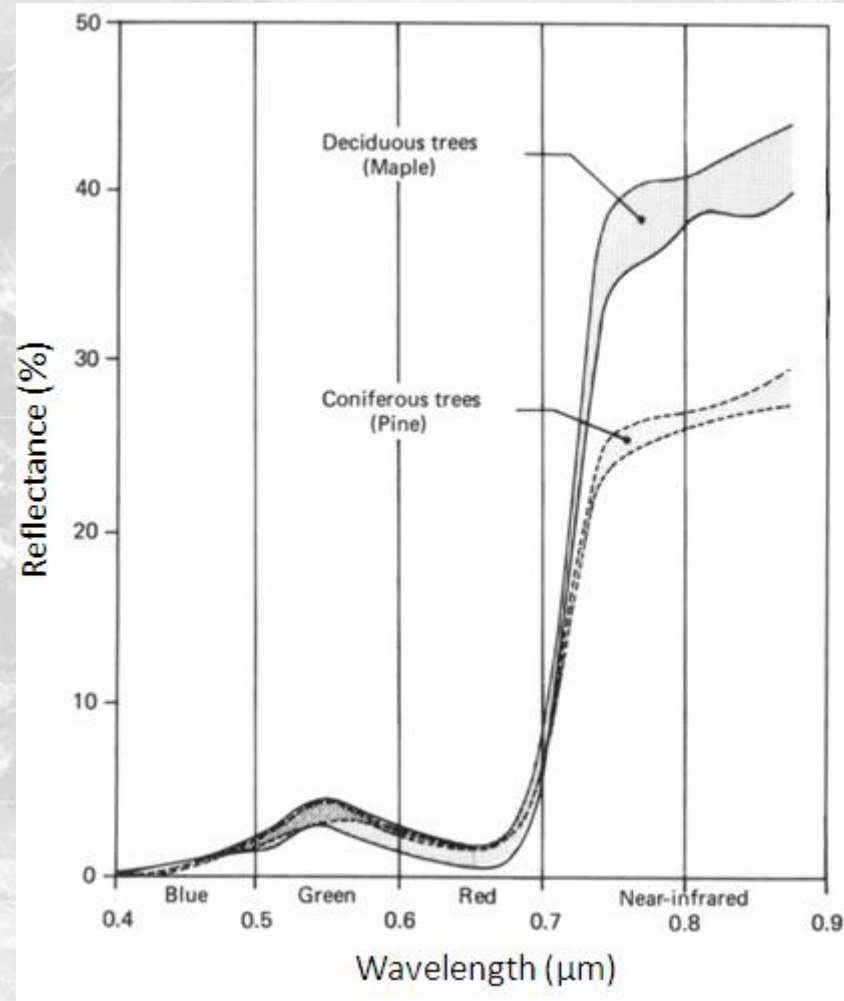
Surface type	Albedo %
Grass	25
Concrete	20
Water	5-70
Fresh snow	80
Forest	5-10
Thick cloud	75
Dark soil	5-10

Large range is due to the specular reflection characteristics.

Spectral Reflectance Curve

Graphical representation of the spectral response over different wavelengths of the electromagnetic spectrum

- Give an insight into the spectral characteristics of different objects
- Used for the selection of a particular wavelength band for remote sensing data acquisition



Reflected Energy in Remote Sensing

Energy reflected from the surface is recorded in remote sensing

Fraction of energy that is reflected / scattered is unique for each material

- Used for distinguishing different features on an image

Within a feature class, energy reflected / emitted / absorbed depends on the wavelength

- Features may be similar and hence indistinguishable using single spectral band
- Their reflectance properties may be different in another spectral band
- Use of multiple wavelength bands helps to further differentiate the features within one class
- Reflected energy from multiple wavelength bands are recorded in multi-spectral remote sensing

Use of Spectral Reflectance in Remote Sensing ...

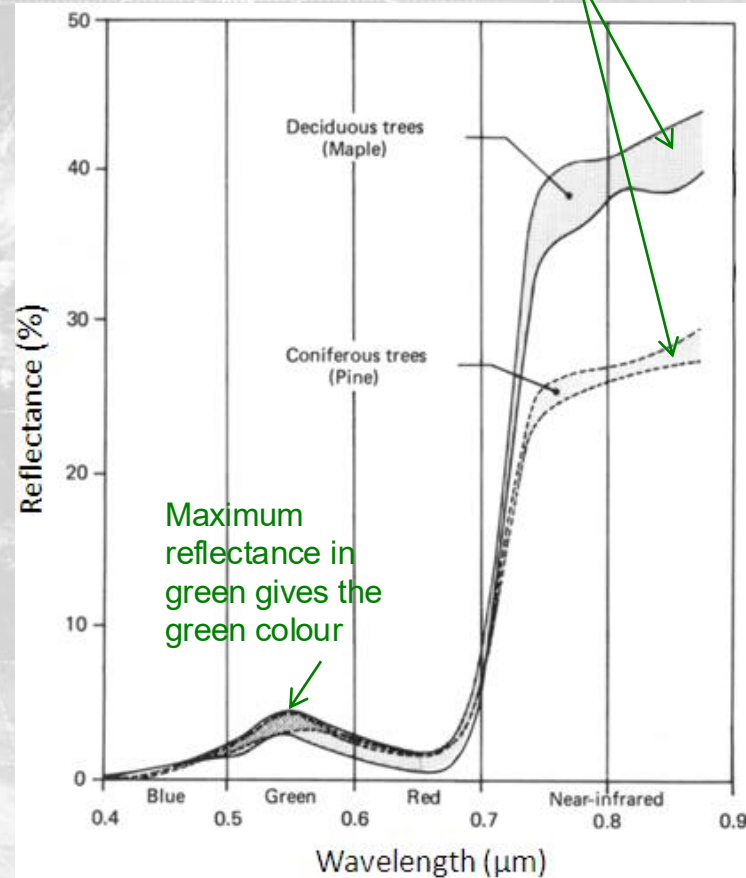
Example:

Generalized spectral reflectance curves for deciduous and coniferous trees

Sensor selection to differentiate deciduous and coniferous trees

- Curves overlap in the visible portion
- Both class will be seen in shades of green
 - Deciduous and coniferous trees cannot be differentiated through visible spectrum
- Spectral reflectance are quite different in NIR
 - Deciduous and coniferous trees can be differentiated through NIR spectrum

Spectral reflectance within one class is not unique, and hence the ranges are shown



Use of Spectral Reflectance in Remote Sensing ...



Panchromatic photograph using reflected sunlight over the visible wavelength

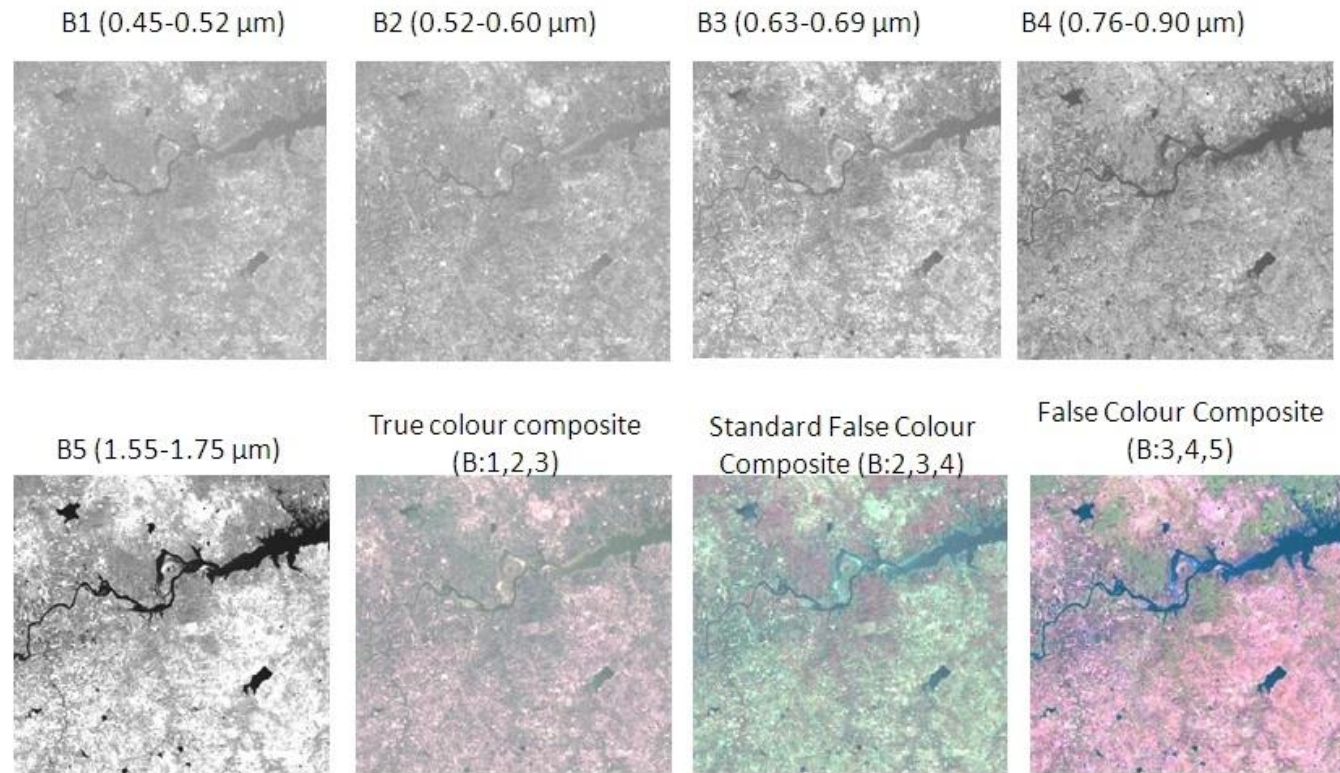


Black and white infrared photograph using reflected sunlight over 0.7 to 0.9 mm wavelength

- Coniferous and deciduous trees are not differentiable
- Deciduous trees show bright signature compared to coniferous trees

(Source: Lillesand et al., 2004)

Use of Spectral Reflectance in Remote Sensing ...



A part of the Krishna River Basin as seen in different bands of the Landsat ETM+ imagery

- Reflectance from water, vegetation and fallow lands are different in different bands
- A combination of more than one spectral band helps to attain better differentiation of these features

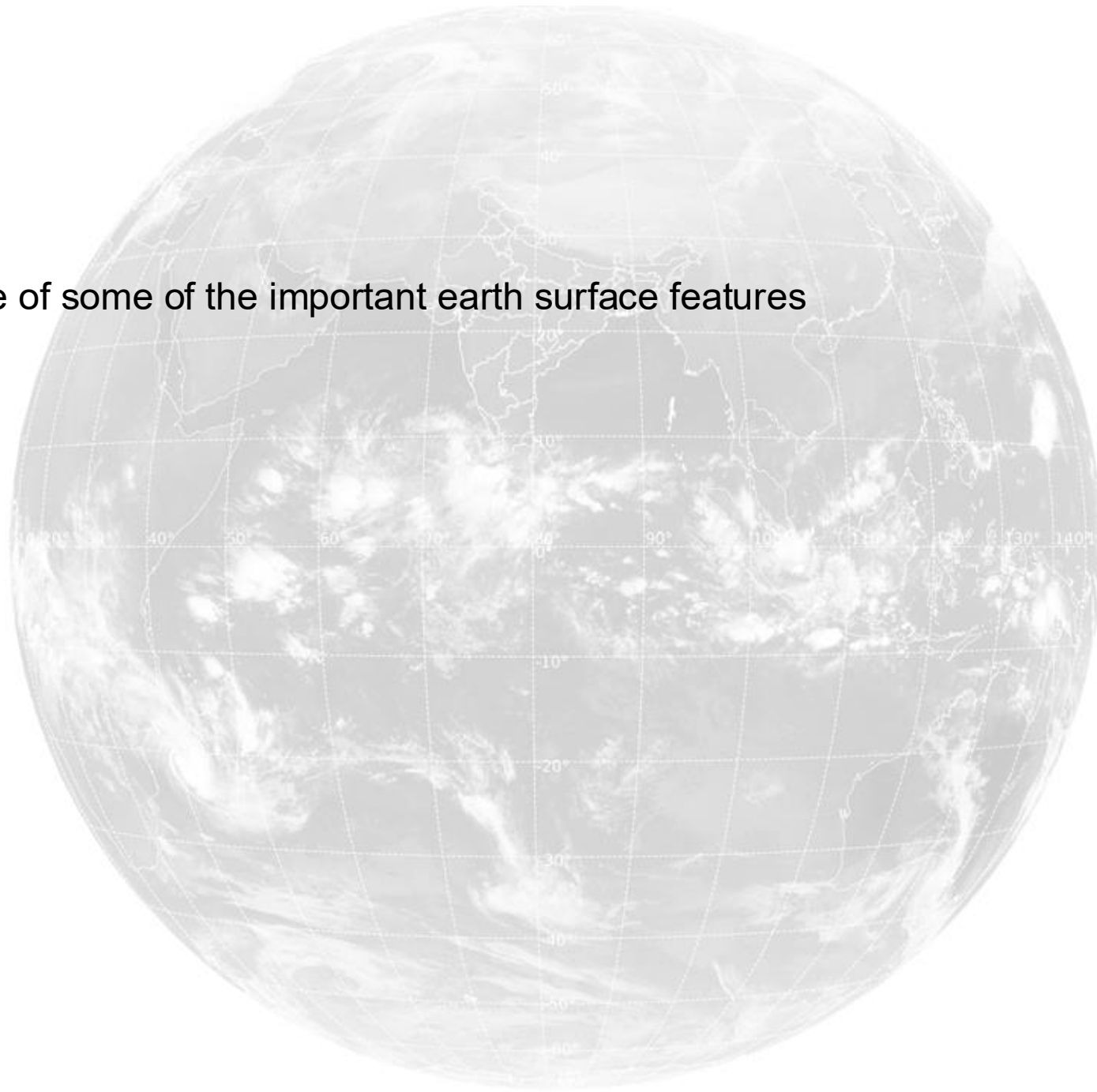
Use of Spectral Reflectance in Remote Sensing...

- Spectral reflectance characteristics of the surface features is used to identify the features and to study their characteristics
- Requires basic understanding of the general reflectance characteristics of different features

Objectives

Spectral reflectance curve of some of the important earth surface features

- Vegetation
- Bare soil
- Water



Spectral Reflectance Curve

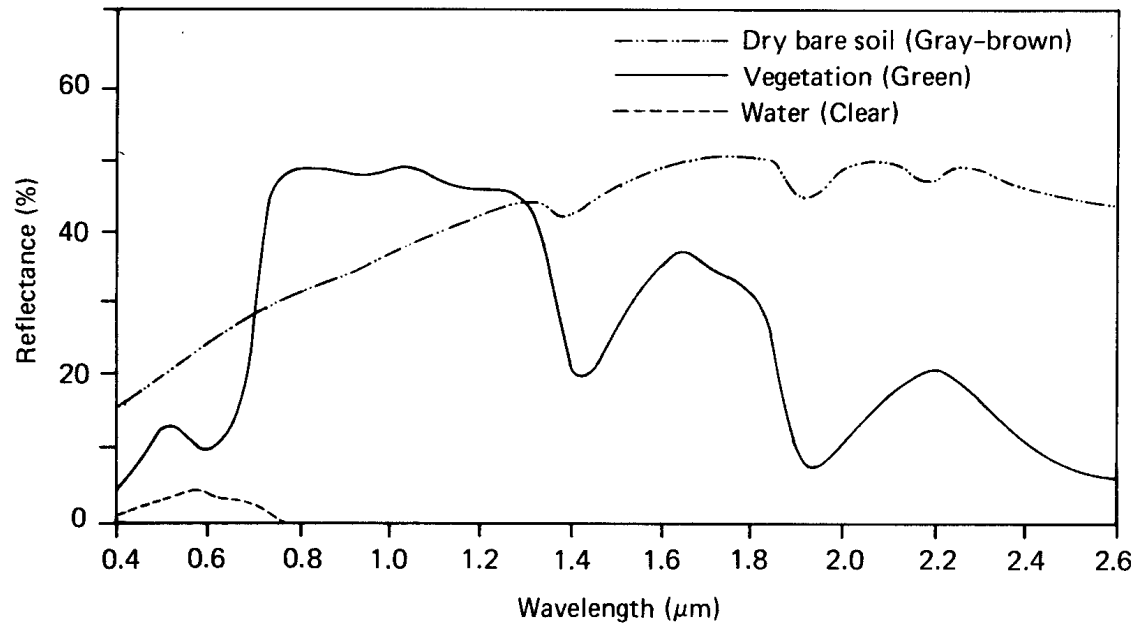
- Spectral Reflectance R_λ

$$R_\lambda = \frac{E_R(\lambda)}{E_I(\lambda)} = \frac{\text{Energy of wavelength } \lambda \text{ reflected from the object}}{\text{Energy of wavelength } \lambda \text{ incident on the object}} \times 100$$

- Spectral Reflectance Curve

- Graphical representation of the spectral response over different wavelengths of the electromagnetic spectrum
 - Gives an insight into the spectral characteristics of different objects
 - Used for the selection of a particular wavelength band for remote sensing data acquisition
- Essential to interpret and analyze an image obtained in any one or multiple wavelengths

Spectral Reflectance Curves...



Typical spectral reflectance curves for vegetation, soil and water (Lillesand et al., 2004)

Average reflectance curves of healthy vegetation, dry barren soil and clear water bodies

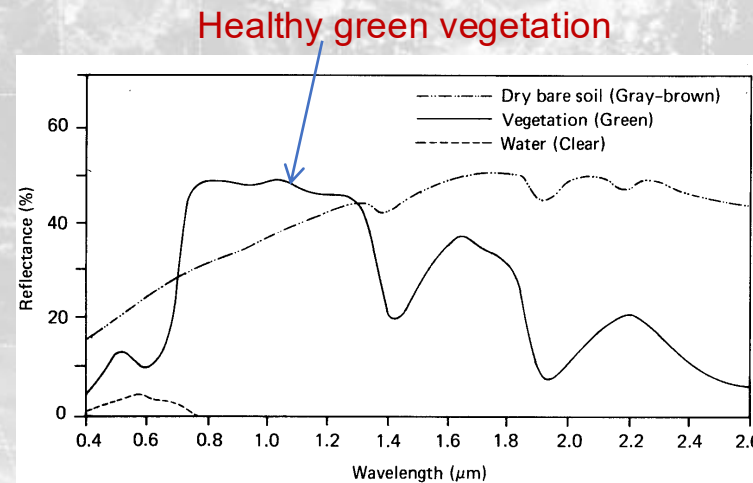
- Reflectance of individual features varies considerably above and below the average
- The average curves demonstrate some fundamental points concerning spectral reflectance

Spectral Reflectance Curve for Vegetation

Spectral reflectance curve for **healthy green vegetation** exhibits the "peak-and-valley" configuration

- Peaks indicate strong reflection in the wavelength bands
- Valleys indicate predominant absorption of the energy in the wavelength band

- Good absorption in visible region (0.4-0.7 μm)
- Absorption reduces and reflection increases in the red/infrared boundary near 0.7 μm
- Reflectance is nearly constant from 0.7-1.3 μm
- Reflectance decreases for longer wavelengths

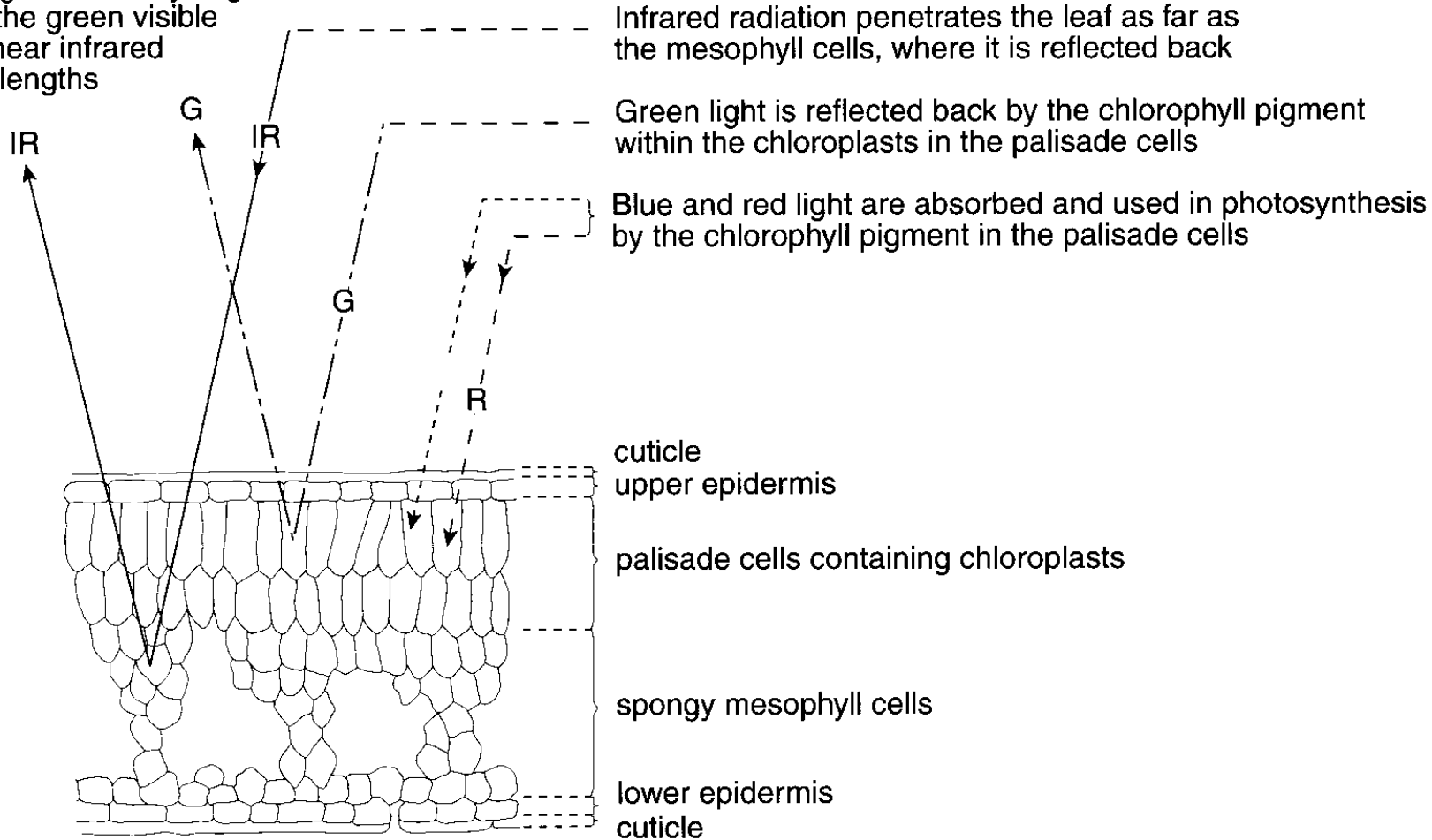


(Source : Lillesand et al., 2004)

Spectral Reflectance Curve for Vegetation...

Spectral response of vegetation depends on the structure of the plant leaves

The electromagnetic response from green healthy vegetation is at the green visible and near infrared wavelengths

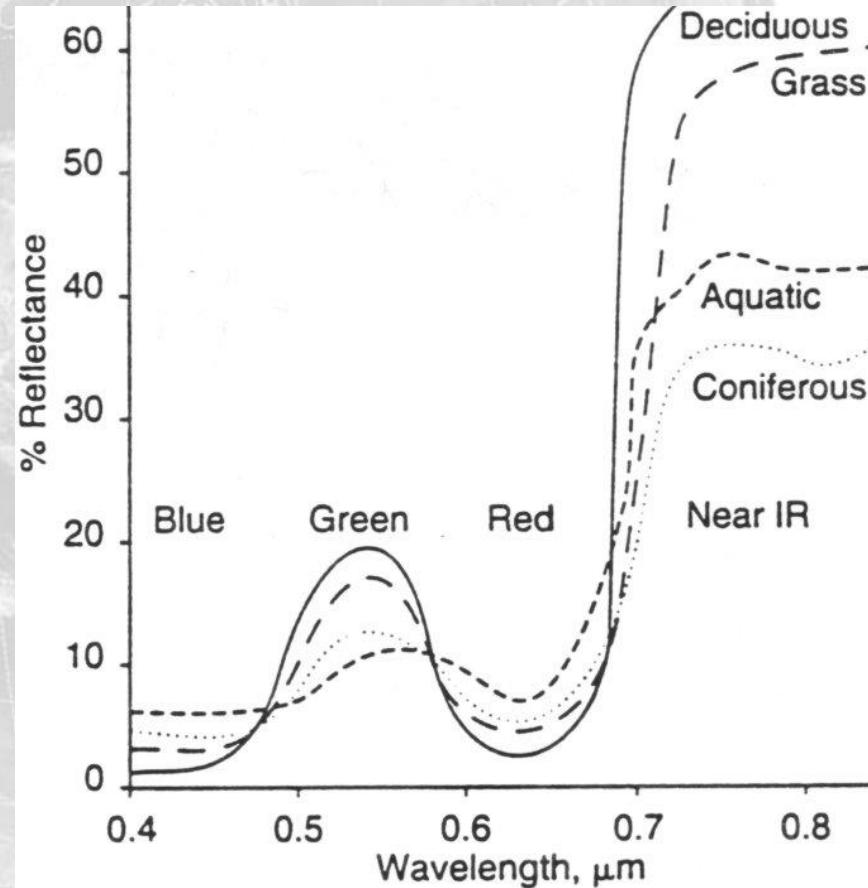


Spectral Reflectance Curve for Vegetation...

- Spectral response of vegetation depends on the structure of the plant leaves
 1. Strongly absorbs the energy in the visible region

Visible spectrum and the Chlorophyll pigments

- The palisade cells containing sacs of green pigment (chlorophyll) strongly absorb energy in the bands centered at 0.45 and 0.67 μm (blue and red)
 - Causes valleys in the visible portion of the curve
- Reflection peaks for green in the visible region
 - Makes our eyes perceive healthy vegetation as green in colour
 - Only 10-15% of the incident energy is reflected in the green band



Spectral reflectance of healthy vegetation in the visible and NIR wavelength bands <http://www.geog.ucsb.edu/>

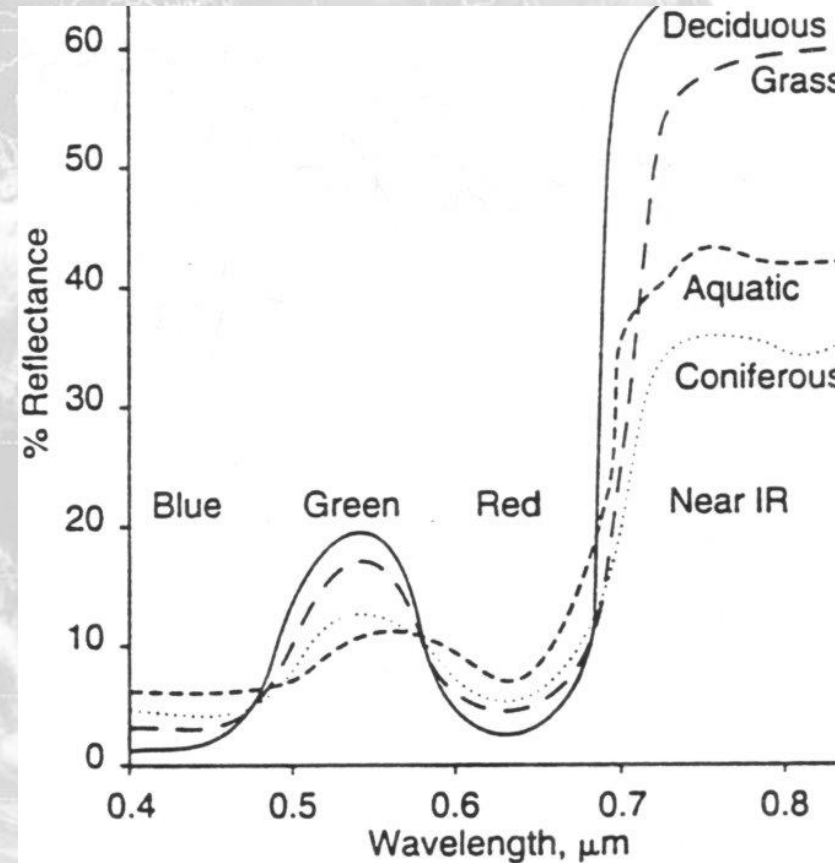
Spectral Reflectance Curve for Vegetation...

Spectral response of vegetation depends on the structure of the plant leaves

2. High reflectance in the reflected IR or NIR region

NIR and the Mesophyll Cells

- IR radiation penetrates the palisade cells and reaches the mesophyll cells
- Mesophyll cells reflect ~ 60% of the NIR radiation reaching this layer
 - At 0.7 μm , the reflectance increases
 - In 0.7-1.3 μm , ~ 50% of the incident energy is reflected
 - Healthy vegetation shows brighter response in the NIR region compared to the green region
- Most of the remaining energy is transmitted
 - Absorption in this spectral region is minimum



Spectral reflectance of healthy vegetation in the visible and NIR wavelength bands <http://www.geog.ucsb.edu/>

Spectral Reflectance Curve for Vegetation...

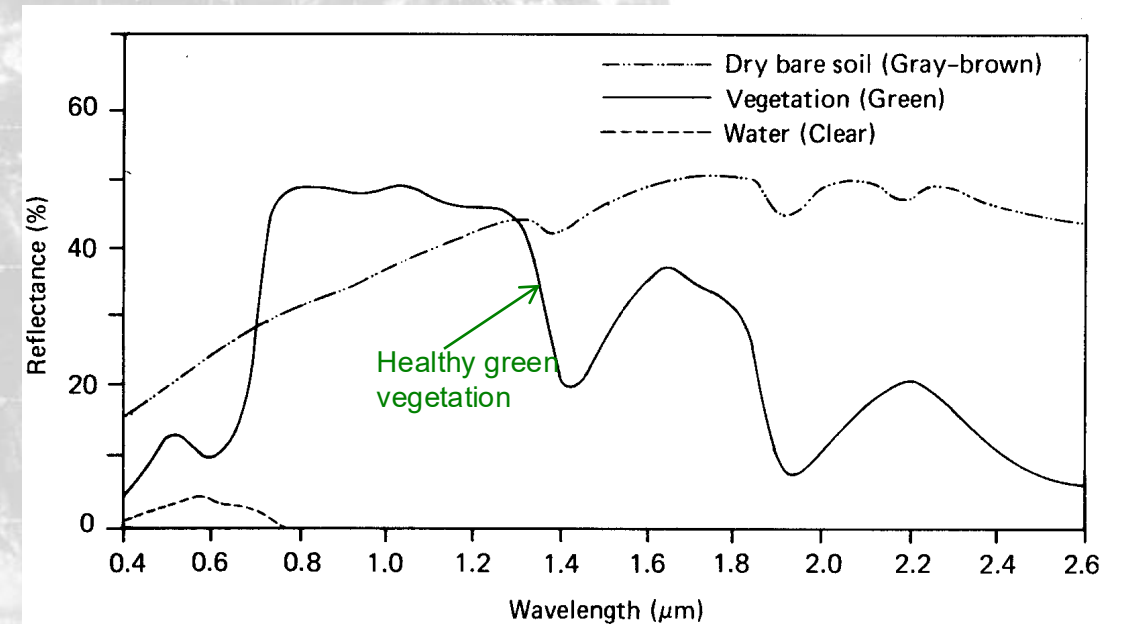
Spectral response of vegetation depends on the structure of the plant leaves

3. At wavelengths beyond 1.3 μm , leaf reflectance is approximately inversely related to the total water present in a leaf

– Total water is a function of both the moisture content and the thickness of a leaf

- A little to no transmittance of energy beyond 1.3 μm
- Energy beyond 1.3 μm is absorbed or reflected
- Dips in reflectance occur at 1.4, 1.9, and 2.7 μm
 - Water in the leaf strongly absorbs the energy at these wavelengths
 - These wavelength regions are referred to as **water absorption bands**

Reflectance peaks occur at 1.6 and 2.2 μm , between the absorption bands



Spectral Reflectance Curve for Vegetation...

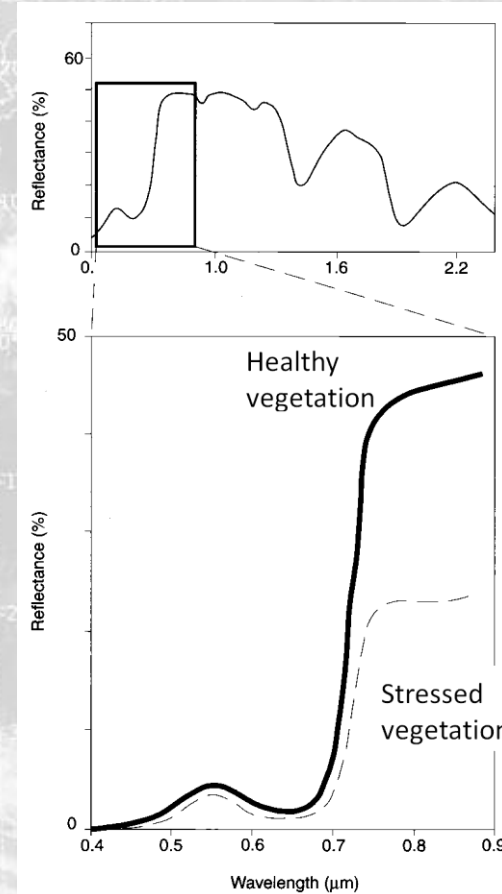
- Spectral response of vegetation depends on the structure of the plant leaves

Healthy vegetation

- Chlorophyll content in the palisade cells absorbs blue and red in the visible region
- Mesophyll cells strongly reflect the NIR radiation

Stressed vegetation

- Decrease in the chlorophyll content
 - Less absorption in the blue and red bands
 - Red and blue bands also get reflected along with the green band, giving yellow or brown colour
- NIR bands are absorbed by the stressed mesophyll cells
 - Causing dark tones in the image



(Source: Gibson, 2000)

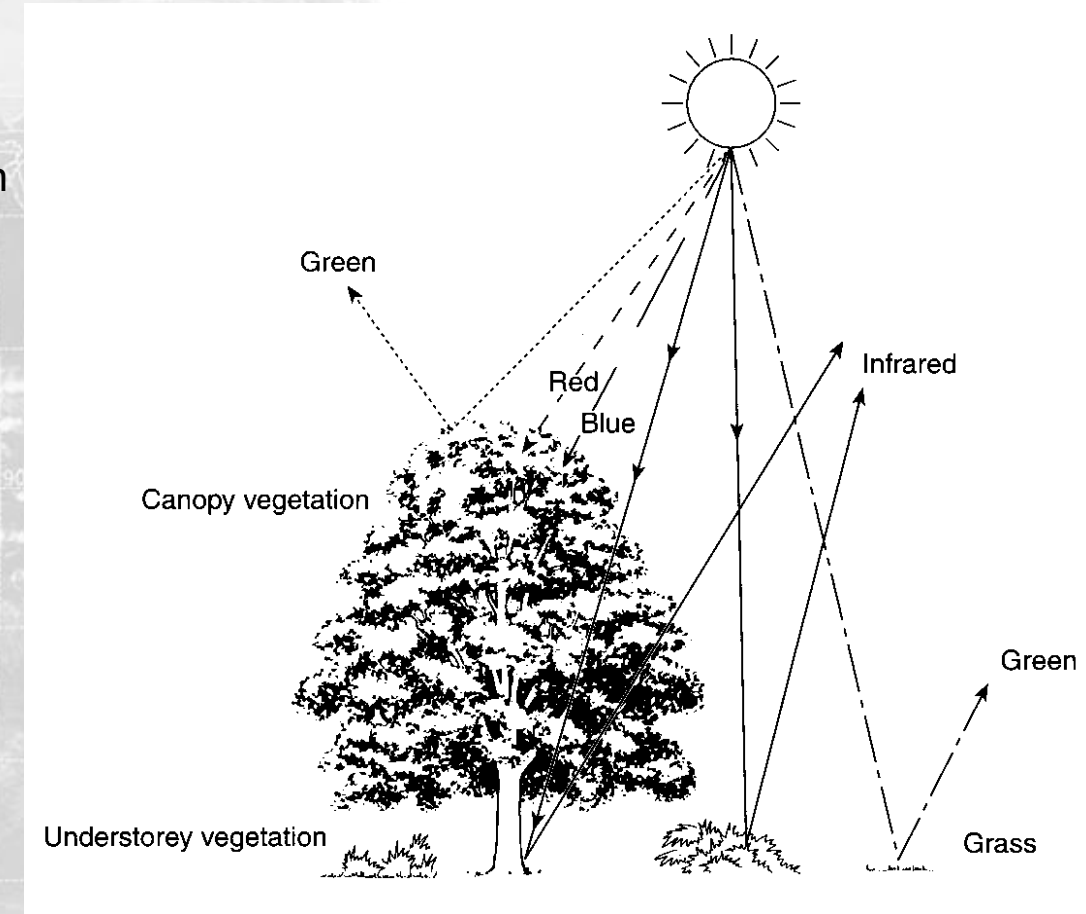
Spectral Reflectance Curve for Vegetation...

Transmittance

- Transmittance of electromagnetic radiation is less in the visible region
- Transmittance increases in the infrared region
- A little to no transmittance of energy beyond $1.3\text{ }\mu\text{m}$

Transmittance and reflectance

- Vegetation canopies display a layered structure
- Energy transmitted by one layer is available for reflection or absorption by the layers below it
 - Total infrared reflection from thicker canopies will be more compared to thin canopy cover
 - From the reflected NIR, the density of the vegetation canopy can thus be interpreted.



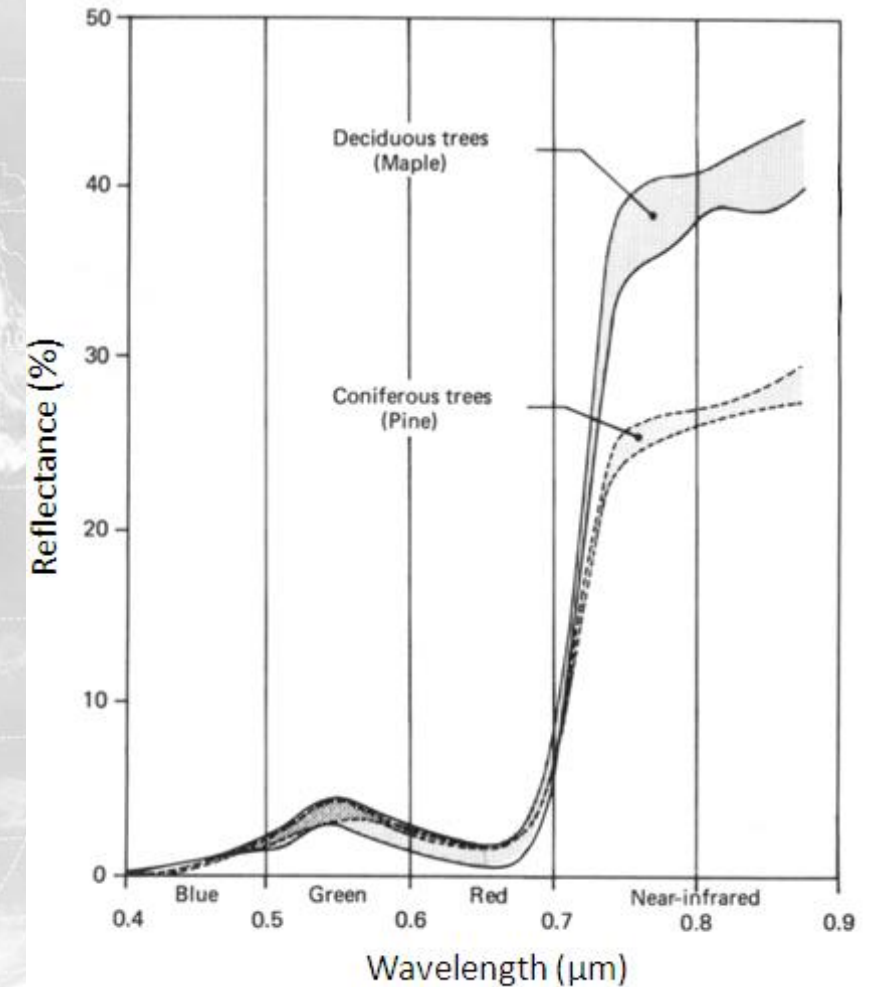
Reflectance from dense forest and thin vegetation canopies (Gibson, 2000)

Spectral Reflectance Curve for Vegetation...

- Total infrared reflection from thicker canopies will be more compared to thin canopy cover

Example:

1. For a densely grown agricultural area, the NIR signature will be more
2. Deciduous and coniferous trees
 - Spectral reflectance may be similar in the green band
 - Deciduous trees show higher reflection in the NIR band



Spectral Reflectance of Soil

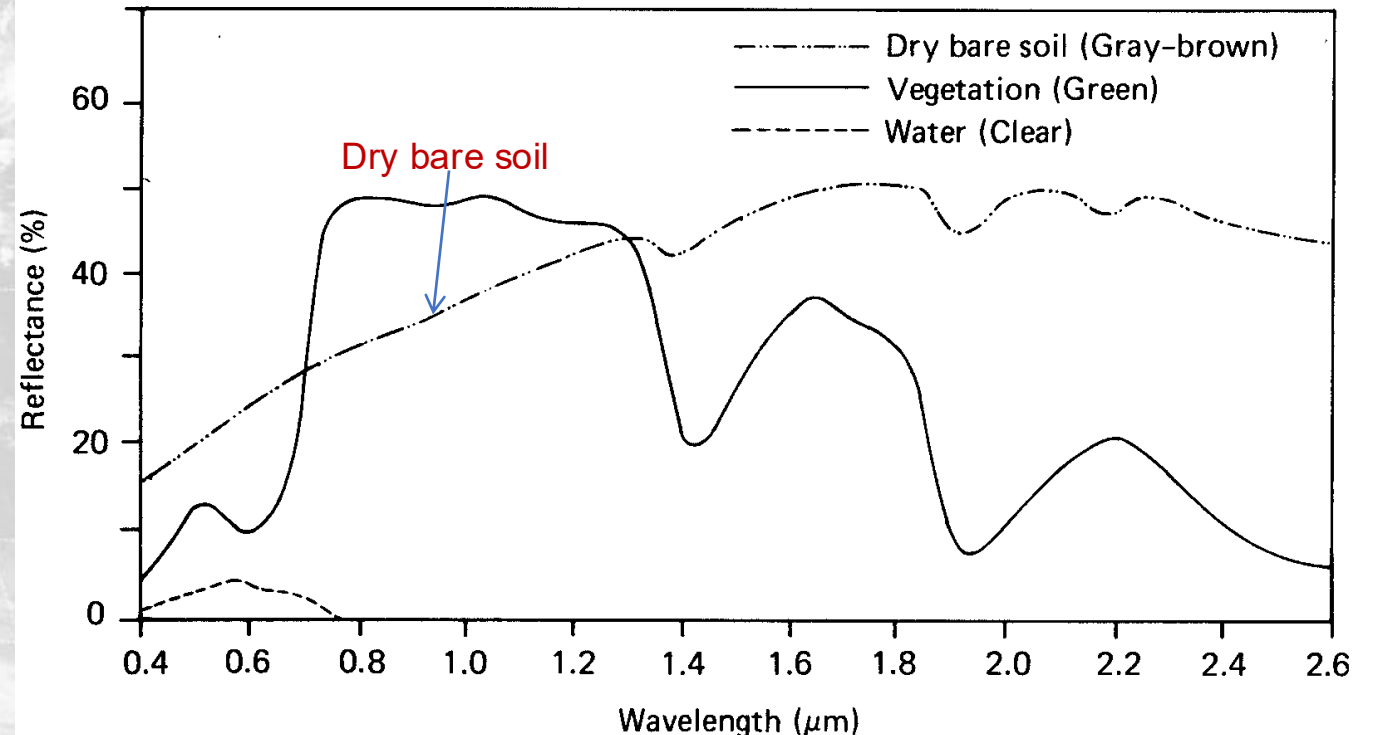
Spectral reflectance curve for soil shows considerably less peak-and-valley variation compared to that for vegetation

–The factors that influence soil reflectance act over less specific spectral bands

Factors affecting soil reflectance

- Moisture content
- Soil texture (proportion of sand, silt, and clay)
- Surface roughness
- Presence of iron oxide and organic matter

These factors are complex, variable, and interrelated



(Source : Lillesand et al., 2004)

Spectral Reflectance of Soil...

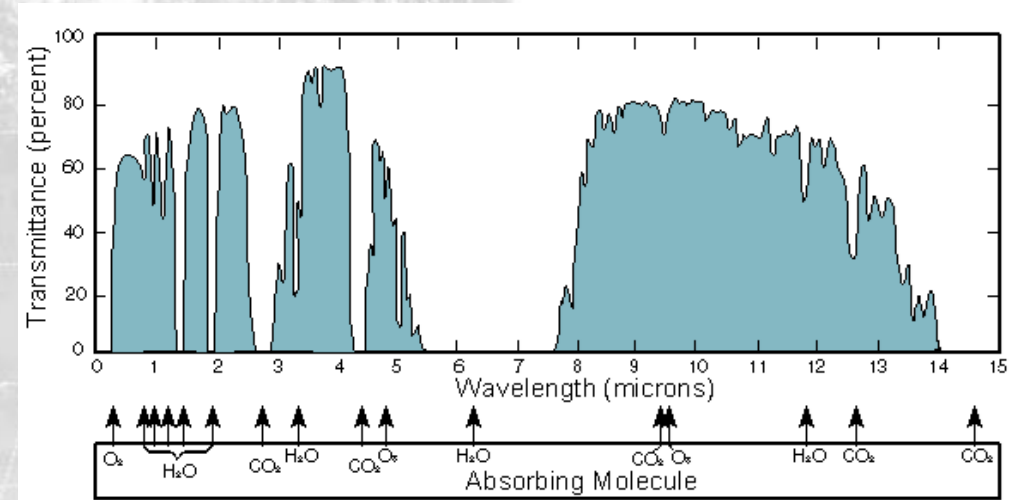
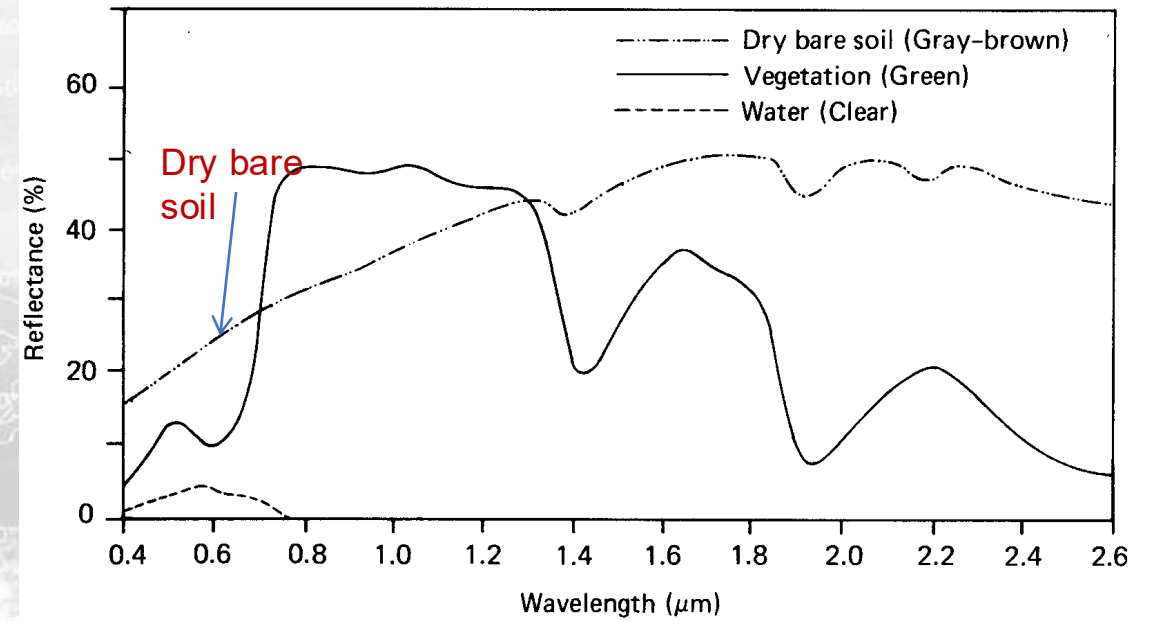
Dry soil

- Coarse texture → Less reflectance → Dark tone
- Fine texture → more reflectance → Light tone

Wet soils display reverse tendency

Moisture content reduces the reflectance

- Coarse texture → better drainage → less moisture content → good reflectance
- Fine texture → poor drainage → more moisture content → poor reflectance
- Water absorption bands are at 1.4, 1.9, and 2.7 μm wavelengths
- Clay soils have hydroxyl ion absorption bands at 1.4 and 2.2 μm .
- Surface roughness reduces the reflectance
- Presence of organic matter reduces the soil reflectance
- Presence of iron oxide significantly decreases soil reflectance, at least in the visible region



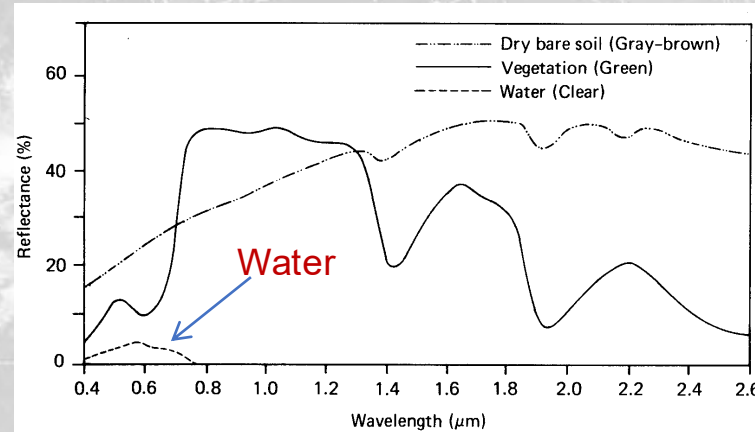
Spectral Reflectance of Water

Water provides a semi-transparent medium for the electromagnetic radiation

Electromagnetic radiations get reflected, transmitted or absorbed in water

Spectral responses varies with

- Wavelength of the radiation
- Physical and chemical characteristics of the water



(Source : Lillesand et al., 2004)

Water in liquid phase

- High reflectance in the visible region between 0.4μm and 0.6μm
- Wavelengths beyond 0.7μm are completely absorbed.

Water in solid phase (ice or snow)

- Good reflection at all visible wavelengths

Spectral Reflectance of Water...

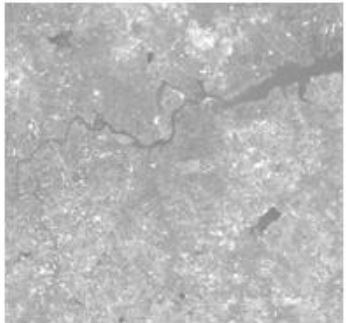
Liquid water

- High reflectance in the visible region between $0.4\mu\text{m}$ and $0.6\mu\text{m}$
- Wavelengths beyond $0.7\mu\text{m}$ are completely absorbed.

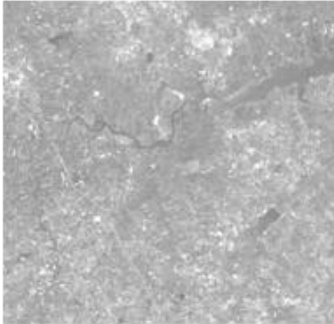
Clear water appears darker in tone in the NIR image

Mapping of water bodies with remote sensing data is done in reflected infrared wavelengths

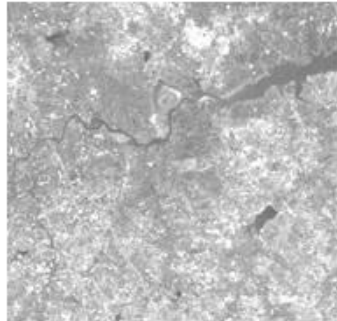
B1 ($0.45\text{--}0.52\mu\text{m}$)



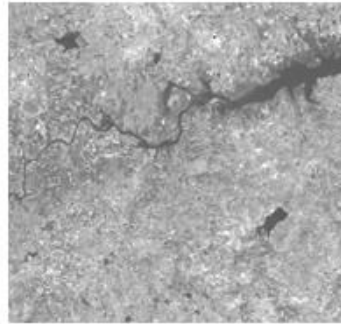
B2 ($0.52\text{--}0.60\mu\text{m}$)



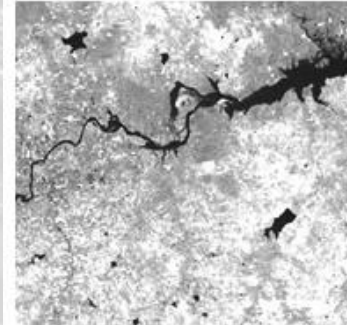
B3 ($0.63\text{--}0.69\mu\text{m}$)



B4 ($0.76\text{--}0.90\mu\text{m}$)



B5 ($1.55\text{--}1.75\mu\text{m}$)



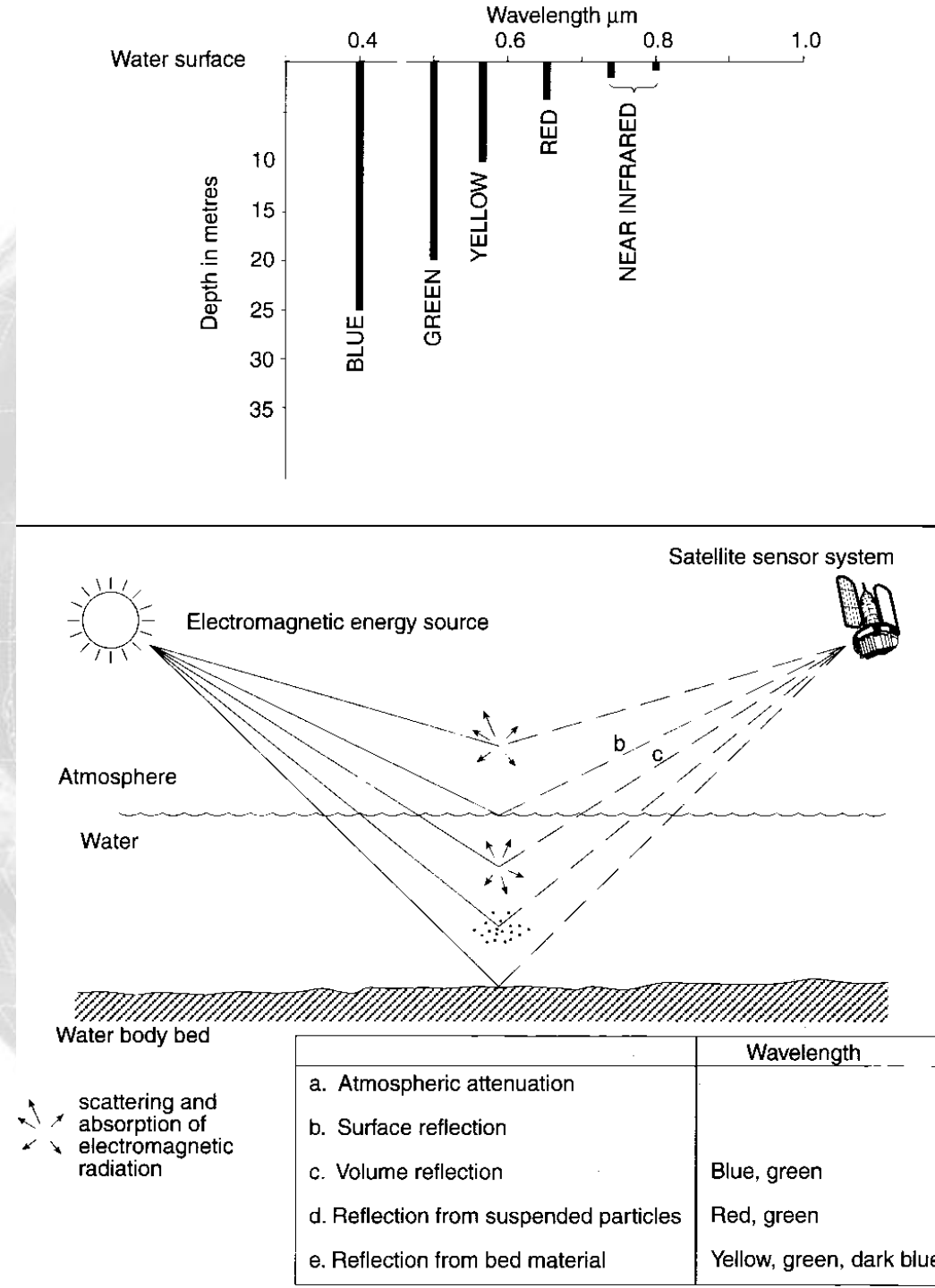
Part of the Krishna River Basin in different bands of the Landsat ETM+ imagery

- The water body appears in dark colour in all bands and displays sharp contrast in the IR bands.

Spectral Reflectance of Water...

- Reflectance from a water body can be from
 - An interaction with the water's surface (specular reflection)
 - An interaction with material suspended in the water
 - An interaction with the bottom surface of the water body

Complex spectral response from a water body
(Gibson, 2000)



Spectral Reflectance of Water...

- Reflectance properties of a water body also depends on the materials present in water
- Clear water
 - Absorbs relatively little energy having wavelengths shorter than $0.6 \mu\text{m}$.
 - High transmittance typifies these wavelengths with a maximum in the blue-green portion of the spectrum.
- Turbidity : Presence of suspended sediments increases visible reflectance
- Chlorophyll : Decrease blue wavelength reflection and increase green wavelength reflection
- Dissolved oxygen concentration, pH, and salt concentration
 - Cannot be observed directly through changes in water reflectance
 - Correlation between these parameters and observed reflectance is used

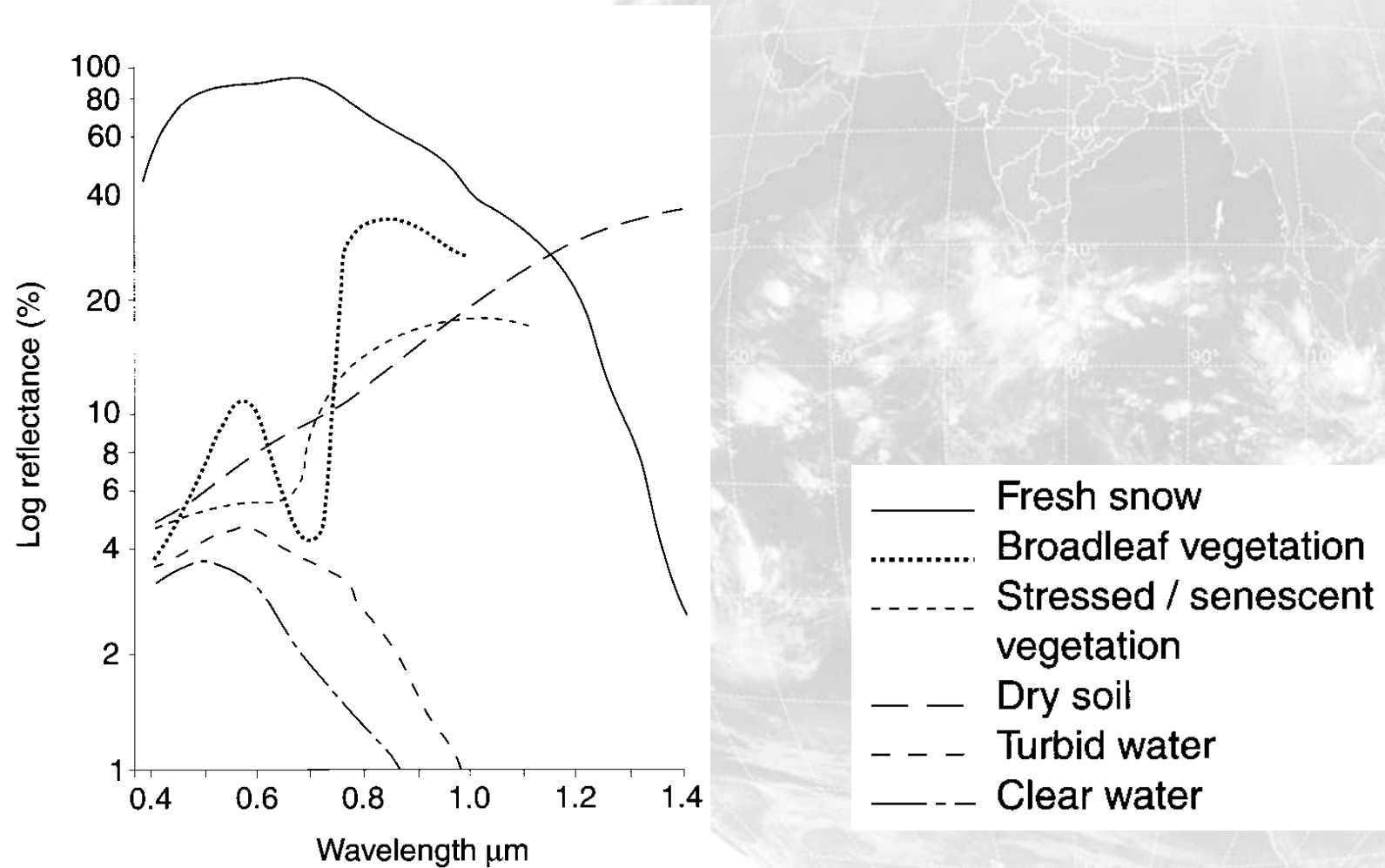
More details read: [remote sensing applications for monitoring water quality related articles](#)

Spectral Reflectance of Water...

- Variation in the spectral reflectance in the **visible region** can be used to differentiate
 - Shallow and deep waters
 - Clear and turbid waters
 - Rough and smooth water bodies
- Reflectance in the **NIR range** are generally used to
 - Delineate the water bodies
 - To study the algal boom and phytoplankton concentration in water

Further details on the spectral characteristics of vegetation, soil, and water can be found in Swain and Davis (1978)

Spectral Reflectance of Some Natural Features



Spectral Reflectance in Remote Sensing

Multi spectral remote sensing

- Multiple sensors are used to sense the reflectance in different wavelength bands
- Reflectance recorded in multiple bands are analysed to find spectral reflectance variation with wavelength
- Using the average spectral reflectance curves as the basic information, the spectral reflectance variation is used to identify the target features.

Multi Spectral Remote Sensing-Example



Aerial photographs of a stadium normal colour

- Artificial turf inside the stadium and the natural vegetation appears in the same colour



Aerial photographs of a stadium in colour IR

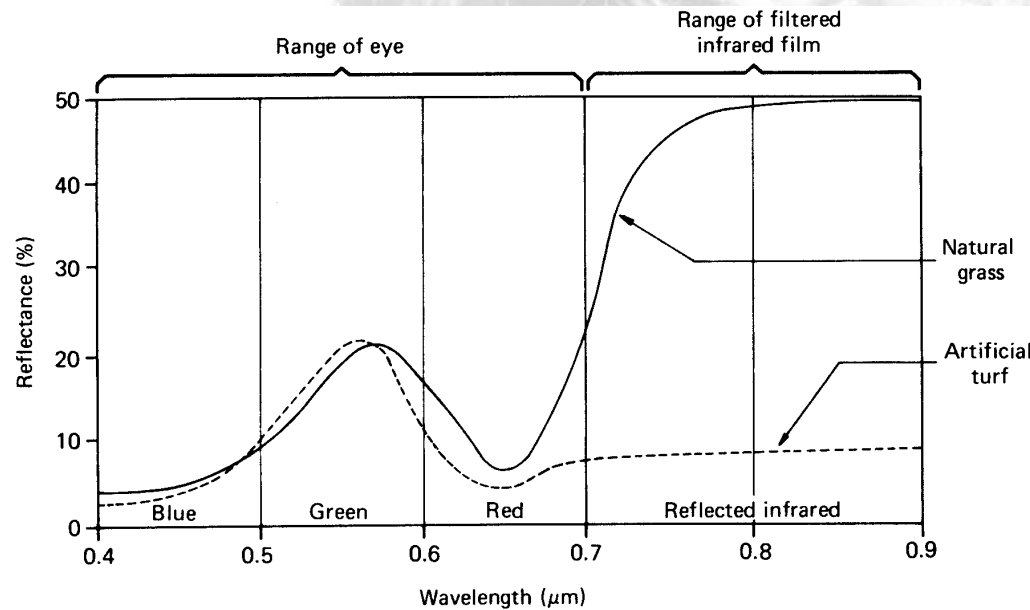
- Artificial turf appears dark, whereas the natural vegetation shows high reflectance in the IR region

(Images are taken from Lillesand et al., 2004)

Multi Spectral Remote Sensing-Example

Artificial turf inside the stadium and the natural vegetation appear in the same colour in the visible region

Artificial turf appears dark, whereas the natural vegetation shows high reflectance in the IR region

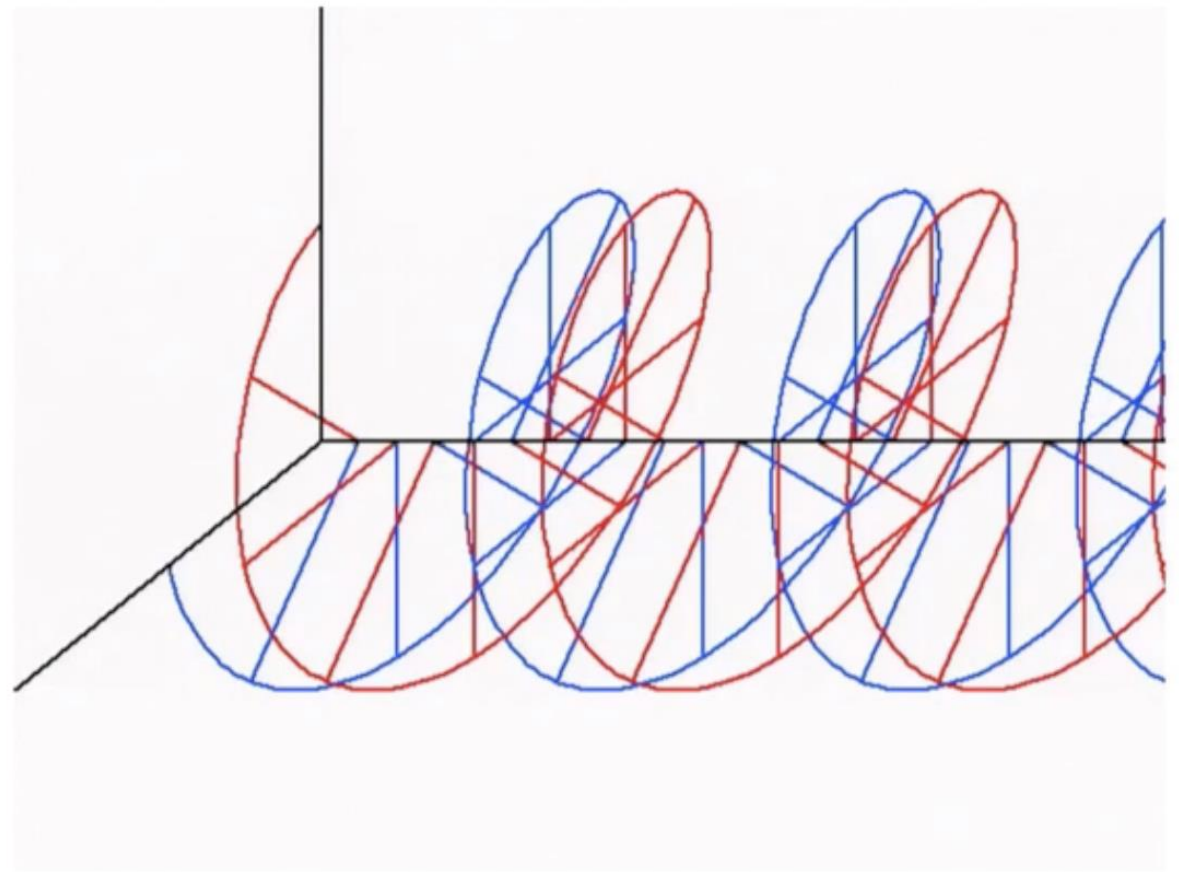
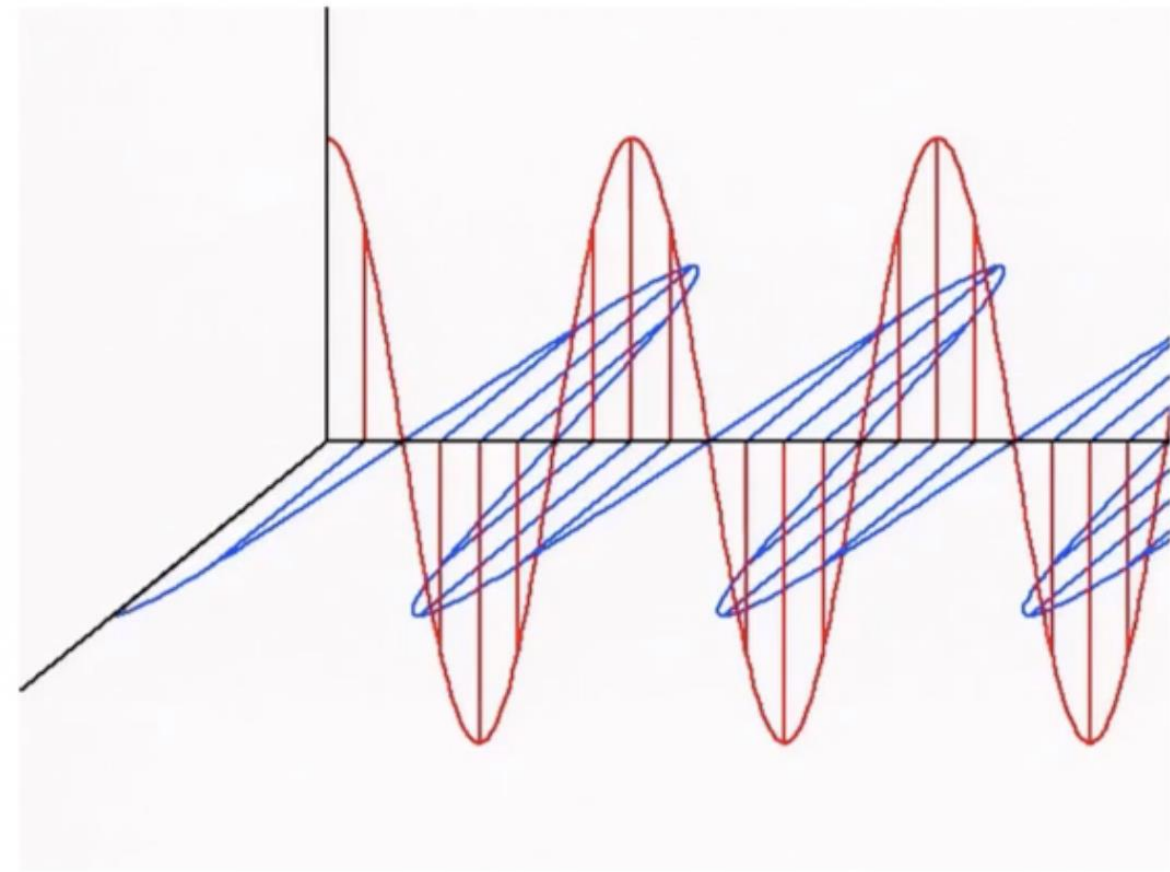


Spectral reflectance curves of the natural vegetation and the artificial turf (From Lillesand et al., 2004)

Polarization of EMR

- Polarization indicates the **direction of the orientation of the electric** field in EMR.
- Sunlight of most other common light sources are randomly polarised (mixture of waves with different polarisation)
- Some of the common polarisations are: **Horizontal, vertical, circular, elliptical**.
- Plane of vibration parallel to plane of incidence – Horizontal Polarisation
- Plane of vibration perpendicular to plane of incidence – Vertical Polarisation.
- A linearly polarised wave is also known as **plane polarised** wave.

Polarization of EMR



Polarization of EMR

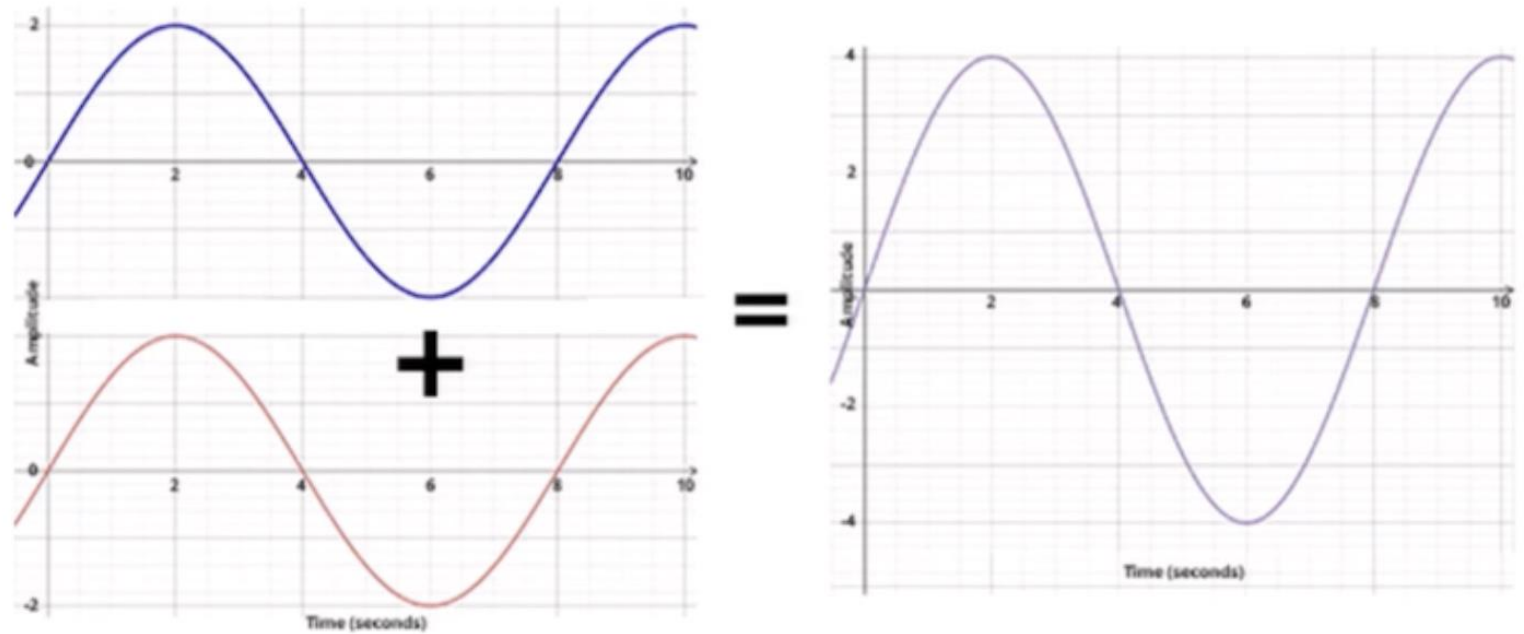
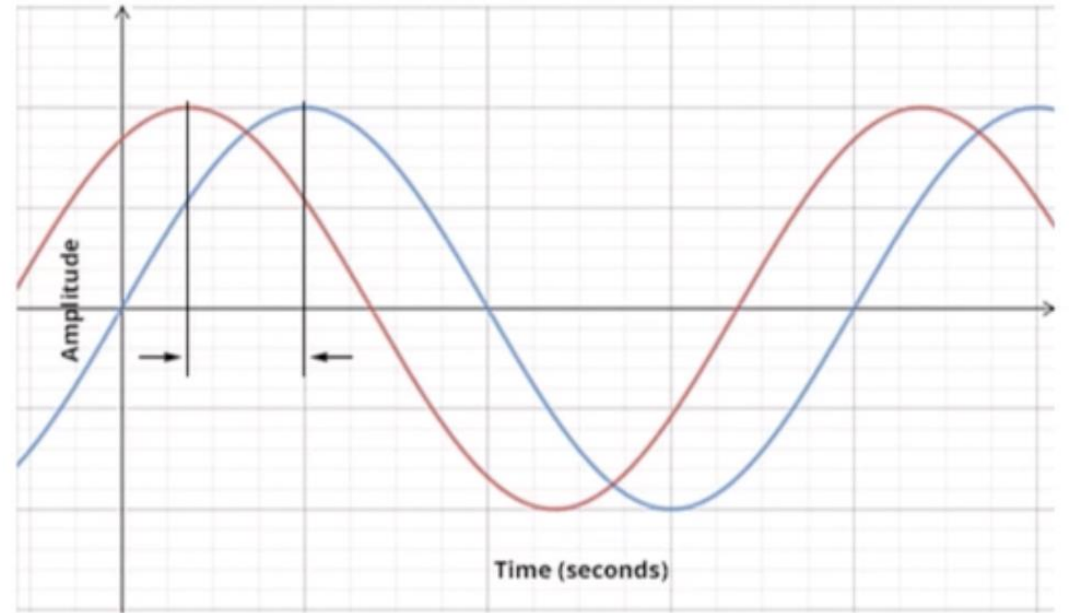
- Interaction of EMR with any matter (say atmosphere, land surface) will affect the polarisation of the EMR.



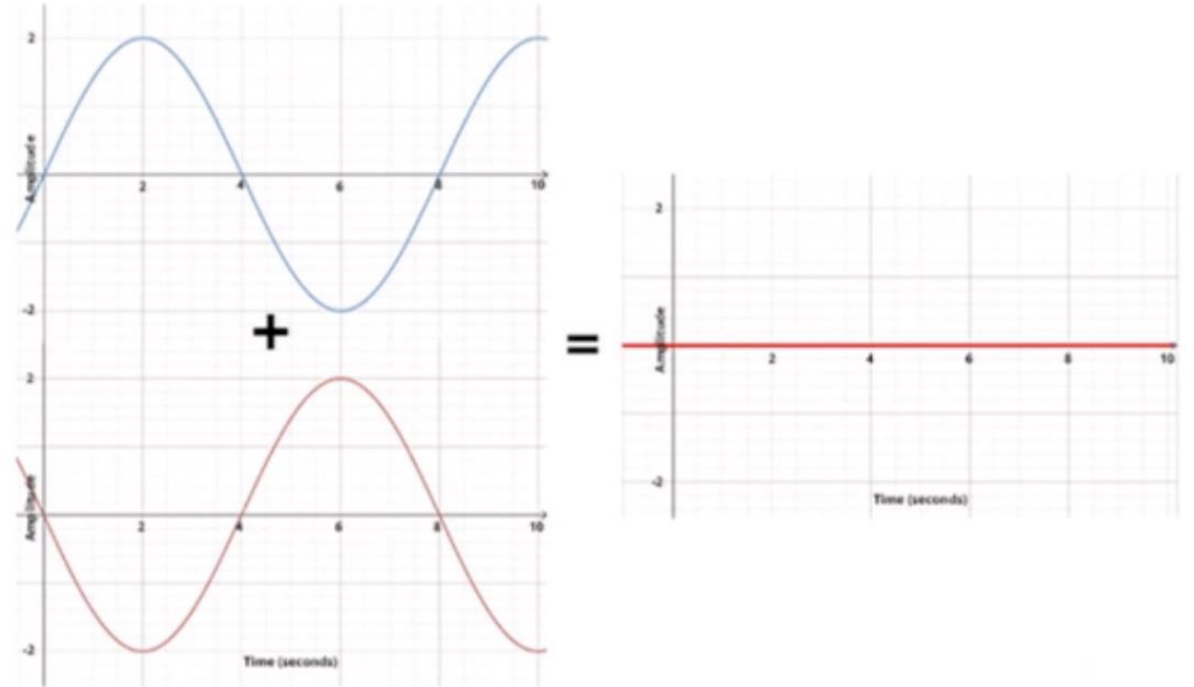
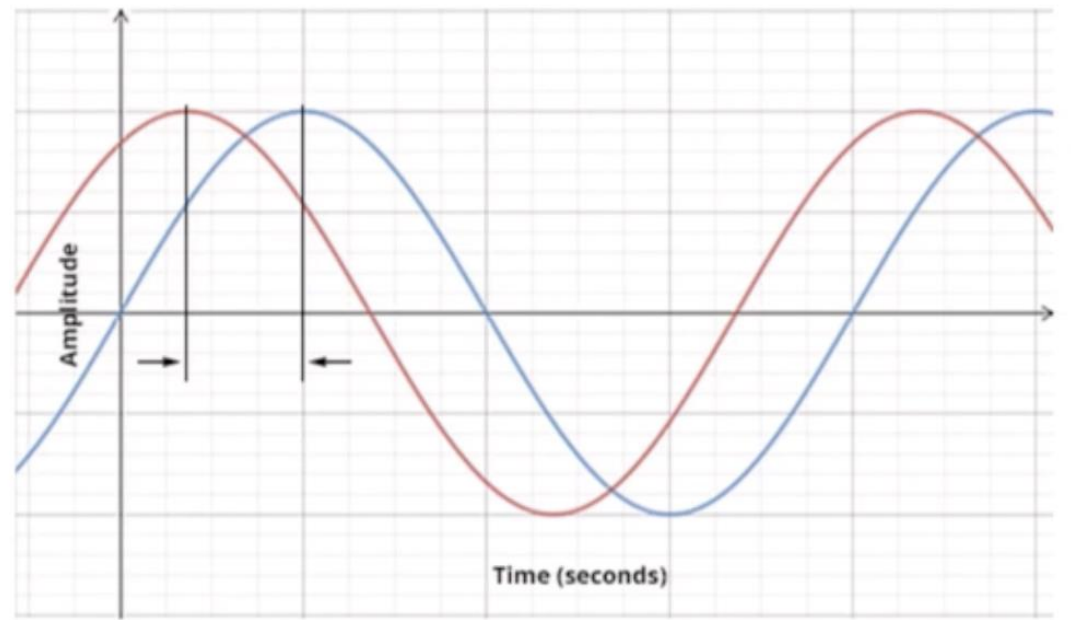
Coherent radiation

- Two waves are said to be coherent if there is a systematic relationship between their **phase**.
- Two waves will be perfectly coherent only if they are **monochromatic** and the waves are **parallel** (i.e. collimated).
- Coherent waves, when they meet, may result in interference.

Coherent radiation



Coherent radiation



PRINCIPLES OF RADIOMETRY

- ✓ Radiometry deals with the quantitative measurement of electromagnetic radiation (*due to emission or interaction with matter*)
- ✓ Deals with transfer of radiation from source to detector through a medium for quantitative evaluation
- ✓ Radiometry → Deals with electromagnetic radiation [*all wavelengths*]
- ✓ Photometry → Light interaction that is detected by human eye [*visible band*]
- ✓ Radiometry is a set of techniques for measuring electromagnetic radiation, including visible light.
- ✓ Radiometry Involves the following steps:
 - 1) Measurement of Geometry
 - 2) Measurement of Spectrum

RADIOMETERS



Crookes Radiometer



Net Radiometer



Pyranometer

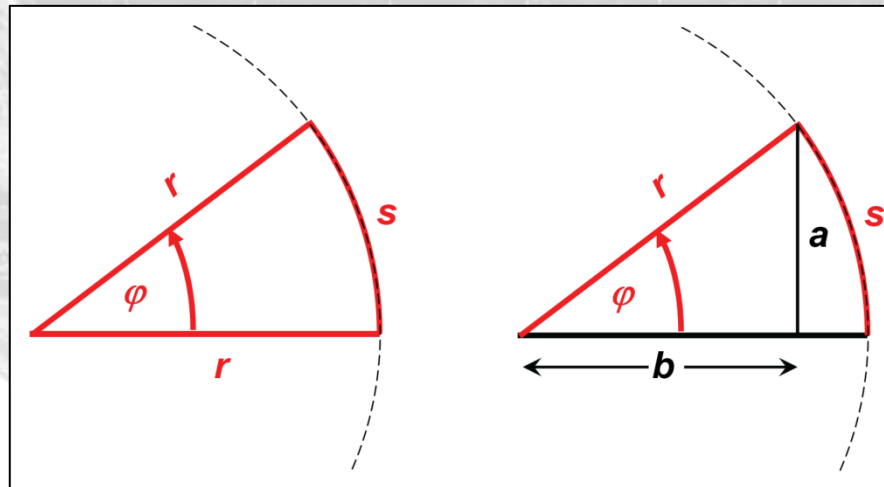


Spectroradiometer

GEOMETRY MEASUREMENT

- ✓ Plane Angle (ϕ) $\rightarrow$ Angle contained by two straight lines meeting at a point
- ✓ Used in 2-D analysis
- ✓ In radians, it is the ratio of arc length (s) to radius (r)
- ✓ For small angles, the curvature of arc length is neglected

$$\phi = \frac{s}{r} \approx \frac{a}{r} = \tan \phi$$



GEOMETRY MEASUREMENT

two-dimensional angle in three-dimensional space that an object subtends

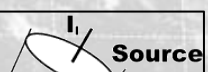
large the object appears to an observer looking from that point.

is (dimensionless)

surface area of object to the square of the radius (A / r^2)

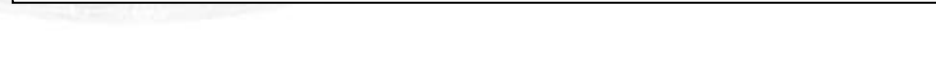
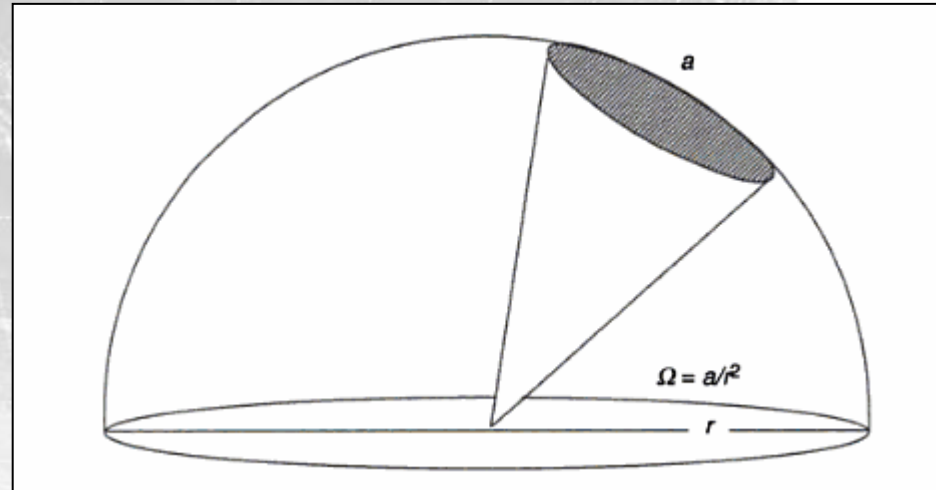
angle subtended by area at a point

solid angle, $\Omega = 4 \pi$



The diagram shows a small surface element labeled I_1 and a light source labeled "Source". A line connects the source to the surface element, and another line is perpendicular to the surface element, illustrating the geometry of light measurement.

-
- The left diagram illustrates the geometry of radiation incident on a differential area element dA on a sphere of radius $r=1$. The normal vector $\mathbf{n}$ is shown. The incident radiation comes from a source at distance r_0 at an angle θ_i relative to the normal. The reflected radiation is shown at an angle θ_r . The solid angle subtended by the source is $d\omega$. The right diagram shows a solid angle Ω subtended by a surface area a at a distance r , with the formula $\Omega = a/r^2$.



the radius of moon = 1.74×10^3 km and the distance between earth and the moon is 3.84×10^5 km

the radius of Sun = 1.496×10^8 km and the distance between sun and earth is distance given is 6.96×10^5 km

Calculate Solid Angle subtended by moon and sun on a point on earth.

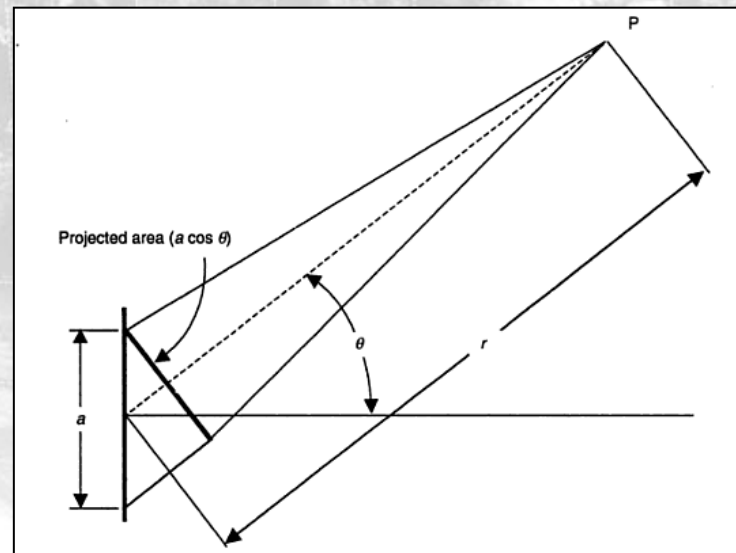
$$\text{Solid Angle} = (A / r^2)$$

$$\text{Moon} = 6.45 \times 10^{-5} \text{ Sr}$$

$$\text{Sun} = 6.79 \times 10^{-5} \text{ Sr}$$

GEOMETRY IN REMOTE SENSING

- ✓ Solid Angle subtended by a flat surface (*rather than curved surface*) is to be considered for RS applications
- ✓ Ex: Solid angle made by ground pixel (*photograph*) at the lense
- ✓ Flat area estimates can be used in place of spherical areas, when solid angle is less than 0.03 Steradians (*Less than 1 % error*)
- ✓ Solid Angle of a surface (*area: a*), whose normal is making an angle ' θ ' in Steradians is given by: $\frac{a \cos \theta}{r^2}$
- ✓ *Projected area has to be considered*



Radiation quantities

Quantity	Usual symbol	Defining equation	Units
Radiant energy	Q		joule
Radiant energy density	W	$W = \frac{dQ}{dV}$	joule/m ³
Radiant flux	Φ	$\Phi = \frac{dQ}{dt}$	watt
Radiant flux density	E (irradiance) M (emittance)	$E, M = \frac{d\Phi}{dA}$	watt/m ²
Radiant intensity	I	$I = \frac{d\Phi}{d\Omega}$	watt/steradian
Radiance	L	$L = \frac{dI}{dA \cos \theta}$	watt/steradian m ²

• All these quantities can also be considered for different wavelengths (e.g. spectral radiant emittance)

RADIOMETRIC MEASUREMENTS

1) RADIANT ENERGY (Q)

- Quantity of energy carried by electromagnetic radiation (stream of photons)
- Calculated by integrating radiant flux (or power) with respect to time
- Measure of capability to perform physical work (heat, change of state, movement, ...) when interacting with matter
- Consists of energy at all wavelength bands
- Expressed in Joules (Ws)
- Radiant energy at a specific wavelength is called as spectral radiant energy (Q_A)

$$Q_A = \frac{dQ}{d\lambda}$$

RADIOMETRIC MEASUREMENTS

2) RADIANT FLUX (RADIANT POWER) - ϕ

- Rate at which, the radiant energy is emitted (or, transmitted, or received) in the form of EM radiation from a point (source) to the surface
- Represents energy per unit time (*Power*)
- Measured in Joules per Sec (or, Watts)

$$\phi = \frac{dQ}{dt}$$

- Spectral radiance flux is the transfer of radiant flux per unit wavelength at a given wavelength λ

$$\phi_{\lambda} = \frac{d\phi}{d\lambda}$$

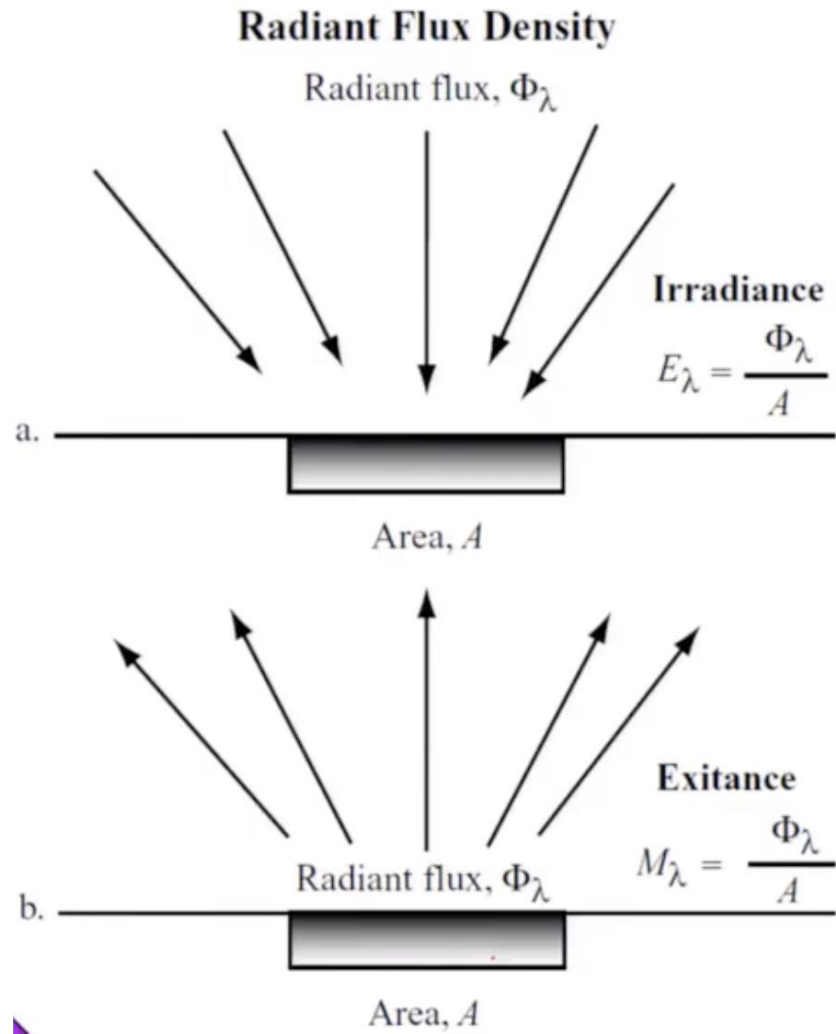
RADIOMETRIC MEASUREMENTS

3) IRRADIANCE (E)

- Measure of radiant flux per unit area
- Expresses in Watts per square meters
- If the surface is not perpendicular to the direction of propagation of EM wave, projected area has to be considered
- Projected Area = $dA \cos \theta$
- Irradiance (E) → Flux density of radiant flux arriving at a surface
- Exitance (M) → Radiant flux emitted (leaving) from a surface
- Exitance from a source is inversely proportional to the square of the radial distance from the source

$$E = \frac{d\Phi}{dA}$$

Irradiance and Emittance



RADIMETRIC MEASUREMENTS

4) RADIAL INTENSITY (I)

- Radiant flux leaving a source per unit solid angle in a given direction
- Expressed in Watts per Steradians

$$I = \frac{d\Phi}{d\omega}$$

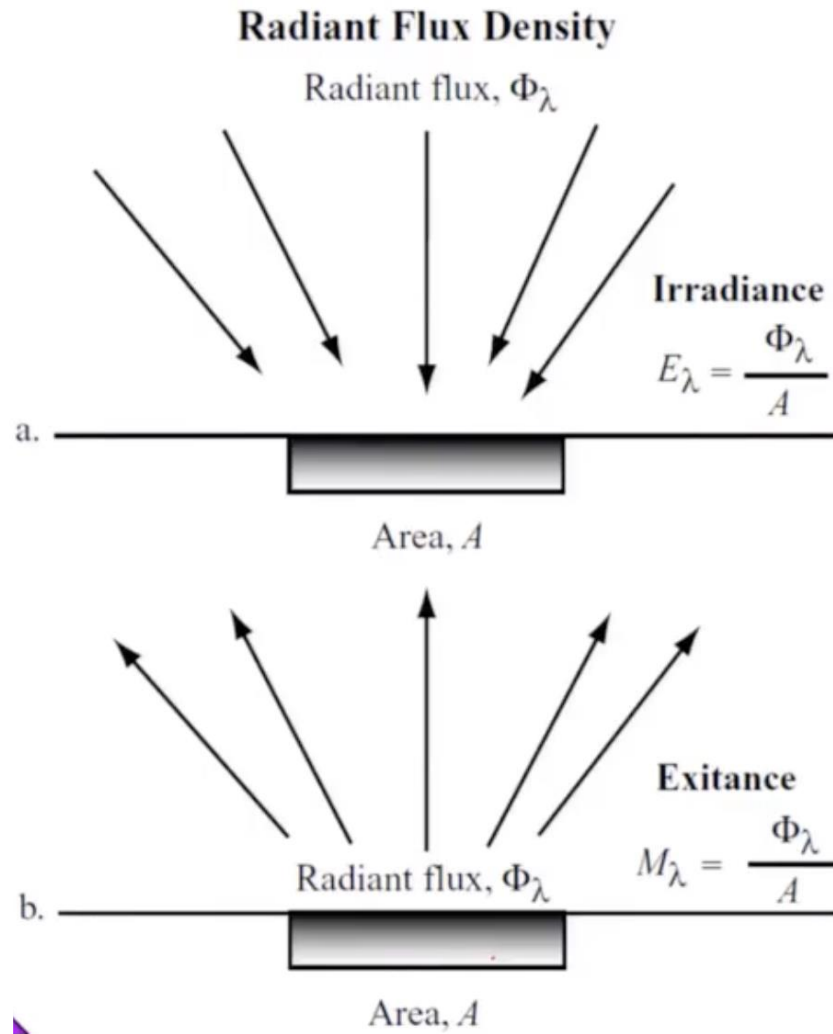
- If a point source radiates equally in all directions (*isotropic*), $I = \Phi / 4\pi$

5) RADIANCE (L)

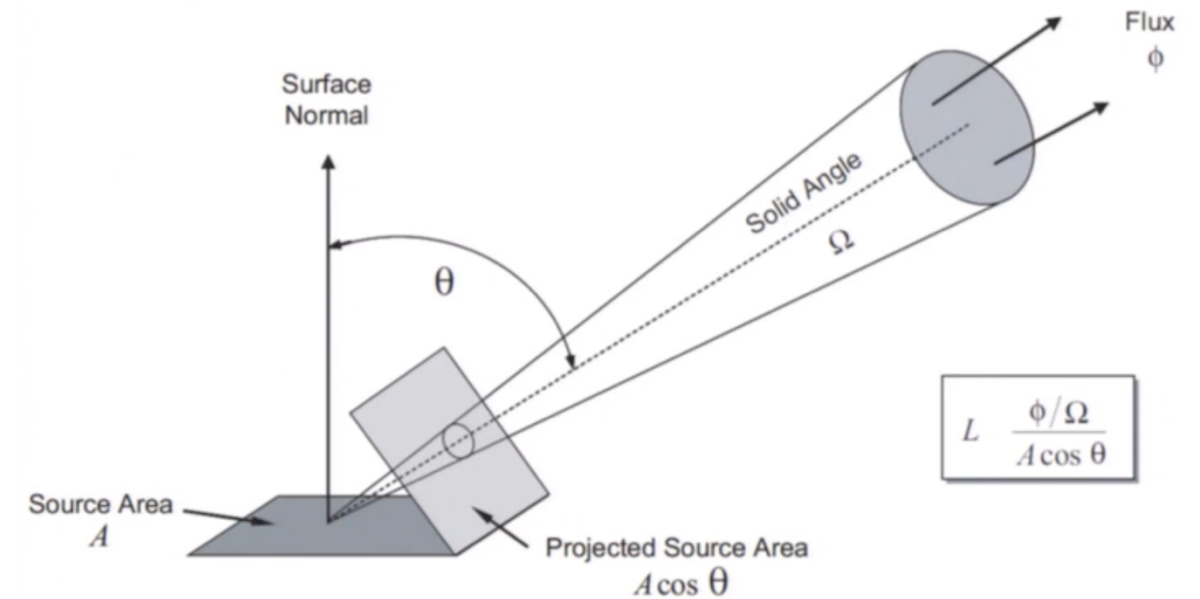
- Radiant flux per unit solid angle leaving an extended source in a given direction per unit projected area in the direction
- Expresses in Watts per Steradians per square meters

$$L = \frac{d^2\Phi}{d\Omega dA \cos \theta}$$

Irradiance and Emittance



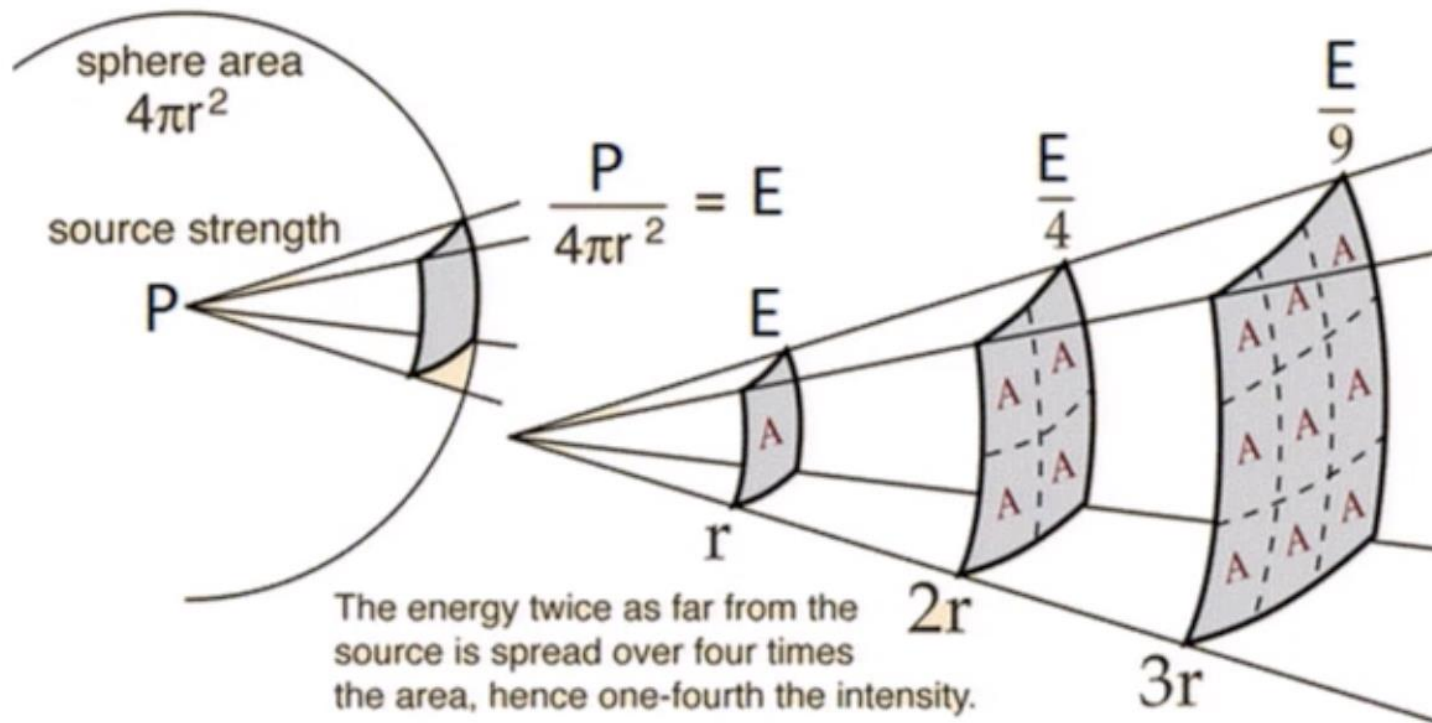
Concept of Radiance



Relationship between radiance and radiant flux density

- Radiant flux density reduces with square of distance whereas
- Radiance is independent of the distance

Units: P in Watt and E in W/m^2



Flux is directionless,
radiance is directional

Relationship between radiance and radiant flux density

Radiant flux density	E (irradiance)	$E, M = \frac{d\Phi}{dA}$	watt/m ²
	M (emittance)		
Radiant intensity	I	$I = \frac{d\Phi}{d\Omega}$	watt/steradian
Radiance	L	$L = \frac{dI}{dA \cos \theta}$	watt/steradian m ²

$$E = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} L \cos \theta d\Omega$$

$$E = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} L \cos \theta \sin \theta d\theta d\varphi$$

$$E = L\pi$$

This integration assumes that the surface is lambertian.

RADIMETRIC MEASUREMENTS

S. No.	Quantity	Symbol	Unit	Definition
1.	Radiant energy	Q	joule	Quantity of energy carried. Is a measure of the radiation to do work.
2.	Radiant flux (radiant power)	Φ	watts	Radiant energy emitted or incident upon a surface per unit time. That is, rate of flow of energy.
3.	Irradiance	E	watts m ⁻²	Radiant flux falling per unit area of the surface.
	Exitance	M	watts m ⁻²	Symbol M is used for radiant flux emitted by unit area.
4.	Radiant intensity	I	watts sr ⁻¹	Radiant flux leaving per unit solid angle, in a specified direction.
5.	Radiance	L	watts m ⁻² sr ⁻¹	Radiant flux per unit projected area and per unit solid angle.

Table 3.1 Radiometric quantities and units.

Radiation quantities

Quantity	Usual symbol	Defining equation	Units
Radiant energy	Q		joule
Radiant energy density	W	$W = \frac{dQ}{dV}$	joule/m ³
Radiant flux	Φ	$\Phi = \frac{dQ}{dt}$	watt
Radiant flux density	E (irradiance) M (emittance)	$E, M = \frac{d\Phi}{dA}$	watt/m ²
Radiant intensity	I	$I = \frac{d\Phi}{d\Omega}$	watt/steradian
Radiance	L	$L = \frac{dI}{dA \cos \theta}$	watt/steradian m ²

- All these quantities can also be considered for different wavelengths (e.g. spectral radiant emittance)

What do satellite measure?

From Energy to Radiance

What is Radiance?

$$I_{\lambda} = dQ/dt/dA/d\lambda/d\Omega$$

I_{λ} : Radiance detected at the satellite sensor for a given wavelength

dQ : Energy coming from the earth/atmosphere (Joules)

dt : Time interval that the satellite detector is sensing energy coming from the earth (Seconds)

dA : Detector area on the satellite that the energy is striking (meter²)

$d\lambda$: Wavelength interval that the satellite is sensing (nano meter)

$d\Omega$: Solid angle through which satellite views the earth (Steradian: Sr)

$$Flux = dQ/dt = \frac{Joules}{Second} = Watts$$

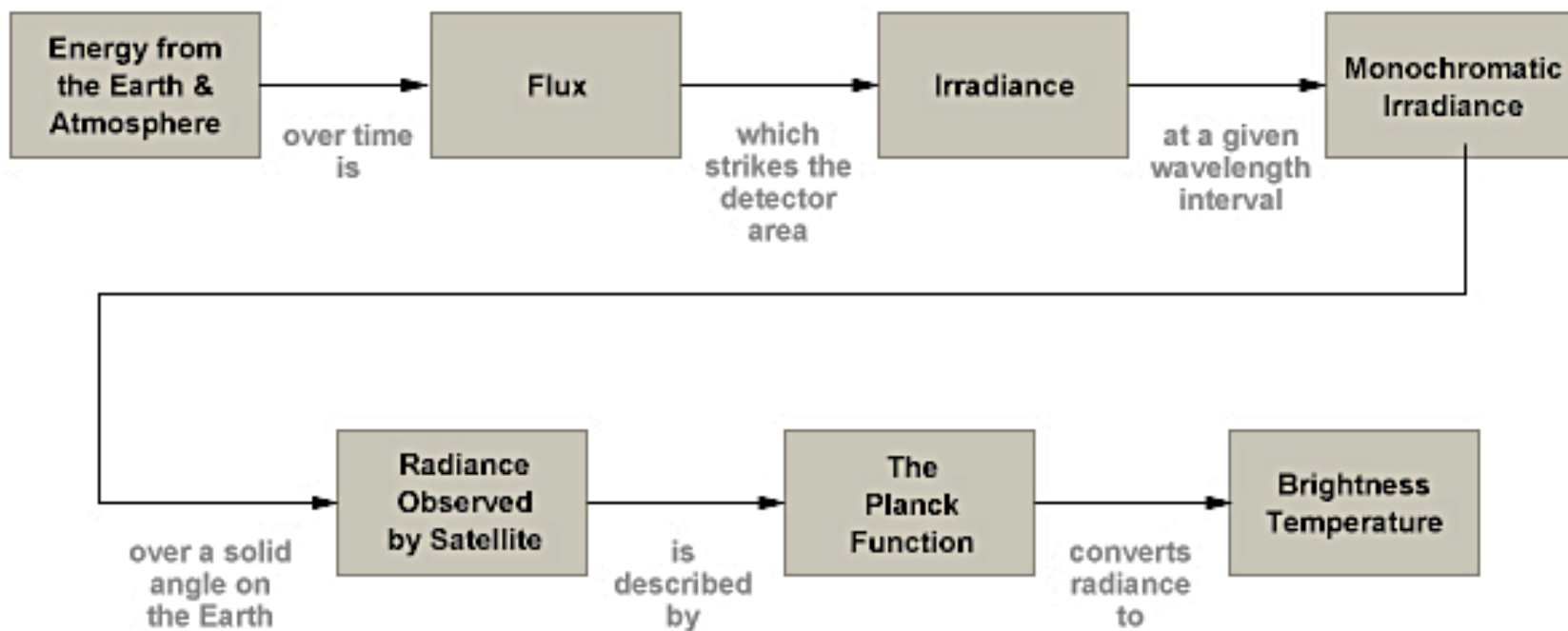
$$Irradiance = dQ/dt/dA = Watts/meter^2$$

$$Monochromatic Irradiance = dQ/dt/dA/d\lambda$$



Footprint grows at large local zenith angle

Summary Chart



©The COMET Program

Inverse Square law

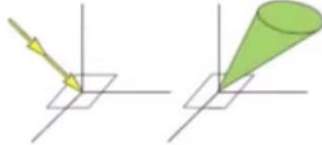
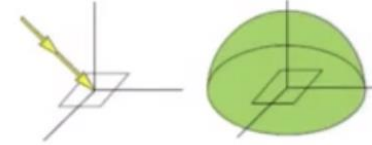
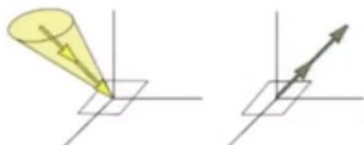
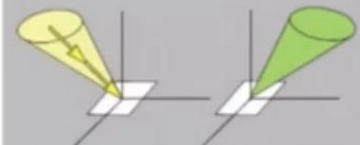
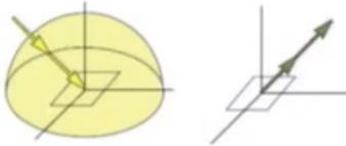
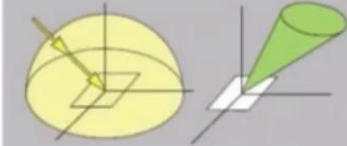
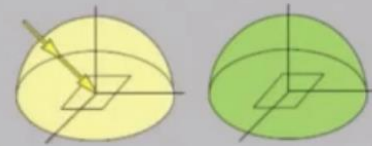
$$E_d = E_s \times \frac{4\pi R_s^2}{4\pi d^2}$$

Calculate the mean solar irradiance at the earth's surface. Assume equivalent black body temperature of sun = 5770 K, radius of sun = 7×10^8 m and mean earth-sun distance = 1.5×10^{11} m

1. Calculate the radiant flux density emitted by sun
 - Stefan boltzman law ($M = \sigma T^4$) = 6.2847×10^7 W/m²
2. Total radiant flux emitted by sun = $M \times \text{area of sun} = P = M \times 4\pi r^2 = 3.8698 \times 10^{26}$ W
3. Irradiance at earth surface / Energy received by earth per unit area = $P / (4\pi d^2) = 1368.67$ W/m² (**Solar constant**)

Directionality of RS measurements

Relation of incoming and reflected radiance terminology used to describe reflectance quantities

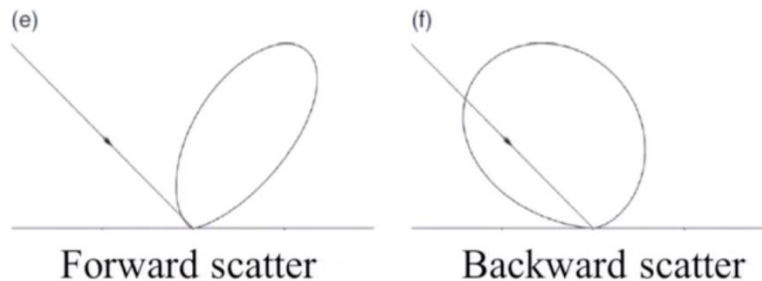
<i>Incoming/Reflected</i>	Directional	Conical	Hemispherical
<i>Directional</i>	Bidirectional CASE 1 	Directional-conical CASE 2 	Directional-hemispherical CASE 3 
<i>Conical</i>	Conical-directional CASE 4 	Biconical CASE 5 	Conical-hemispherical CASE 6 
<i>Hemispherical</i>	Hemispherical-directional CASE 7 	Hemispherical-conical CASE 8 	Bi-hemispherical CASE 9 

Reflectance and albedo

$$\rho_{\lambda} = \frac{\Phi_{\text{reflected}_{\lambda}}}{\Phi_{i_{\lambda}}} \quad \text{Reflectance}$$

- The observed reflectance may either be directional or conical or hemispherical.
- The difference between these may be higher for non-lambertian surfaces.
- For most of the surface studies we need bi-hemispherical reflectance.

Nature of reflecting surfaces



Few objects have a primary direction of reflection



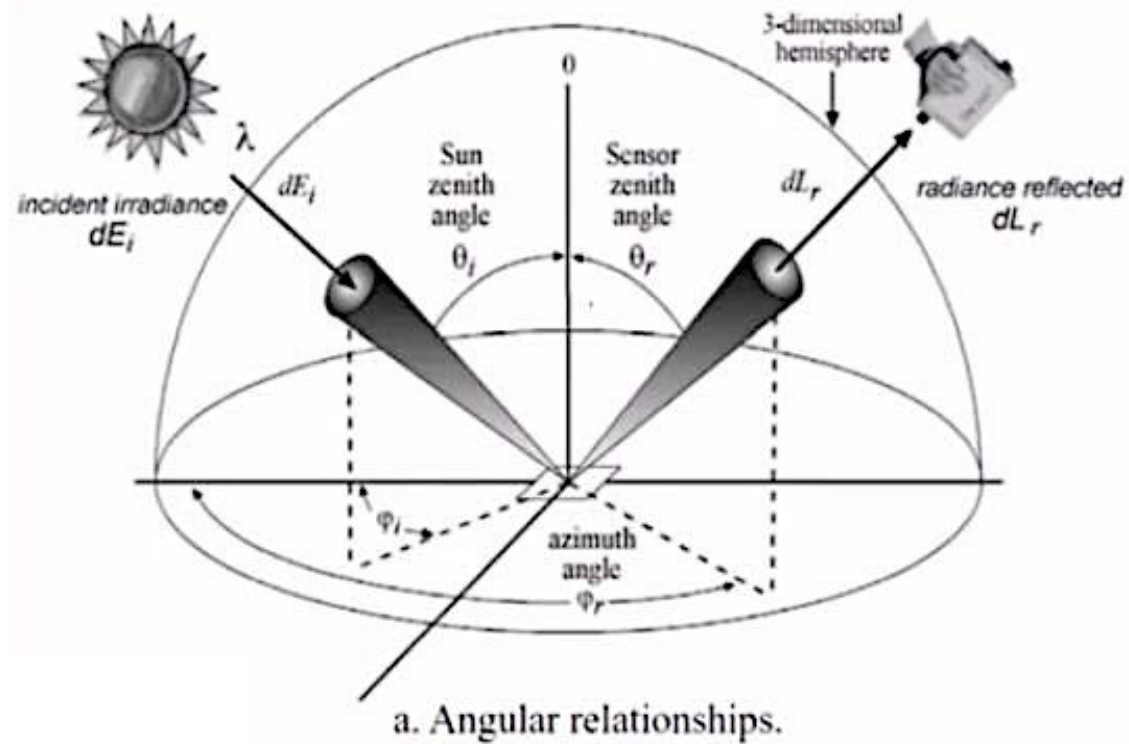
Image courtesy: Prof. Schaff's webpage
https://www.umb.edu/spectralmass/terra_aqua_modis/modis



Reflectance and albedo

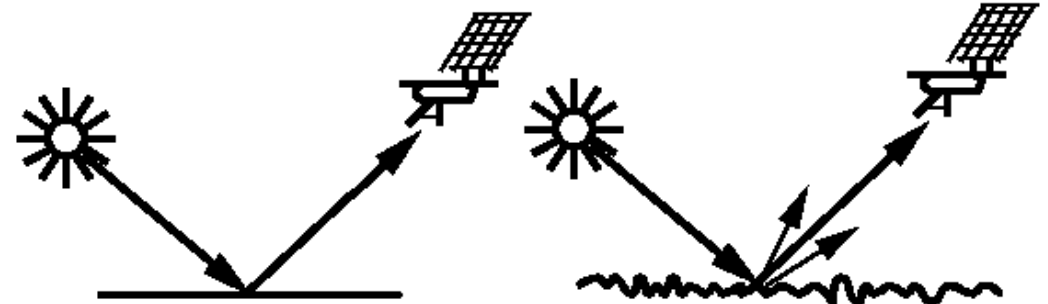
- Directional hemispherical reflectance (or) black sky albedo.
- Bi-hemispherical reflectance (or) albedo.
- Albedo can also be spectral or broadband.
- When pure diffuse light only considered for bi-hemispherical reflectance, it is known as white sky albedo.
- The actual albedo will be a linear combination of white sky and black sky albedo based on fractional of diffused skylight.

Source-object-sensor geometry



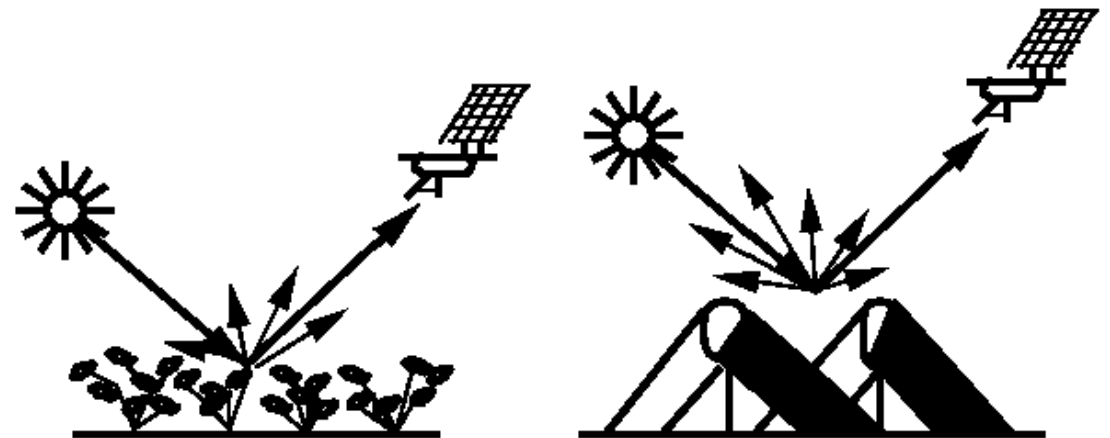
Bidirectional Reflectance Distribution Functions: Causes

Wolfgang Lucht, 1997



Mirror BRDF:
specular reflectance

Rough water surface BRDF:
sunglint reflectance



Volume scattering BRDF:
leaf/vegetation reflectance

Gap-driven BRDF (Forest):
shadow-driven reflectance

BRDF

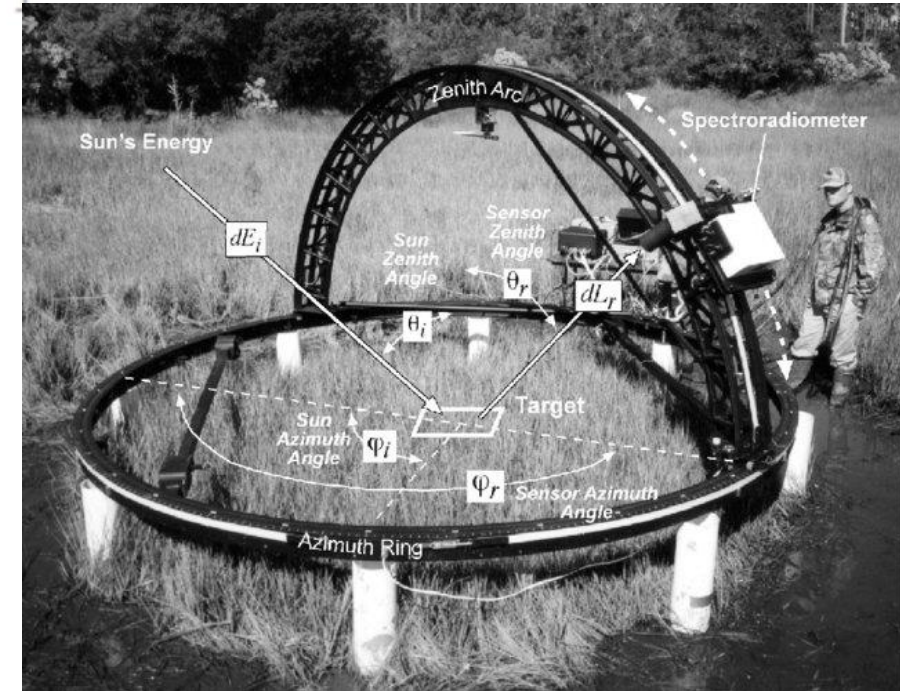
- Bidirectional Reflectance Distribution Function
- Is a measure of reflection of light from a given direction in another direction in the hemisphere.
- Is an intrinsic reflection property of the object.

$$\text{BRDF} = f_r(\theta_i, \phi_i; \theta_r, \phi_r) = \frac{dL_r(\theta_i, \phi_i; \theta_r, \phi_r)}{dE_i(\theta_i, \phi_i)} [\text{sr}^{-1}]$$



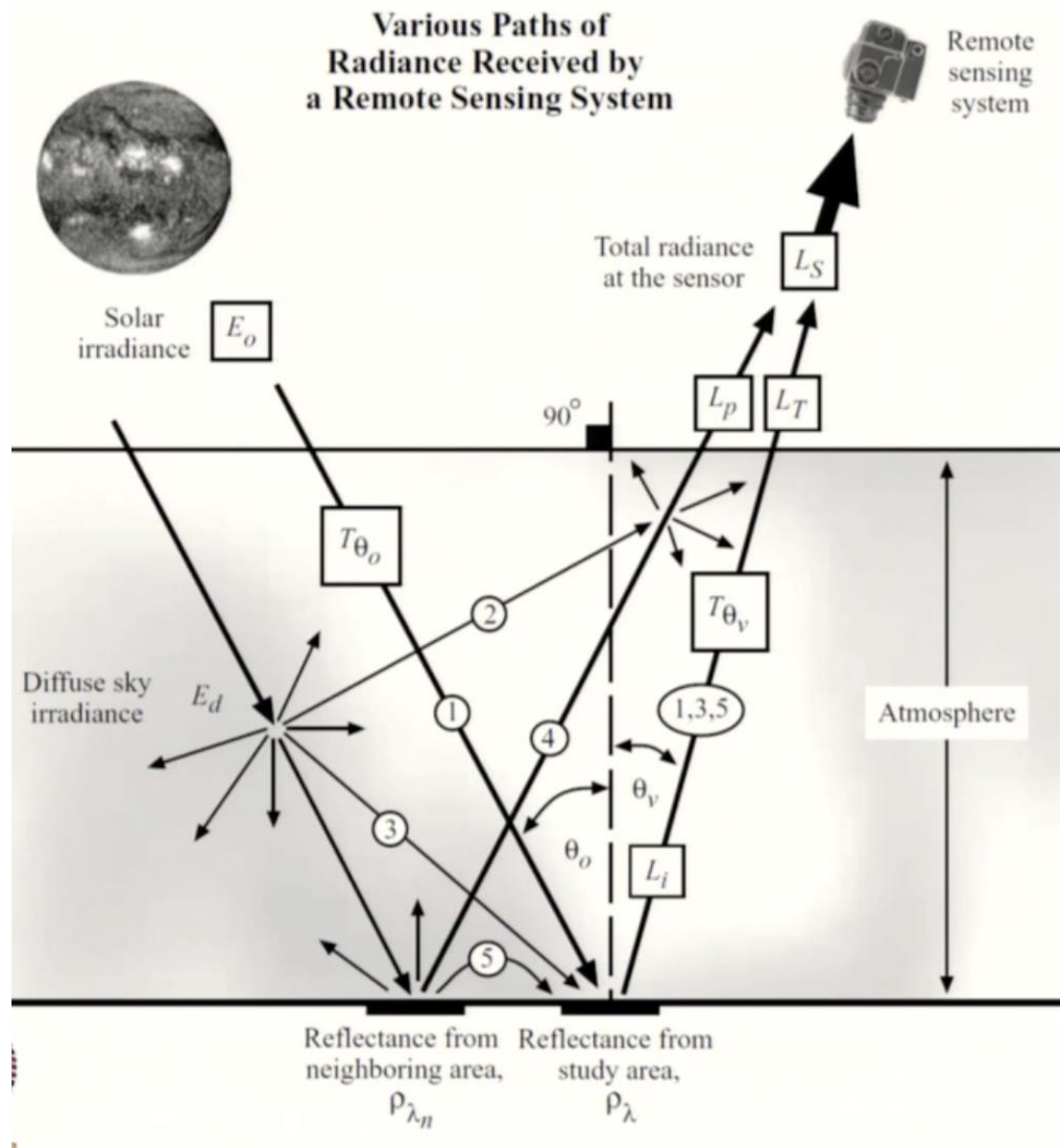
What is the BRDF value for a true Lambertian surface?

$$E=L \times \pi \rightarrow \text{BRDF} = 1/\pi$$



Reflectance and BRDF

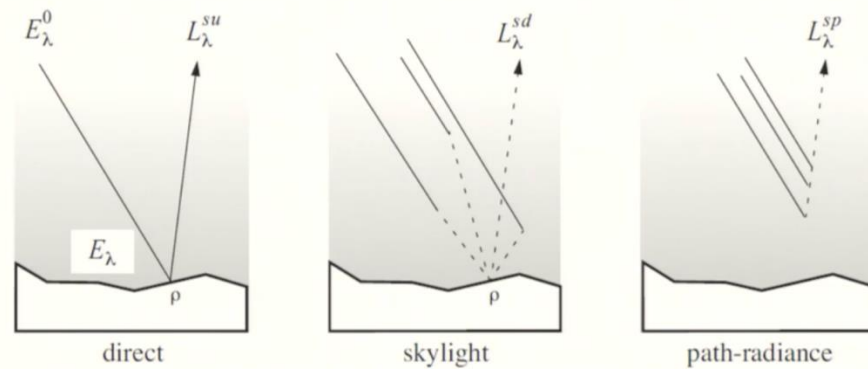
- Remember: BRDF and Reflectance are **different**!
- **BRDF is a function**, when **integrated** over corresponding solid angles (incident and reflected) and viewing geometry (elevation and azimuth angles) **will give us reflectance**.
- Reflectance has no units. BRDF has a unit of sr^{-1} .



Various paths of energy to RS system

1. Direct radiance
2. Path radiance
3. Reflected diffuse
4. Reflection from neighboring features
5. Multiple reflections

Radiation reaching the sensor at shorter wavelengths



$$L_{\lambda}^s = L_{\lambda}^{su} + L_{\lambda}^{sd} + L_{\lambda}^{sp}$$

Calculating radiance reaching the sensor in visible, NIR and SWIR bands

Solar spectral irradiance
reaching the TOA

$$E_{\lambda}^0 = \frac{M_{\lambda}}{\pi} \times \frac{\text{area solar disk}}{(\text{distance-to-earth})^2}$$

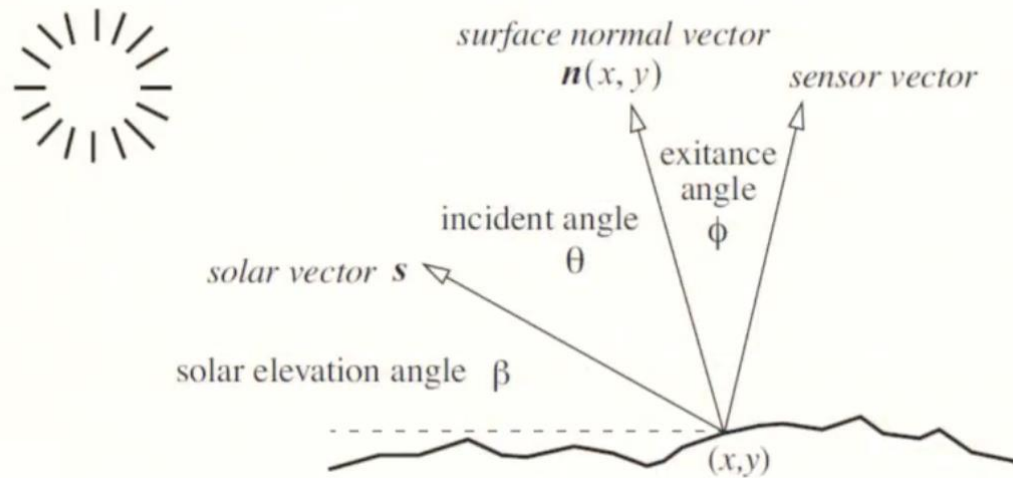
Solar spectral irradiance
reaching the surface

$$E_{\lambda} = \tau_s(\lambda) E_{\lambda}^0$$

Calculating radiance reaching the sensor in visible, NIR and SWIR bands

earth's surface:

$$E_{\lambda}(x, y) = \tau_s(\lambda) E_{\lambda}^0 \mathbf{n}(x, y) \cdot \mathbf{s}$$
$$= \tau_s(\lambda) E_{\lambda}^0 \cos[\theta(x, y)]$$



Calculating radiance reaching the sensor in visible, NIR and SWIR bands

Interaction with the surface

$$\begin{aligned} L_{\lambda}(x, y) &= \rho(x, y, \lambda) \frac{E_{\lambda}(x, y)}{\pi} \\ \text{earth's surface:} \quad &= \rho(x, y, \lambda) \frac{\tau_s(\lambda) E_{\lambda}^0}{\pi} \cos[\theta(x, y)] \end{aligned}$$

Reflected radiation reaching the sensor

$$\begin{aligned} L_{\lambda}^{su} &= \tau_v(\lambda) L_{\lambda} \\ \text{at-sensor:} \quad &= \rho(x, y, \lambda) \frac{\tau_v(\lambda) \tau_s(\lambda) E_{\lambda}^0}{\pi} \cos[\theta(x, y)] . \end{aligned}$$

Calculating radiance reaching the sensor in visible, NIR and SWIR bands

Diffuse radiation reaching the sensor (Component 2)

$$\text{at-sensor: } L_{\lambda}^{sd} = F(x, y) \rho(x, y, \lambda) \frac{\tau_v(\lambda) E_{\lambda}^d}{\pi}.$$

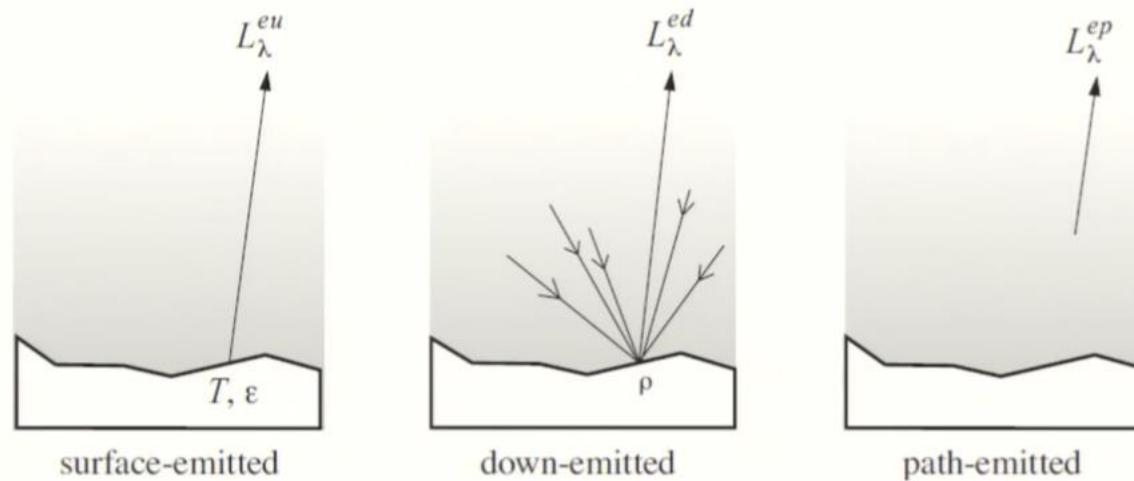
$F(x, y)$ = Fraction of sky

Total at sensor radiance in shorter wavelengths

at-sensor:

$$\begin{aligned} L_{\lambda}^s(x, y) &= L_{\lambda}^{su}(x, y) + L_{\lambda}^{sd}(x, y) + L_{\lambda}^{sp} \\ &= \rho(x, y, \lambda) \frac{\tau_v(\lambda) \tau_s(\lambda) E_{\lambda}^0}{\pi} \cos[\theta(x, y)] + F(x, y) \rho(x, y, \lambda) \frac{\tau_v(\lambda) E_{\lambda}^d}{\pi} + L_{\lambda}^{sp} \\ &= \rho(x, y, \lambda) \frac{\tau_v(\lambda)}{\pi} \{ \tau_s(\lambda) E_{\lambda}^0 \cos[\theta(x, y)] + F(x, y) E_{\lambda}^d \} + L_{\lambda}^{sp}. \end{aligned}$$

Radiation reaching the sensor at longer (MWIR and LWIR) wavelengths



$$L_{\lambda}^e = L_{\lambda}^{eu} + L_{\lambda}^{ed} + L_{\lambda}^{ep}$$

Radiation reaching the sensor at longer (MWIR and LWIR) wavelengths

earth's surface: $L_{\lambda}(x, y) = \varepsilon(x, y, \lambda) \frac{M_{\lambda}[T(x, y)]}{\pi}$

at-sensor:

$$\begin{aligned} L_{\lambda}^{eu}(x, y) &= \tau_v(\lambda) L_{\lambda}(x, y) \\ &= \varepsilon(x, y, \lambda) \frac{\tau_v(\lambda) M_{\lambda}[T(x, y)]}{\pi} \end{aligned}$$

Other components reaching the sensor at MWIR and LWIR

Reflected downward emitted reaching the sensor (Component 2)

at-sensor: $L_{\lambda}^{ed} = F(x, y, \lambda) \rho(x, y, \lambda) \frac{\tau_v(\lambda) M_{\lambda}^a}{\pi}$

Total at sensor radiance in MWIR and LWIR wavelengths

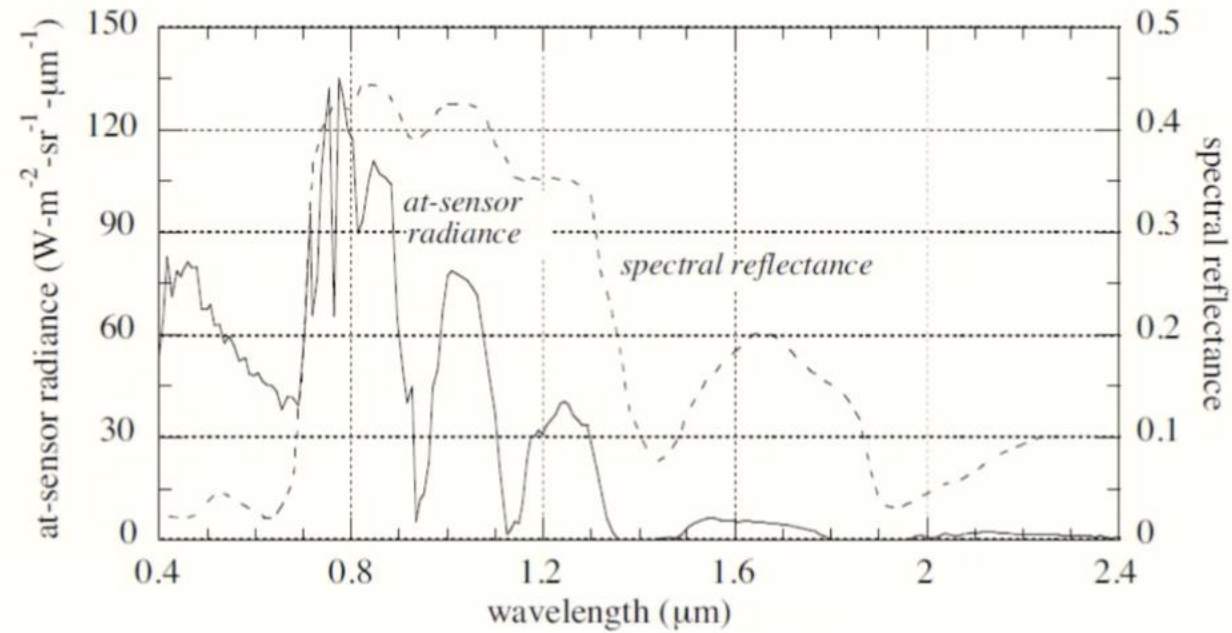
at-sensor:

$$\begin{aligned} L_{\lambda}^e(x, y) &= L_{\lambda}^{eu} + L_{\lambda}^{ed} + L_{\lambda}^{ep} \\ &= \varepsilon(x, y, \lambda) \frac{\tau_v(\lambda)}{\pi} [M_{\lambda}[T(x, y)]] + F(x, y, \lambda) \rho(x, y, \lambda) \frac{\tau_v(\lambda) M_{\lambda}^a}{\pi} + L_{\lambda}^{ep} \end{aligned}$$

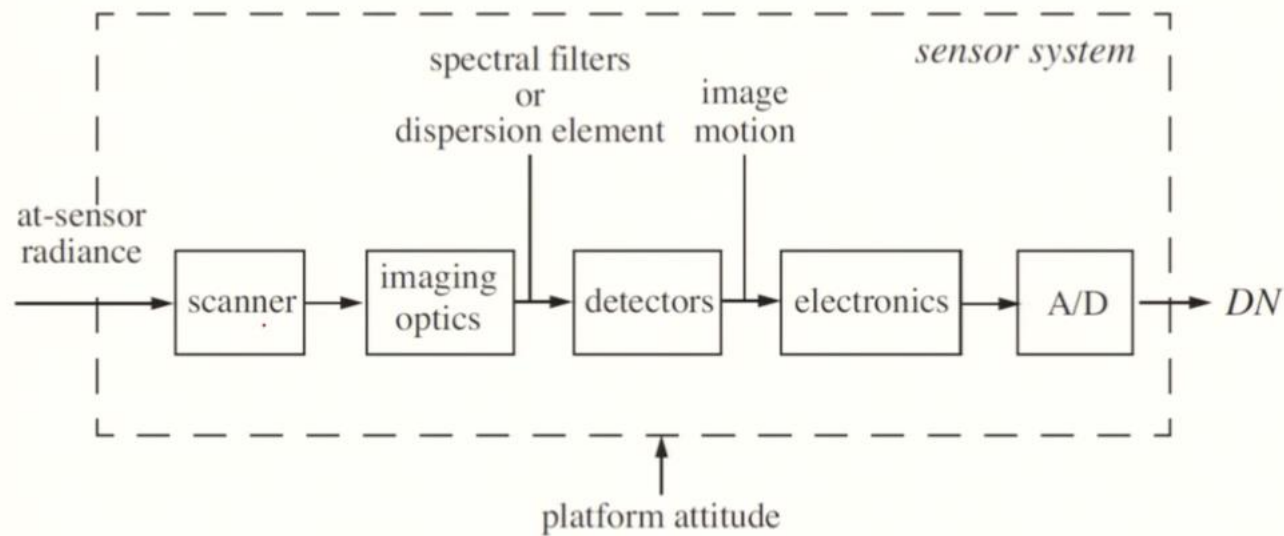
Example

For green grass

The at-sensor radiance must be corrected for atmospheric, solar and topographic effects before comparing with spectral libraries

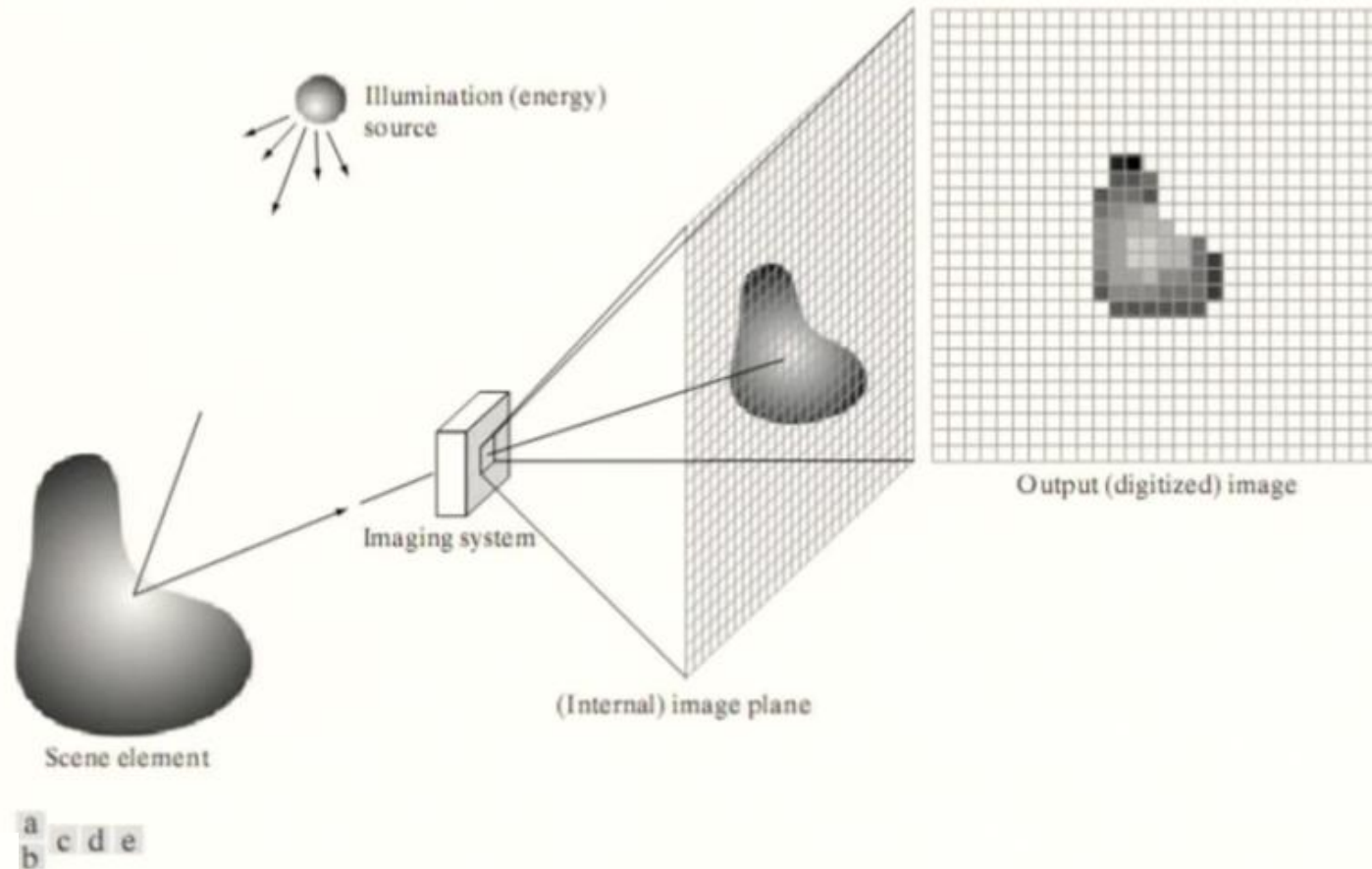


Typical RS data acquisition process

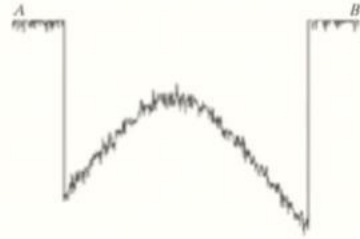
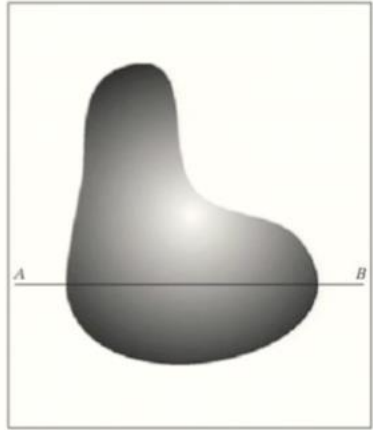


However, please note that, in addition to DN, many satellite products are now available for various geophysical parameters itself (e.g. surface reflectance, Land Surface Temperature etc.). These products belongs to [level 2](#), [level 3](#) or [level 4](#) level processing.

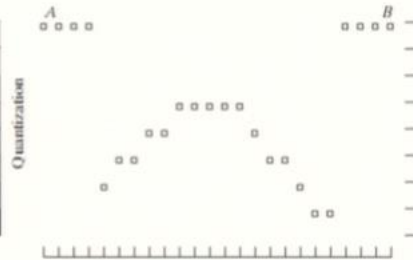
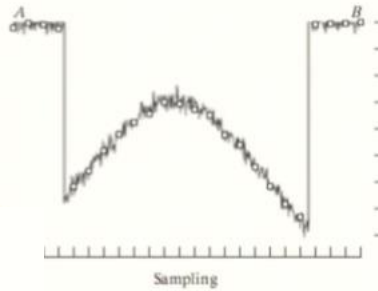
Image formation process



Sampling and Quantization

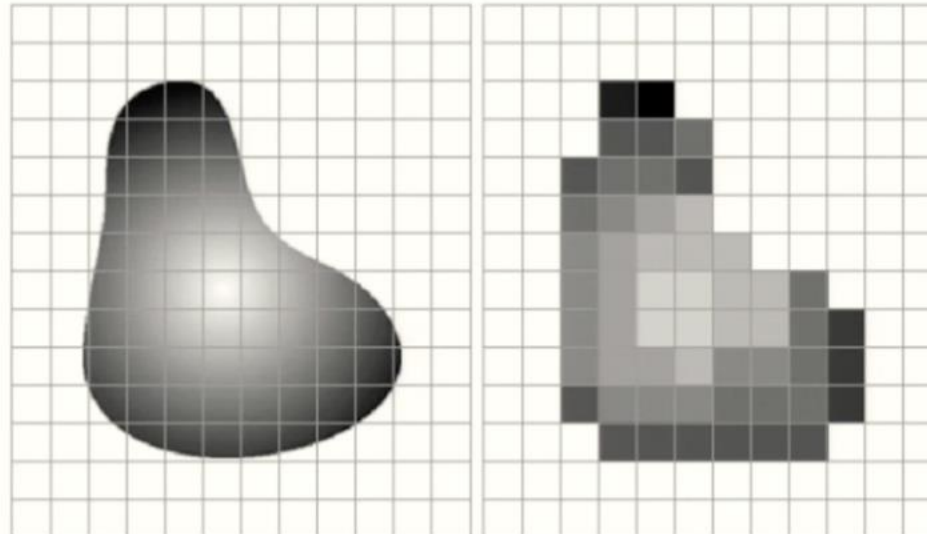


The received radiance is rescaled, converted to an integer and then stored as DN (**Digital Number**) in the image.
e.g. Landsat Level-1 data products will have DN values stored in them.



Quantization

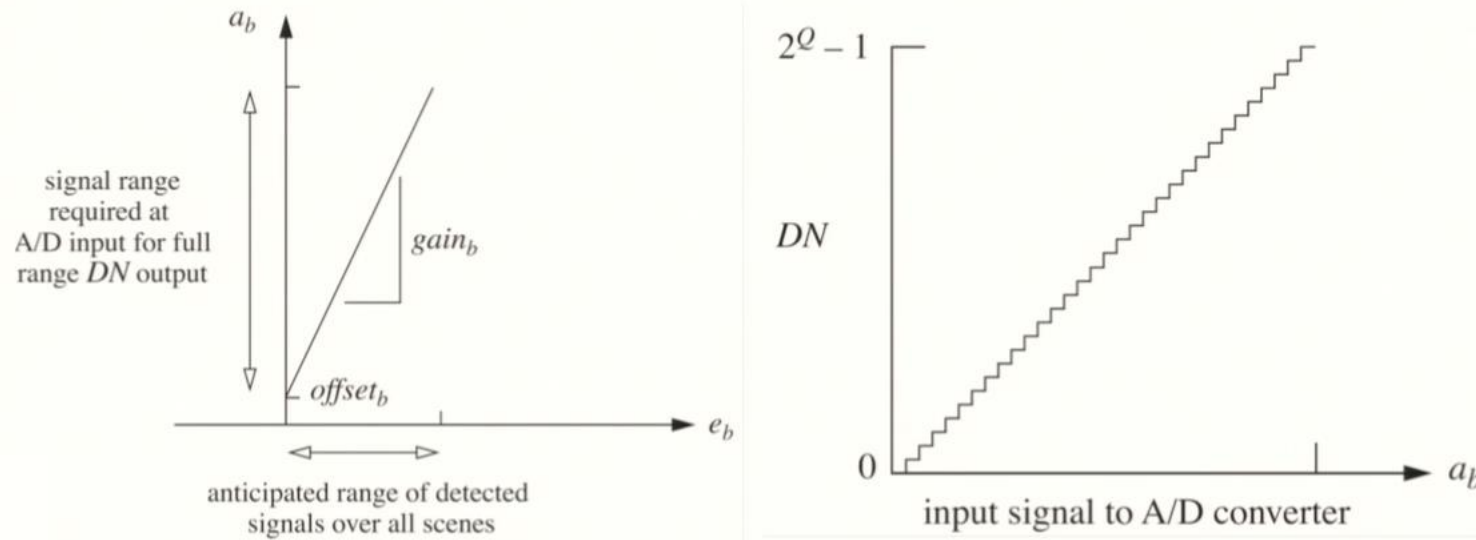
Sampling



The number of bits used for quantization is an indicator of the **radiometric resolution** of the sensor.

Radiometric Calibration

- Radiometric calibration is done in order to relate the observed radiance with the DN.
- Each RS system will have a band specific values to convert the observed radiance to DN.



- The basic conversion followed is:

$$Q_{cal} = Gain(L_{TOA} - L_{TOA,min}) + Q_{cal,min}$$

$$Gain = \frac{Q_{cal,max} - Q_{cal,min}}{L_{TOA,max} - L_{TOA,min}}$$

Gain will have units of counts per unit radiance.

Conversion of DN back to radiance

- As users, we will get DN in our images (for L1 data). For several applications, we need to convert this DN to radiance before getting any useful information.

$$L_{TOA} = \frac{L_{TOA,max} - L_{TOA,min}}{Q_{cal,max} - Q_{cal,min}} (Q_{cal} - Q_{cal,min}) + L_{TOA,min}$$

Confusingly, the scaling factor here (actually, $1/G$) is also referred as [Gain](#).

In recent datasets, this conversion is represented as a linear function with the form as:

$$L_{TOA} = M_L \times Q_{cal} + A_L$$

Where, M_L and A_L are multiplicative and additive factors respectively.

Conversion of radiance to reflectance

- In visible, NIR and SWIR bands, we are interested in obtaining the reflectance of the surface.
- The TOA radiance can be converted to TOA reflectance as:

$$\rho_{TOA} = \frac{\pi * L_{sat} * d^2}{E_{sun} * \cos(\theta_s)}$$

Here, E_{sun} is the solar constant, however at the wavelength corresponding to the sensor's spectral bandwidth and weighted by sensor's spectral response.

d is the earth-sun distance factor (in Astronomical units). Since E_{sun} is calculated based on average earth sun distance (= 1 AU) , we need this factor.

θ_s is the solar zenith angle.

In the numerator, we have converted the sensor radiance to reflected radiant flux density using [conversion factor \$\pi\$](#) (assuming Lambertian surface).

Conversion of radiance to reflectance

- What we are really interested is to remove the effects of atmosphere and get surface reflectance.

Please recall what we have learnt in the previous lecture: The radiance reaching the sensor in visible, NIR and SWIR wavelengths is:

at-sensor:

$$\begin{aligned} L_{\lambda}^s(x, y) &= L_{\lambda}^{su}(x, y) + L_{\lambda}^{sd}(x, y) + L_{\lambda}^{sp} \\ &= \rho(x, y, \lambda) \frac{\tau_v(\lambda) \tau_s(\lambda) E_{\lambda}^0}{\pi} \cos[\theta(x, y)] + F(x, y) \rho(x, y, \lambda) \frac{\tau_v(\lambda) E_{\lambda}^d}{\pi} + L_{\lambda}^{sp} \\ &= \rho(x, y, \lambda) \frac{\tau_v(\lambda)}{\pi} \{ \tau_s(\lambda) E_{\lambda}^0 \cos[\theta(x, y)] + F(x, y) E_{\lambda}^d \} + L_{\lambda}^{sp} . \end{aligned}$$

On rearranging the previous equation, we get the equation for surface reflectance.

$$\rho_{surf} = \frac{\pi * (L_{sat} - L_{path})}{\tau_v \left(\frac{E_{sun}}{d^2} * \cos(\theta_s) * \tau_s + E_{down} \right)}$$

Remember, this is for a horizontal surface assuming Lambertian reflecting nature!

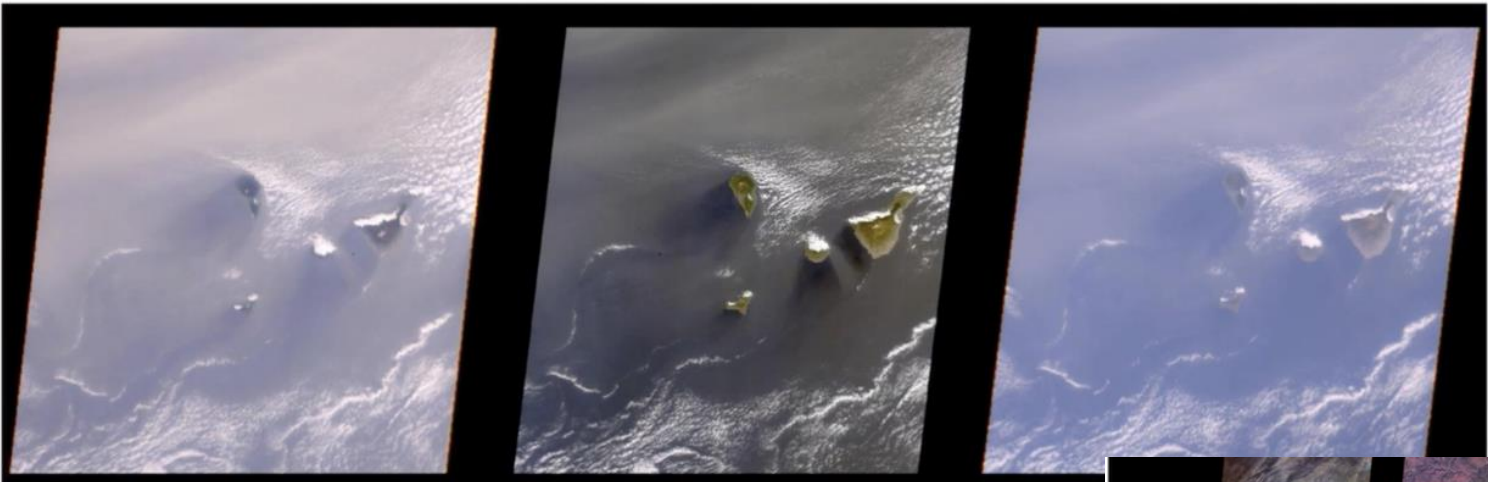
Removing the effects of atmosphere

The atmospheric effects can be reduced/removed using

1. A radiative transfer model (RTM) with field observed atmospheric observation at the time of satellite overpass (ideal),
2. A RTM with modelled atmospheric physical and chemical characteristics (practical),
3. Atmospheric correction using natural or artificial reflectance targets within the images (used in calibration exercises)
4. Image based atmospheric correction (practical with limitations).

Change in atmosphere transmissivity and surface reflectance with viewing angle

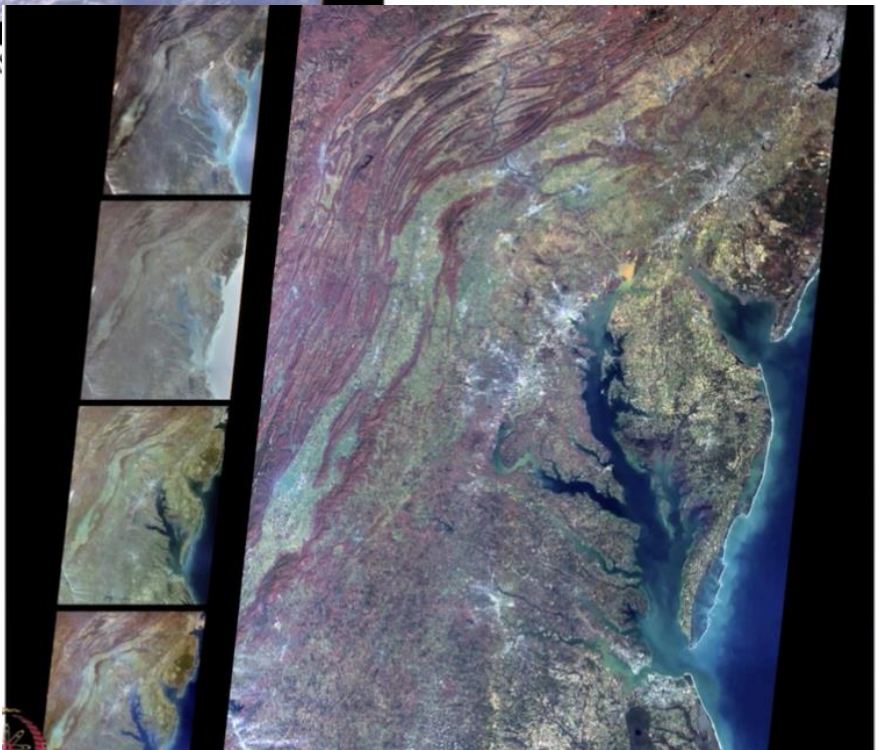
Image acquired on 29th February, 2000 over Canary Islands



MISR, 70.5° Forward looking

MISR, Nadir looking

MISR, 70.5° Backward looking



Change in surface reflectance with viewing angle

Image acquired on 24th March, 2000 over parts of USA

Simple image based atmospheric correction

- One of the popular image based correction method is the Dark Object Subtraction (DOS) method.
- The DOS method corrects the image for the path radiance alone.
- The DOS method assumes that some objects within each band image will be perfectly dark and will have near zero radiance and whatever radiance observed over those pixels is essentially due to path radiance.
- That estimated path radiance will be subtracted from the satellite observed radiance for all the pixels.
- Care must be taken while choosing the dark pixels in different bands as choosing different pixels may yield varied results.

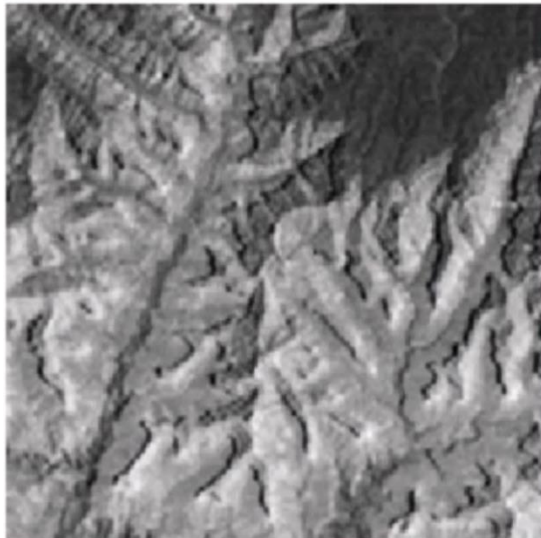
Caution!

- Image based models are simplified models of the atmospheric effects.
- Many models are available each having its own assumption.
- The performance of the models will vary spatially and temporally (e.g. some models may not perform well over humid climates).
- Though the models offer practical solutions, keep in mind that the final corrected reflectance will still contain errors!

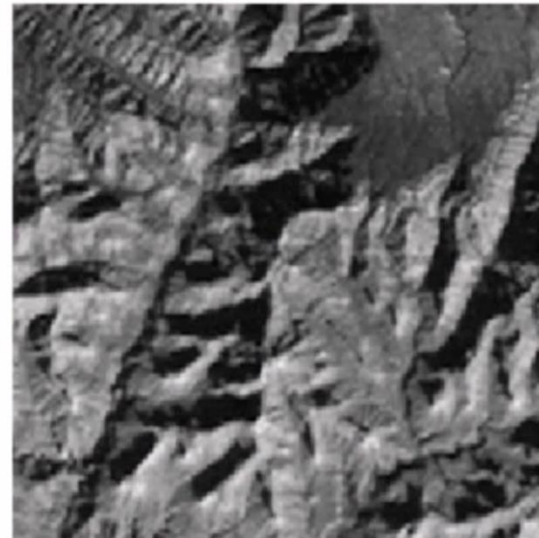
Topographic correction

- The SR obtained over hilly areas, will have the effects of topography. i.e. some portions will be shadowed and some portions will be under direct sun exposure.
- For such images, the SR in the image will be different from the value that would be if the area is flat and horizontal.
- Hence, it is advised to perform topographic correction on images acquired over areas with large variation in topography.

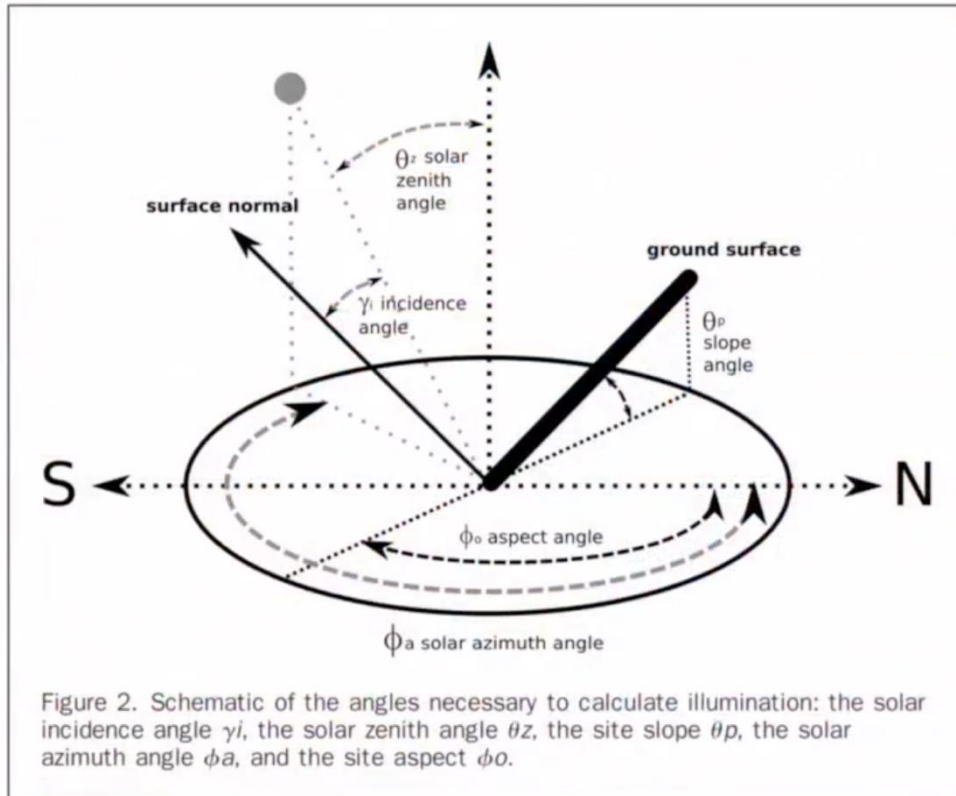
Topographic and sun angle effect



June 11, 1981



October 20, 1980



Goslee S. C. 2012. Topographic corrections of satellite data for regional monitoring. *Photogrammetric Engineering and Remote Sensing*, 78, 973–981.

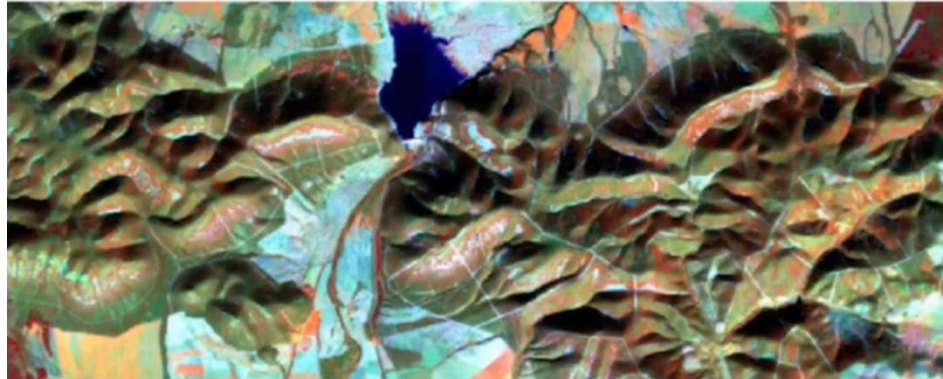
Caution!

Satellite view angle is not considered here, assuming **nadir** viewing sensors.

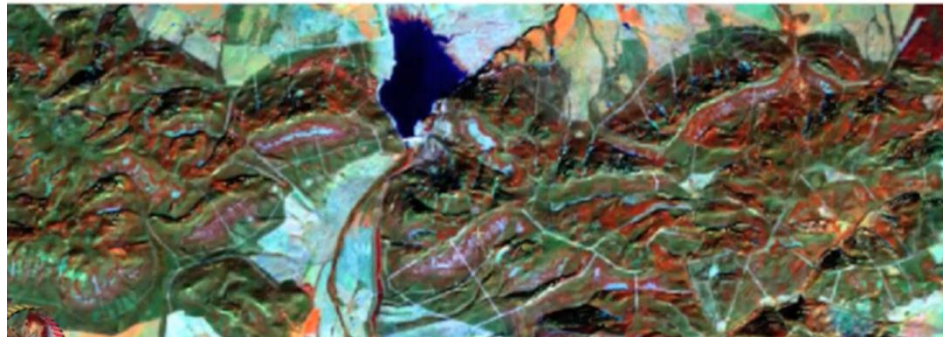
$$\cos \gamma_i = \cos \theta_s \cos \eta_i + \sin \theta_s \sin \eta_i \cos(\phi_a - \phi_o)$$

Topographic correction models

- There are different models, which correct the image for topographic effects.
- The models either assume the surface to be Lambertian or non-Lambertian.
- The models use elevation information (Digital Elevation Model, DEM) to correct for the topographic effects.
- Most of these models correct the data for variation in angle of illumination.
- Please remember, that these correction methods also can only partially correct areas under high illumination and shadows.
- Sometimes, there can be over correction due to the assumption of Lambertian surface.



Topographic correction



Hantson and Chuvieco (2011), Evaluation of different topographic correction methods for Landsat imagery, International Journal of Applied Earth Observation and Geoinformation, 13, 691–700

Two simple models

$$\rho_{\lambda,h,i} = \rho_{\lambda,i} \left(\frac{\cos \theta_s}{\cos \gamma_i} \right)$$

- **Cosine correction model** which assumes the surface as Lambertian.
- Corrects the reflectance only for illumination geometry.
- Correction is independent of wavelength.

$$\rho_{\lambda,h,i} = \rho_{\lambda,i} \left(\frac{\cos \theta_s + c_\lambda}{\cos \gamma_i + c_\lambda} \right)$$

$$c_\lambda = \left(\frac{b_\lambda}{m_\lambda} \right)$$

- **C-correction model** tries to overcome this band independent correction to some degree
- An empirical correction factor c is introduced, which captures the relationship between reflectance in a band with illumination geometry.

$$\rho_{\lambda,i} = b_\lambda + m_\lambda \cos \gamma_i$$

This equation will be obtained using linear regression.